



# Electrical and Electronic Measurements

Detailed Solutions

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**GATE Electrical Engineering**

**PrepFusion<sup>TM</sup>**

Previous Year Questions Solutions

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## SOLUTIONS

# I

## Error Analysis

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### Topics

1. Static Characteristics of Measuring Device
2. Types of Error and Uncertainties of error

PrepFusion

**Q1.25.1**
**NAT**
**2025**
**1M**
**Ans: 5**


Given:

$$R_1 = 100 \, \Omega, \quad R_2 = 900 \, \Omega$$

Each resistor has a maximum error of 5%.

We need the maximum percentage error in the equivalent resistance when the resistors are connected in parallel.

**Method 1**

For two resistors in parallel,

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

Taking logarithms,

$$\ln R = \ln R_1 + \ln R_2 - \ln(R_1 + R_2)$$

Differentiating,

$$\frac{dR}{R} = \frac{dR_1}{R_1} + \frac{dR_2}{R_2} - \frac{dR_1 + dR_2}{R_1 + R_2}$$

For maximum error, use absolute values:

$$\frac{\Delta R}{R} = \frac{\Delta R_1}{R_1} + \frac{\Delta R_2}{R_2} - \frac{\Delta R_1 + \Delta R_2}{R_1 + R_2}$$

Now,

$$\Delta R_1 = 0.05 \times 100 = 5 \, \Omega, \quad \Delta R_2 = 0.05 \times 900 = 45 \, \Omega$$

Substituting,

$$\begin{aligned} \frac{\Delta R}{R} &= \frac{5}{100} + \frac{45}{900} - \frac{5 + 45}{100 + 900} \\ &= 0.05 + 0.05 - 0.05 = 0.05 \end{aligned}$$

Hence, the maximum percentage error is

5

**Method 2**

Nominal equivalent resistance:

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{100 \times 900}{100 + 900} = \frac{90000}{1000} = 90 \, \Omega$$

For maximum resistance, take both resistors at their maximum values:

$$R_{1,\max} = 100 + 5 = 105 \, \Omega$$

$$R_{2,\max} = 900 + 45 = 945 \, \Omega$$

Then,

$$R_{\max} = \frac{105 \times 945}{105 + 945} = \frac{99225}{1050} = 94.5 \, \Omega$$

So,

$$\Delta R = 94.5 - 90 = 4.5 \, \Omega$$

Percentage error:

$$\frac{\Delta R}{R} \times 100 = \frac{4.5}{90} \times 100 = 5$$

Thus,

5

### Key Points

- Logarithmic differentiation is a convenient method for error propagation in composite expressions.

### Mistakes to be avoided

- While using logarithmic differentiation, ensure the negative sign for the denominator term is retained.
- Do not directly add  $5\% + 5\%$ ; the denominator term also contributes to the error expression.

**Q1.22.1**

MCQ

2022

1M

Ans: B



For a balanced Wheatstone bridge,

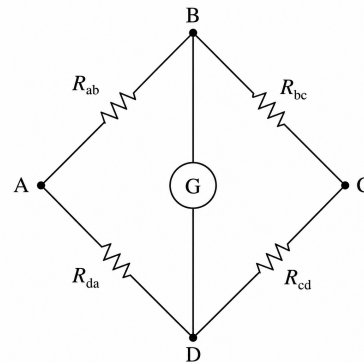


Fig. Solution Q1.22.1

$$\frac{R_{BC}}{R_{AB}} = \frac{R_{CD}}{R_{DA}}$$

Substituting the given nominal values,

$$R_{AB} = 1 \text{ k}\Omega = 1000 \Omega, \quad R_{BC} = 100 \Omega, \quad R_{DA} = 300 \Omega$$

$$R_{CD} = \frac{100 \times 300}{1000} = 30 \Omega$$

Now,

$$R_{CD} = \frac{R_{BC} R_{DA}}{R_{AB}}$$

For multiplication/division,

$$\frac{\Delta R_{CD}}{R_{CD}} = \frac{\Delta R_{BC}}{R_{BC}} + \frac{\Delta R_{DA}}{R_{DA}} + \frac{\Delta R_{AB}}{R_{AB}}$$

Given percentage errors:

$$R_{AB} : 2.1\%, \quad R_{BC} : 0.5\%, \quad R_{DA} : 0.4\%$$

Therefore,

$$\% \text{ error} = 2.1 + 0.5 + 0.4 = 3.0\%$$

Absolute error:

$$\Delta R_{CD} = \frac{3}{100} \times 30 = 0.9 \Omega$$

Hence,

$$R_{CD} = 30 \Omega \pm 0.9 \Omega$$

Therefore, the correct option is

$$\boxed{\text{B. } 30 \Omega \pm 0.9 \Omega}$$

## Key Points

- For multiplication/division operations, percentage errors add directly.

**Q1.20.1**

**NAT**

**2020**

**2M**

**Ans: 5 to 6**



The diode current equation is, taking  $\eta = 1$ ,

$$I = I_0 e^{V_D/V_T}$$

We need the percentage uncertainty in current due to uncertainties in  $V_T$  and  $V_D$ .

Step 1: Take logarithm

Taking natural logarithm,

$$\ln I = \ln I_0 + \frac{V_D}{V_T}$$

Step 2: Sensitivity with respect to  $V_T$

Differentiating w.r.t  $V_T$ ,

$$\frac{1}{I} \frac{\partial I}{\partial V_T} = -\frac{V_D}{V_T^2}$$

Thus,

$$\left| \frac{\partial I}{\partial V_T} \right| = \frac{V_D}{V_T^2} I$$

Step 3: Sensitivity with respect to  $V_D$

Differentiating w.r.t  $V_D$ ,

$$\frac{1}{I} \frac{\partial I}{\partial V_D} = \frac{1}{V_T}$$

Hence,

$$\left| \frac{\partial I}{\partial V_D} \right| = \frac{I}{V_T}$$

Step 4: Use uncertainty propagation formula

For independent uncertainties,

$$W_I = \sqrt{\left( \frac{\partial I}{\partial V_T} W_{V_T} \right)^2 + \left( \frac{\partial I}{\partial V_D} W_{V_D} \right)^2}$$

Dividing by  $I$ ,

$$\frac{W_I}{I} = \sqrt{\left( \frac{V_D}{V_T^2} W_{V_T} \right)^2 + \left( \frac{1}{V_T} W_{V_D} \right)^2}$$

Step 5: Substitute values

$$V_T = 29 \text{ mV} = 0.029 \text{ V}$$

$$W_{V_T} = 2 \text{ mV} = 0.002 \text{ V}$$

$$V_D = 0.02 \text{ V}$$

$$W_{V_D} = 1 \text{ mV} = 0.001 \text{ V}$$

Therefore,

$$\frac{W_I}{I} = \sqrt{\left( \frac{0.02}{(0.029)^2} (0.002) \right)^2 + \left( \frac{0.001}{0.029} \right)^2}$$

$$\frac{W_I}{I} = \sqrt{0.002262 + 0.001189}$$

$$= \sqrt{0.003451} = 0.0587$$

Percentage uncertainty:

$$0.0587 \times 100 = 5.87\%$$

5.87

### Key Points

- If measured quantities x,y have uncertainties/errors, then how to calculate the uncertainty in a derived quantity q:

- Addition:**

$$q = mx + ny \rightarrow \delta q = \sqrt{(m\delta x)^2 + (n\delta y)^2}$$

- Subtraction:**

$$q = mx - ny \rightarrow \delta q = \sqrt{(m\delta x)^2 + (n\delta y)^2}$$

- Multiplication:**

$$q = mxy \rightarrow \frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

- Division:**

$$q = m\frac{x}{y} \rightarrow \frac{\delta q}{q} = \sqrt{\left(\frac{\delta x}{x}\right)^2 + \left(\frac{\delta y}{y}\right)^2}$$

- Power:**

$$q = x^m y^n \rightarrow \frac{\delta q}{q} = \sqrt{\left(m\frac{\delta x}{x}\right)^2 + \left(n\frac{\delta y}{y}\right)^2}$$

- Logarithm:**

$$q = \ln(mx) \rightarrow \delta q = \frac{\delta x}{|x|}$$

- Exponential:**

$$q = e^{mx} \rightarrow \frac{\delta q}{q} = m\delta x$$

**Q1.17.1**

**NAT**

**2017**

**1M**

**Ans: 4.0 to 4.1**



Given measurements:

$$V = 220 \text{ V} \pm 1\%, \quad I = 5 \text{ A} \pm 1\%, \quad W = 555 \text{ W} \pm 2\%$$

For a single-phase load,

$$\cos \phi = \frac{W}{VI}$$

For multiplication/division, add the worst case errors of measured value:

$$\% \Delta(\cos \phi) = \% \Delta W + \% \Delta V + \% \Delta I$$

Substituting the given values,

$$\% \Delta(\cos \phi) = 2 + 1 + 1 = 4\%$$

Hence, the worst-case percentage error in power factor is

4.0

**Q1.17.2**
**MCQ**
**2017**
**1M**
**Ans: A**


Let the equivalent resistance of two parallel resistors be

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

where  $R_1, R_2$  are measured resistances and  $\Delta R_1, \Delta R_2$  are their uncertainties.

If a quantity depends on multiple variables,

$$R = f(R_1, R_2),$$

then the small change in  $R$  is

$$dR = \frac{\partial R}{\partial R_1} dR_1 + \frac{\partial R}{\partial R_2} dR_2.$$

For uncertainties,

$$\Delta R = \frac{\partial R}{\partial R_1} \Delta R_1 + \frac{\partial R}{\partial R_2} \Delta R_2.$$

But since uncertainties are independent random quantities, we use RMS combination:

$$(\Delta R)^2 = \left( \frac{\partial R}{\partial R_1} \Delta R_1 \right)^2 + \left( \frac{\partial R}{\partial R_2} \Delta R_2 \right)^2.$$

Thus,

$$\Delta R = \pm \sqrt{\left( \frac{\partial R}{\partial R_1} \Delta R_1 \right)^2 + \left( \frac{\partial R}{\partial R_2} \Delta R_2 \right)^2}.$$

Therefore, the correct option is

$$\text{A. } \pm \sqrt{\left( \frac{\partial R}{\partial R_1} \Delta R_1 \right)^2 + \left( \frac{\partial R}{\partial R_2} \Delta R_2 \right)^2}$$

### Key Points

#### • Why RMS method is used:

Suppose,

$$q = x + y$$

and both measurements have random uncertainties:

$$\delta x, \quad \delta y$$

If we directly add uncertainties,

$$\delta q = \delta x + \delta y$$

it assumes both errors always occur in the same direction simultaneously.

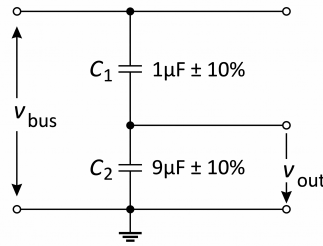
- Direct addition gives the **worst-case error**, but practical measurement errors are random in nature.
- One uncertainty may be positive while another may be negative, causing partial cancellation. Hence, uncertainties are combined statistically using RMS method.

**Q1.15.1**
**NAT**
**2015**
**1M**
**Ans: 11.75 to 12.25**


Given,

$$V_{\text{bus}} = 100 \text{ kV rms}, \quad C_1 = 1 \mu\text{F} \pm 10\%, \quad C_2 = 9 \mu\text{F} \pm 10\%$$

The output is taken across  $C_2$ .



**Fig. Solution Q1.15.1**

For series capacitors,

$$V_{\text{out}} = V_{\text{bus}} \frac{X_{C2}}{X_{C1} + X_{C2}}$$

Since  $X_C = \frac{1}{\omega C}$ ,

$$V_{\text{out}} = V_{\text{bus}} \frac{\frac{1}{C_2}}{\frac{1}{C_1} + \frac{1}{C_2}} = V_{\text{bus}} \frac{C_1}{C_1 + C_2}$$

To maximize  $V_{\text{out}}$ , take  $C_1$  maximum and  $C_2$  minimum:

$$C_{1,\text{max}} = 1.1 \mu\text{F}, \quad C_{2,\text{min}} = 8.1 \mu\text{F}$$

Thus,

$$V_{\text{out,max}} = 100 \frac{1.1}{1.1 + 8.1} = 100 \frac{1.1}{9.2} = 11.9565 \text{ kV}$$

11.96

### Mistakes to be avoided

- While writing voltage division expression across capacitor keep in mind that it is capacitive reactance and not capacitance.

**Q1.06.1**
**MCQ**
**2006**
**2M**
**Ans: D**


Given,

$$w = \frac{z}{xy}$$

For multiplication/division, maximum relative uncertainties add:

$$\frac{\Delta w}{w} = \frac{\Delta x}{x} + \frac{\Delta y}{y} + \frac{\Delta z}{z}$$

For  $x$  (Given  $x=80$ ):

$$\Delta x = 0.5\% \times 80 = 0.4$$

$$\frac{\Delta x}{x} = \frac{0.4}{80} = 0.005 = 0.5\%$$

For  $y$  (Given  $y=20$ ):

$$\Delta y = 1\% \times 100 = 1$$

$$\frac{\Delta y}{y} = \frac{1}{20} = 0.05 = 5\%$$

For  $z$  (Given  $z=50$ ):

$$\Delta z = 1.5\% \times 50 = 0.75$$

$$\frac{\Delta z}{z} = \frac{0.75}{50} = 0.015 = 1.5\%$$

Thus,

$$\frac{\Delta w}{w} = 0.5\% + 5\% + 1.5\% = 7\%$$

Therefore, the correct option is

**D.  $\pm 7.0\%$**

### Mistakes to be avoided

- For  $y$ , the uncertainty is specified as **percentage of full-scale value**, not percentage of the measured reading.

**Q1.99.1**

MCQ

1999

2M

Ans: A

● ● ○

For a PMMC voltmeter, the **figure of merit (sensitivity)** is

$$S = \frac{1}{I_{fs}}$$

$S$  = sensitivity in  $\Omega/V$ ,  $I_{fs}$  = full-scale deflection current

**Given:**

Both meters are

$$100 \mu A$$

Converting into ampere,

$$100 \mu A = 100 \times 10^{-6} A = 10^{-4} A$$

**Step 1: Calculate sensitivity**

$$S = \frac{1}{10^{-4}} = 10^4 \Omega/V = 10 \text{ k}\Omega/V$$

both use the same meter movement:

$$100 \mu A$$

Therefore, the correct option is

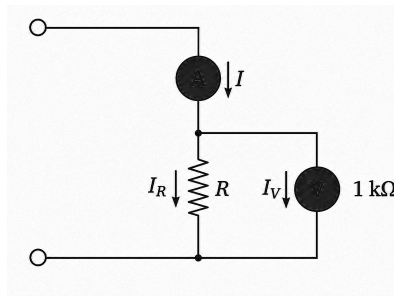
**A.  $10 \text{ k}\Omega/V$  and  $10 \text{ k}\Omega/V$**

### Key Points

- Sensitivity depends only on the **full-scale deflection current**  $I_{fs}$ , not on the voltage range.
- Even if voltmeters have different ranges such as:

$$10 \text{ V} \quad \text{and} \quad 100 \text{ V}$$

they will have the same sensitivity if the meter movement current  $I_{fs}$  is same.

**Q1.97.1**
**NAT**
**1997**
**2M**
**Ans: 111.11, 10**

**Fig. Solution Q1.97.1**
**Method 1**

Given:

$$I = 1 \text{ A}, \quad V = 100 \text{ V}, \quad R_V (\text{Voltmeter resistance}) = 1 \text{ k}\Omega = 1000 \Omega$$

The ammeter measures total current:

$$I = I_R + I_V$$

Voltmeter current:

$$I_V = \frac{V}{R_V} = \frac{100}{1000} = 0.1 \text{ A} \Rightarrow I_R = 1 - 0.1 = 0.9 \text{ A}$$

Actual resistance:

$$R = \frac{V}{I_R} = \frac{100}{0.9} = 111.11 \Omega$$

Measured resistance:

$$R_m = \frac{V}{I} = \frac{100}{1} = 100 \Omega$$

Percentage error:

$$\% \text{ error} = \frac{111.11 - 100}{111.11} \times 100 = 10\%$$

**Method 2**

Given:

$$V = 100 \text{ V}, \quad I = 1 \text{ A}, \quad R_V = 1000 \Omega$$

Using current division,

$$I_R = I \cdot \frac{R_V}{R + R_V}$$

Also,

$$V = I_R R$$

Substituting:

$$100 = \left( 1 \cdot \frac{1000}{R + 1000} \right) R \Rightarrow 100(R + 1000) = 1000R$$

$$100R + 100000 = 1000R \Rightarrow 900R = 100000 \Rightarrow R = 111.11 \Omega$$

Measured resistance:

$$R_m = \frac{V}{I} = \frac{100}{1} = 100 \Omega$$

Percentage error:

$$\% \text{ error} = \frac{111.11 - 100}{111.11} \times 100 = 10\%$$

$$R = 111.11 \Omega, \quad \% \text{ error} = 10\%$$

**Q1.95.1** NAT 1995 1M Ans: Guard Ring ● ○ ○

In an ideal parallel plate capacitor, electric field lines are perfectly straight and confined between the plates. Ideally,

$$E = \frac{V}{d}, \quad C = \frac{\epsilon A}{d}$$

These relations assume all electric field lines remain inside the plate area.

**Fringing Effect**

At the edges of capacitor plates, electric field lines spread outward instead of remaining straight.

**How guard ring helps**

A guard ring is a conducting ring placed around the sensing electrode and kept at the same potential as the main electrode.

With guard ring, field lines terminate on the guard ring instead of spreading into space.

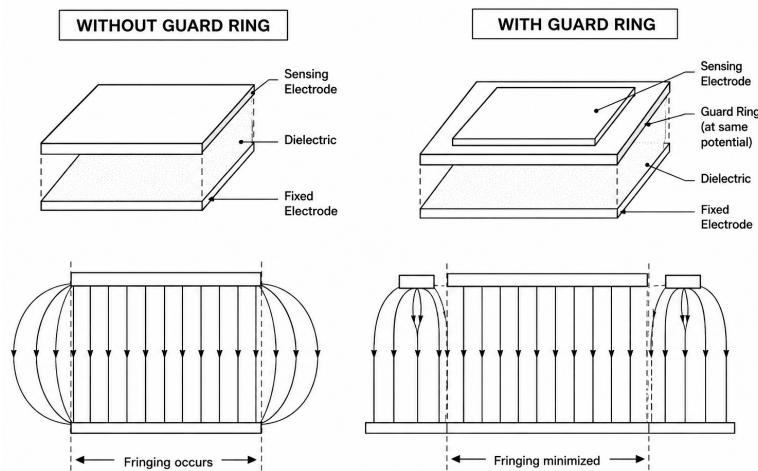


Fig. Solution Q1.95.1

Therefore answer is:

**Fringing Effect**

**Key Points**

- **Simple intuition:** Guard ring “captures” stray field lines before they spread outward, so the capacitor behaves more like an ideal parallel plate capacitor.
- Due to fringing:
  - effective plate area increases,
  - capacitance changes slightly,
  - measurement becomes inaccurate.

**Q1.95.2** MCQ 1995 1M Ans: D ● ○ ○

Accuracy is specified as

$$\pm(\% \text{ of full scale})$$

We compare percentage error while measuring 1 A.

For  $M_1$ :

$$\Delta I = \frac{0.10}{100} \times 20 = 0.02 \text{ A} \Rightarrow \% \text{ error} = \frac{0.02}{1} \times 100 = 2\%$$

For  $M_2$ :

$$\Delta I = \frac{0.20}{100} \times 10 = 0.02 \text{ A} \Rightarrow \% \text{ error} = 2\%$$

For  $M_3$ :

$$\Delta I = \frac{0.50}{100} \times 5 = 0.025 \text{ A} \Rightarrow \% \text{ error} = \frac{0.025}{1} \times 100 = 2.5\%$$

For  $M_4$ :

$$\Delta I = \frac{1.00}{100} \times 1 = 0.01 \text{ A} \Rightarrow \% \text{ error} = 1\%$$

Minimum error occurs for  $M_4$

Therefore, the correct option is

**D.  $M_4$**

### Key Points

- When accuracy is specified as  $\pm(\%$  of full scale), the absolute error depends on the full-scale range, not on the measured reading.
- If accuracy is specified as  $\pm(\%$  of full scale), then absolute error is:

$$\Delta I = \frac{\% \text{ accuracy}}{100} \times (\text{full scale value})$$

- Lower full-scale range generally gives better accuracy for small measurements.

### Mistakes to be avoided

- Do not calculate error using percentage of the measured value; it is given as percentage of full scale.
- Do not directly give answer on basis of accuracy data, sometimes it can be a trap.

**Q1.94.1**

MCQ

1994

1M

Ans: B



**Precision** means repeated measurements are close to each other, i.e., good repeatability.

Example:

10.1, 10.1, 10.1, 10.1

These measurements are highly precise.

**Accuracy** means the measured value is close to the true value.

Example:

True value = 12, Measured value = 12.01

This measurement is highly accurate.

A measurement can be precise but inaccurate.

Example:

True value = 12

Measured Values = 10.1, 10.1, 10.1

These readings are **highly precise**, but **inaccurate**.

because they are far from the true value.

Therefore,

Precision does NOT guarantee accuracy

So the answer is:

**B. False**

**Q1.92.1**
**MCQ**
**1992**
**1M**
**Ans: D**


Resistance measured using voltmeter-ammeter method:

$$R = \frac{V}{I}$$

For maximum possible error in division:

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

Given:

$$\frac{\Delta V}{V} = 2.4\%, \quad \frac{\Delta I}{I} = 1.0\%$$

Therefore,

$$\frac{\Delta R}{R} = 2.4\% + 1.0\% = 3.4\%$$

Hence, maximum possible percentage error is

3.4%

Therefore, the correct option is

**D. 3.4%**

### Key Points

- For multiplication/division, maximum fractional errors add directly:

$$\frac{\Delta q}{q} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

### Mistakes to be avoided

- Do not use RSS method when the question asks for maximum possible error.

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## DEBUG INFO

### Chapter I

Error Analysis

**Total Questions : 13**

MCQ : 7

MSQ : 0

NAT : 6



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## SOLUTIONS

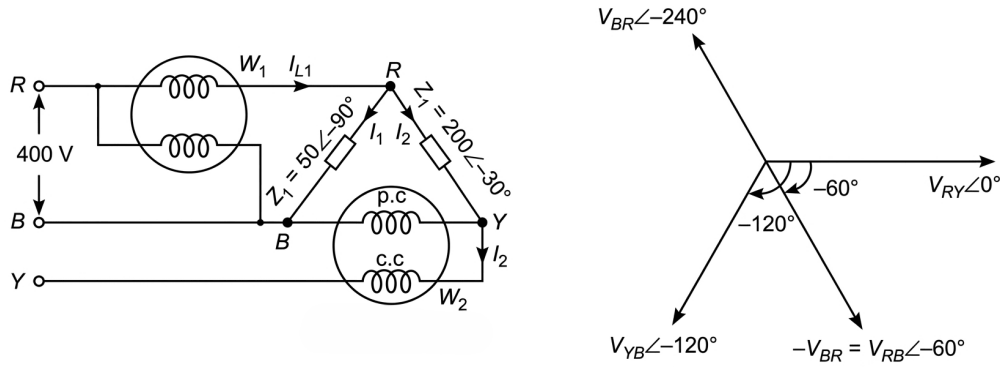


# Analog Measuring Instruments

### Topics

1. PMMC Type Instruments
2. Moving Iron & Dynamometer Type Instruments
3. Dynamometer Type Wattmeter
4. LPF Wattmeter
5. Two Wattmeter Measurement
6. Energy Meter
7. Hot wire Instruments & Electrostatic Voltmeters

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**Q2.24.1**
**NAT**
**2024**
**2M**
**Ans: 692.80**

**Fig. Solution Q2.24.1**
**Given:**

$$Z_1 = 50 \angle -90^\circ \Omega, \quad Z_2 = 200 \angle -30^\circ \Omega$$

**Line voltages:**

$$V_{RY} = 400 \angle 0^\circ, \quad V_{YB} = 400 \angle -120^\circ, \quad V_{BR} = 400 \angle -240^\circ$$

Hence,

$$V_{RB} = -V_{BR} = 400 \angle -60^\circ$$

**Current through  $Z_1$ :**

$$I_1 = \frac{V_{RB}}{Z_1} = \frac{400 \angle -60^\circ}{50 \angle -90^\circ} = 8 \angle 30^\circ$$

**Current through  $Z_2$ :**

$$I_2 = \frac{V_{RY}}{Z_2} = \frac{400 \angle 0^\circ}{200 \angle -30^\circ} = 2 \angle 30^\circ$$

**Current through current coil of  $W_1$ :**

$$I_{L1} = I_1 + I_2 = 8 \angle 30^\circ + 2 \angle 30^\circ = 10 \angle 30^\circ$$

**For  $W_1$ :**

$$V_{PC} = V_{RB} = 400 \angle -60^\circ$$

$$W_1 = 400 \times 10 \cos(-60^\circ - 30^\circ) = 4000 \cos(-90^\circ) = 0$$

**For  $W_2$ :**

$$V_{YB} = 400 \angle -120^\circ, \quad I_2 = 2 \angle 30^\circ$$

$$W_2 = 400 \times 2 \cos(-120^\circ - 30^\circ) = 800 \cos(-150^\circ) = -692.8 \text{ W}$$

**Difference:**

$$|W_1 - W_2| = |0 - (-692.8)|$$

$$\boxed{692.80 \text{ W}}$$

### Key Points

- In wattmeter calculations,

$$W = VI \cos \theta$$

where  $\theta$  is the angle between pressure coil voltage and current coil current.

### Mistakes to be avoided

- Read Question carefully, it says find "magnitude" of difference.

Q2.23.1

MCQ

2023

2M

Ans: C

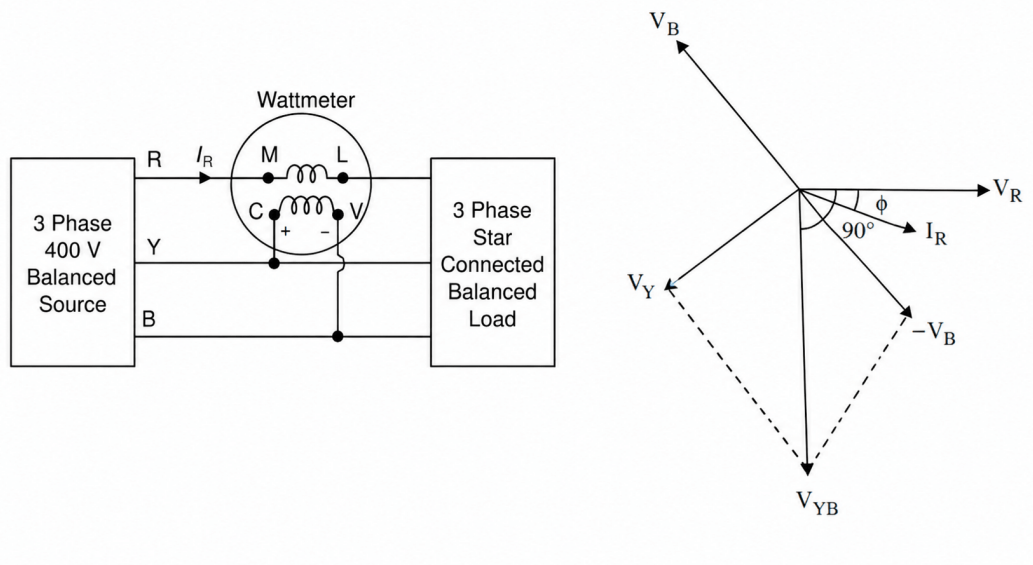


Fig. Solution Q2.23.1

#### Observation from wattmeter connection:

The current coil (CC) is connected in R-line, hence wattmeter measures:

$$I_{CC} = I_R$$

The pressure coil (PC) is connected between Y and B, therefore:

$$V_{PC} = V_{YB}$$

Thus wattmeter reading is:

$$W = V_{YB} I_R \cos(\text{angle between } V_{YB} \text{ and } I_R)$$

#### Why angle becomes $90^\circ - \phi$ :

Assume phase sequence:

$$R \rightarrow Y \rightarrow B$$

Let,

$$V_R = V_p \angle 0^\circ, \quad V_Y = V_p \angle -120^\circ, \quad V_B = V_p \angle 120^\circ$$

Line voltage:

$$V_{YB} = V_Y - V_B = \sqrt{3} V_p \angle -90^\circ$$

Therefore, angle between  $V_{YB}$  and  $I_R$  becomes:  $90^\circ - \phi$

Hence wattmeter reads:

$$W = V_L I_L \cos(90^\circ - \phi) = V_L I_L \sin \phi$$

#### Substituting values:

$$-400 = 400 \times 2 \times \sin \phi \Rightarrow -400 = 800 \sin \phi$$

$$\sin \phi = -\frac{1}{2} \Rightarrow \phi = -30^\circ$$

Negative angle indicates leading power factor.

Therefore,

$$PF = \cos 30^\circ = 0.866$$

Hence, the correct option is

C. 0.866 leading

### Key Points

- In this connection, the current coil carries phase current  $I_R$  while the pressure coil measures line voltage  $V_{YB}$ .

Since  $V_{YB}$  is shifted by  $90^\circ$  from phase voltage  $V_R$ , the angle in wattmeter becomes:

$$90^\circ - \phi$$

Hence wattmeter reads:

$$W = V_L I_L \cos(90^\circ - \phi) = V_L I_L \sin \phi$$

which is the expression for reactive power:

$$Q = V_L I_L \sin \phi$$

Therefore, this wattmeter connection measures **reactive power**.

#### Q2.20.1

NAT

2020

1M

Ans: 1.732

☒ ☐ ☐

From KCL:

$$I_1 = I_2 + I_3$$

Given:

$$I_2 = 1 \angle 10^\circ, \quad I_3 = 1 \angle 70^\circ$$

Thus,

$$I_1 = 1 \angle 10^\circ + 1 \angle 70^\circ$$

Converting into rectangular form:

$$I_1 = (\cos 10^\circ + j \sin 10^\circ) + (\cos 70^\circ + j \sin 70^\circ)$$

$$= 1.3268 + j1.1133$$

Magnitude:

$$|I_1| = \sqrt{(1.3268)^2 + (1.1133)^2} = \sqrt{3}$$

$$1.732 \text{ A}$$

### Mistakes to be avoided

- Do not directly add magnitudes of phasor currents; add them in rectangular form.
- The phase angle  $40^\circ$  is irrelevant because an ammeter shows only the magnitude of current.

Q2.19.1

MCQ

2019

2M

Ans: D

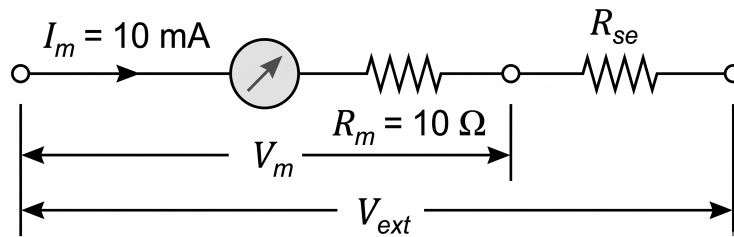


Fig. Solution Q2.19.1

#### Method 1: Direct voltage equation

Given:

$$R_m = 10 \Omega, \quad I_m = 10 \text{ mA} = 0.01 \text{ A}$$

Required voltmeter range:

$$V_{ext} = 100 \text{ V}$$

For full-scale deflection:

$$V_{ext} = I_m(R_m + R_{se})$$

Substituting,

$$100 = 0.01(10 + R_{se}) \Rightarrow 10 + R_{se} = \frac{100}{0.01} = 10000$$

$$R_{se} = 10000 - 10 = 9990 \Omega$$

Hence,

$$R_{se} = 9990 \Omega$$

#### Method 2: Multiplying factor

Voltage across original meter:

$$V_m = I_m R_m = 10 \text{ mA} \times 10 \Omega = 0.1 \text{ V}$$

Thus original meter range is:

$$0 \rightarrow 100 \text{ mV}$$

Required range:

$$0 \rightarrow 100 \text{ V}$$

Multiplying factor:

$$m = \frac{V_{ext}}{V_m} = \frac{100}{0.1} = 1000$$

Series resistance:

$$R_{se} = R_m(m - 1) = 10(1000 - 1) = 9990 \Omega$$

Therefore, the correct option is

$$\text{D. } 9990 \Omega$$

#### Key Points

- To convert a galvanometer into a voltmeter, a high series resistance is added.

- To convert a galvanometer into an ammeter, a low shunt resistance is connected in parallel.

**Q2.18.1**

MCQ

2018

1M

Ans: D



Given that **one wattmeter reads half of the other** and both readings are **positive**, let

$$W_1 = \frac{1}{2} W_2$$

Let,

$\phi$  = power factor angle

#### Method 1

For **Two-wattmeter method**,

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Let  $W_1 = x$  and  $W_2 = 2x$ . Then,

$$\tan \phi = \sqrt{3} \frac{x - 2x}{x + 2x} = \sqrt{3} \frac{-x}{3x} = -\frac{1}{\sqrt{3}}$$

Hence,

$$\phi = 30^\circ$$

and therefore,

$$\text{PF} = \cos 30^\circ = 0.866$$

#### Method 2

For a balanced load,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \quad W_2 = V_L I_L \cos(30^\circ - \phi)$$

Given  $W_1 = \frac{1}{2} W_2$ ,

$$\cos(30^\circ + \phi) = \frac{1}{2} \cos(30^\circ - \phi)$$

Using expansion and simplifying,

$$\tan \phi = \frac{1}{\sqrt{3}} \Rightarrow \phi = 30^\circ$$

Thus,

$$\text{PF} = \cos 30^\circ$$

Therefore, the correct option is

**D. 0.866**

#### Key Points

- If both wattmeter readings are positive, then:

$$\phi < 60^\circ$$

#### Mistakes to be avoided

- While using

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

maintain correct sign convention.

Q2.18.2

MCQ

2018

2M

Ans: A

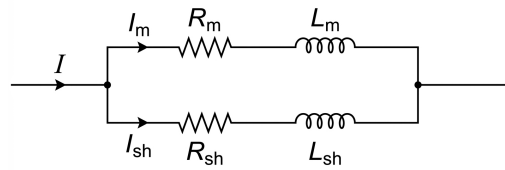


Fig. Solution Q2.18.2

**Given:** Moving iron ammeter range 0–1 A,  $R_m = 50 \text{ m}\Omega$ ,  $L_m = 0.1 \text{ mH}$ . Required extended range: 0–10 A. For **frequency-independent** operation, the meter branch and shunt branch must satisfy

$$\frac{R_m}{L_m} = \frac{R_{sh}}{L_{sh}}$$

$$\text{Time constant, } \tau = \frac{L_m}{R_m} = \frac{0.1 \times 10^{-3}}{50 \times 10^{-3}} = 2 \times 10^{-3} \text{ s} = 2 \text{ ms}$$

For full-scale current,

$$I_m = 1 \text{ A}, \quad I = 10 \text{ A} \Rightarrow I_{sh} = 9 \text{ A}$$

Since the voltage across meter branch and shunt branch is same,

$$I_m R_m = I_{sh} R_{sh}$$

$$1 \times 50 \text{ m}\Omega = 9 R_{sh} \Rightarrow R_{sh} = \frac{50}{9} \text{ m}\Omega = 5.55 \text{ m}\Omega$$

Hence,

$$\tau = 2 \text{ ms}, \quad R_{sh} = 5.55 \text{ m}\Omega$$

Therefore, the correct option is

$$\text{A. } 2, 5.55$$

### Key Points

- Equal time constants mean both branches react equally to frequency change. Therefore, current division remains constant with frequency. Hence, the meter reads correctly at all frequencies.

Q2.17.1

MCQ

2017

2M

Ans: D

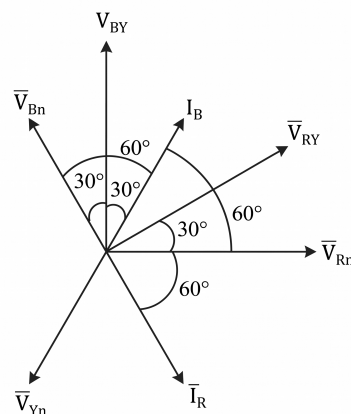


Fig. Solution Q2.17.1

Given:

$$S = 3464 \text{ VA}, \quad V_L = 400 \text{ V}$$

Using,

$$S = \sqrt{3} V_L I_L \Rightarrow 3464 = \sqrt{3} \times 400 \times I_L \Rightarrow I_L = 5 \text{ A}$$

Hence,

$$I_R = I_Y = I_B = 5 \text{ A}$$

Initially, with switch at N: Each wattmeter reads phase power: 577.35 W, therefore

$$\text{Total Power, } P = 3 \times 577.35 = 1732 \text{ W}$$

$$P = \sqrt{3} V_L I_L \cos \phi \Rightarrow 1732 = \sqrt{3} \times 400 \times 5 \cos \phi$$

$$1732 = 3464 \cos \phi \Rightarrow \cos \phi = 0.5 \Rightarrow \phi = 60^\circ$$

Switch moved to position Y, now pressure coils use line voltages.

For  $W_1$ :

$$\angle(V_{RY}, I_R) = 90^\circ \Rightarrow W_1 = 400 \times 5 \cos 90^\circ = 0$$

For  $W_2$ :

When switch moves from N to Y, the pressure coil of  $W_2$  gets short-circuited.

Hence,

$$V_{PC(2)} = 0 \Rightarrow W_2 = 0$$

For  $W_3$ :

$$\angle(V_{BY}, I_B) = 30^\circ \Rightarrow W_3 = 400 \times 5 \cos 30^\circ = 1732 \text{ W}$$

Therefore,

$$W_1 = 0, \quad W_2 = 0, \quad W_3 = 1732 \text{ W}$$

Hence, the correct option is

$$\text{D. } W_1 = W_2 = 0 \text{ and } W_3 = 1732 \text{ W}$$

### Mistakes to be avoided

- Do not use phase voltage in place of line voltage in three-phase power formula.
- Carefully identify pressure coil connection after switch movement, do not only use phasor diagram to calculate power.

Q2.16.1

NAT

2016

2M

Ans: 0.312



Given,

$$v(t) = 100 + 10 \sin \omega t - 5 \sin 3\omega t$$

**Moving-coil voltmeter (PMMC):**

A moving-coil instrument responds to the **average value**.

Since,

$$\overline{\sin \omega t} = 0, \quad \overline{\sin 3\omega t} = 0$$

Hence,

$$V_1 = 100 \text{ V}$$

**Moving-iron voltmeter:**

A moving-iron instrument responds to the **RMS value**.

Therefore,

$$V_2 = \sqrt{100^2 + \left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2}$$

$$= \sqrt{10000 + 50 + 12.5} = \sqrt{10062.5} = 100.312 \text{ V}$$

Hence,

$$V_2 - V_1 = 100.312 - 100 = 0.312 \text{ V}$$

0.31

### Key Points

- PMMC instruments respond to the average value, while moving-iron instruments respond to RMS value.
- **Visual difference:** Moving-iron instrument scale is non-linear, whereas PMMC instrument scale is linear.

**Q2.16.2** NAT 2016 2M Ans: 2 ● ○ ○

**Given:**

Meter constant:

1200 rev/kWh

This means,

1200 revolutions = 1 kWh

Given:

20 revolutions in 30 s

Energy consumed in 20 revolutions:

$$E = \frac{20}{1200} = \frac{1}{60} = 0.01667 \text{ kWh}$$

Time:

$$30 \text{ s} = \frac{30}{3600} = \frac{1}{120} \text{ h}$$

Power:

$$P = \frac{E}{t} = \frac{1/60}{1/120} = 2 \text{ kW}$$

Hence,

2

### Key Points

- Meter constant gives the number of revolutions per unit energy:

$$\text{Meter constant} = \frac{\text{revolutions}}{\text{energy}}$$

### Mistakes to be avoided

- Be careful with units as time is given in seconds and Energy in kWh

**Q2.15.1** NAT 2015 1M Ans: 0.222 ☒ ☐ ☐

Given:

Ammeter range = 0 – 50 A

$V = 0.1$  V at full scale

Required new range: 0-500 A

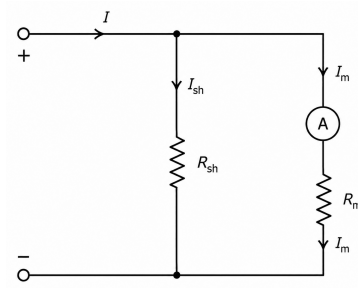


Fig. Solution Q2.15.1

Hence,

$$I_m = 50 \text{ A}, \quad I = 500 \text{ A}$$

Shunt current:

$$I_{sh} = 500 - 50 = 450 \text{ A}$$

Since meter and shunt are in parallel,

$$V_{sh} = V_m = 0.1 \text{ V}$$

Therefore,

$$R_{sh} = \frac{V}{I_{sh}} = \frac{0.1}{450} = 2.22 \times 10^{-4} \Omega$$

Converting into milliohms,

$$R_{sh} = 0.222 \text{ m}\Omega$$

Hence,

0.22

#### Mistakes to be avoided

- Final answer is asked in  $\text{m}\Omega$ , so read question carefully.

**Q2.15.2** MCQ 2015 1M Ans: C ☒ ☒ ☐

#### P. Permanent Magnet Moving Coil (PMMC)

For PMMC instrument,

$$T_d \propto BI$$

Torque depends on current direction.

For AC, current reverses every half cycle, hence torque also reverses every half cycle. Therefore,

$$T_{\text{avg}} = 0$$

Thus PMMC works only for:

DC only

Hence,

$$P \rightarrow 1$$

#### Q. Moving iron connected through Current Transformer (CT)

For moving iron instrument,

$$T_d \propto I^2$$

Hence,

$$(+I)^2 = (-I)^2$$

So MI instrument alone can measure AC and DC.

But CT works only on AC because:

$$e = N \frac{d\phi}{dt}$$

For DC,

$$\frac{d\phi}{dt} = 0$$

Thus,

MI + CT works only on AC

Hence,

$$Q \rightarrow 2$$

### R. Rectifier instrument

A rectifier instrument uses:

AC/DC → Rectifier → PMMC

Rectifier converts AC into DC before feeding PMMC.

Thus it can measure:

AC and DC

Hence,

$$R \rightarrow 3$$

### S. Electrodynamometer

For electrodynamometer,

$$T_d \propto I_1 I_2$$

For AC,

$$(-I_1)(-I_2) = I_1 I_2$$

Hence torque remains positive for both AC and DC.

Therefore,

AC and DC

Hence,

$$S \rightarrow 3$$

Final matching:

$$P \rightarrow 1, \quad Q \rightarrow 2, \quad R \rightarrow 3, \quad S \rightarrow 3$$

Therefore, the correct option is

C. P-1, Q-2, R-3, S-3

### Key Points

- **CT** → Current Transformer. Used to step down high AC current to a measurable low current for protection and measurement. Works only with AC (requires changing magnetic flux).

### Q2.15.3

MCQ

2015

1M

Ans: A



Given:

$$P = 5 \text{ kW}, \quad \text{PF} = 0.707 \Rightarrow \cos \phi = 0.707 \Rightarrow \phi = 45^\circ$$

#### Method 1

For two-wattmeter method,

$$W_1 + W_2 = P = 5$$

and

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Substituting  $\phi = 45^\circ$ ,

$$1 = \sqrt{3} \frac{W_1 - W_2}{5} \Rightarrow W_1 - W_2 = \frac{5}{\sqrt{3}} = 2.887$$

Now,

$$W_1 + W_2 = 5, \quad W_1 - W_2 = 2.887$$

$$2W_1 = 7.887 \Rightarrow W_1 = 3.944 \text{ kW}$$

$$W_2 = 5 - 3.944 = 1.056 \text{ kW}$$

#### Method 2

For a balanced load,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \quad W_2 = V_L I_L \cos(30^\circ - \phi)$$

With  $\phi = 45^\circ$ ,

$$W_1 : W_2 = \cos 75^\circ : \cos 15^\circ = 0.2588 : 0.9659 = 1 : 3.732$$

Also,

$$W_1 + W_2 = 5$$

Let  $W_1 = x$ , then  $W_2 = 3.732x$ . So,

$$x + 3.732x = 5 \Rightarrow x = 1.056$$

$$W_1 = 1.056 \text{ kW}, \quad W_2 = 3.944 \text{ kW}$$

**A. 3.94 kW and 1.06 kW**

### Q2.15.4

NAT

2015

2M

Ans: 0.15

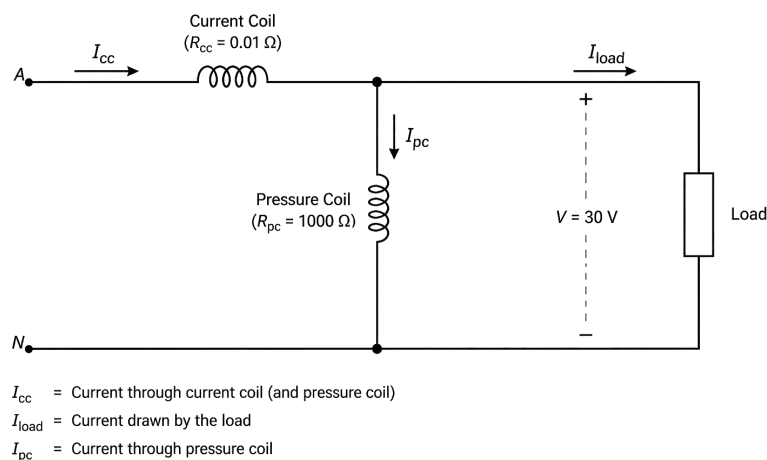


Fig. Solution Q2.15.4

From the figure, the current coil (CC) is in series with the load and the pressure coil (PC) is connected across the load only. Therefore, the wattmeter measures

$$P_m = P_{\text{load}} + P_{\text{PC}}$$

**True load power:**

$$P_{\text{true}} = VI \cos \phi = 30 \times 25 \times 0.8 = 600 \text{ W}$$

**Pressure coil power loss:**

$$P_{\text{PC}} = \frac{V^2}{R_P} = \frac{30^2}{1000} = 0.9 \text{ W}$$

Hence,

$$P_m = 600 + 0.9 = 600.9 \text{ W}$$

**Percentage error:**

$$\% \text{ error} = \frac{600.9 - 600}{600} \times 100 = 0.15\%$$

0.15

### Key Points

- Since the added loss is positive, the measured power is slightly higher than the true power.

### Mistakes to be avoided

- Do not include current coil loss here; only pressure coil loss affects the reading.

**Q2.14.1**

NAT

2014

1M

Ans: 2

● ○ ○

**Given:**

$$\cos \phi = 0.2, \quad V = 150 \text{ V}, \quad I = 10 \text{ A}$$

Full-scale reading:

150 divisions

Maximum measurable power:

$$P_{\text{max}} = VI \cos \phi = 150 \times 10 \times 0.2 = 300 \text{ W}$$

Thus, 150 divisions correspond to 300 W.

Therefore, multiplying factor:

$$\text{MF} = \frac{300}{150} = 2 \text{ W/division}$$

Hence,

2

### Key Points

- LPF wattmeters are commonly used in the **open-circuit test of transformers** because transformer no-load power factor is very low.

**Q2.14.2**
**NAT**
**2014**
**1M**
**Ans: 0.8**
☒ ☐ ☐
**Given:**

$$W_1 = 100 \text{ W}, \quad W_2 = 250 \text{ W}$$

For two-wattmeter method,

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Taking the larger wattmeter first,

$$\begin{aligned} \tan \phi &= \sqrt{3} \frac{250 - 100}{250 + 100} \\ &= 1.732 \times \frac{150}{350} = 0.742 \end{aligned}$$

Hence,

$$\phi = \tan^{-1}(0.742) \approx 36.6^\circ$$

Therefore,

$$\text{PF} = \cos \phi = \cos 36.6^\circ \approx 0.8$$

0.8

**Q2.14.3**
**MCQ**
**2014**
**1M**
**Ans: C**
☒ ☐ ☐
**Given:**

$$W_1 = 2W_2$$

For two-wattmeter method,

$$\tan \phi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Substituting  $W_1 = 2W_2$ ,

$$\tan \phi = \sqrt{3} \frac{2W_2 - W_2}{2W_2 + W_2} = \sqrt{3} \frac{W_2}{3W_2} = \frac{1}{\sqrt{3}}$$

Hence,

$$\phi = \tan^{-1} \left( \frac{1}{\sqrt{3}} \right) = 30^\circ$$

Converting into radians,

$$30^\circ = \frac{\pi}{6}$$

Hence, the correct option is

C.  $\phi = \frac{\pi}{6}$

### Key Points

- In AC circuits, the **load impedance angle** is the same as the **power factor angle**:

$$Z = |Z| \angle \phi$$

### Q2.14.4

MCQ

2014

2M

Ans: A



For a moving-iron voltmeter, the reading is the RMS value.

Given,

$$v(t) = \begin{cases} 1 + \sin \omega t, & 0 \leq \omega t < 6\pi \\ -1 + \sin \omega t, & 6\pi \leq \omega t < 12\pi \end{cases}$$

Let

$$\theta = \omega t$$

Then,

$$V_{\text{rms}}^2 = \frac{1}{12\pi} \left[ \int_0^{6\pi} (1 + \sin \theta)^2 d\theta + \int_{6\pi}^{12\pi} (-1 + \sin \theta)^2 d\theta \right]$$

Expanding,

$$(1 + \sin \theta)^2 = 1 + 2 \sin \theta + \sin^2 \theta$$

$$(-1 + \sin \theta)^2 = 1 - 2 \sin \theta + \sin^2 \theta$$

Therefore,

$$V_{\text{rms}}^2 = \frac{1}{12\pi} \left[ \int_0^{6\pi} (1 + 2 \sin \theta + \sin^2 \theta) d\theta + \int_{6\pi}^{12\pi} (1 - 2 \sin \theta + \sin^2 \theta) d\theta \right]$$

Now,

$$\int_0^{6\pi} \sin \theta d\theta = \int_{6\pi}^{12\pi} \sin \theta d\theta = 0$$

Also,

$$\int_0^{6\pi} 1 d\theta = 6\pi, \quad \int_0^{6\pi} \sin^2 \theta d\theta = 3\pi$$

Similarly,

$$\int_{6\pi}^{12\pi} 1 d\theta = 6\pi, \quad \int_{6\pi}^{12\pi} \sin^2 \theta d\theta = 3\pi$$

Hence,

$$V_{\text{rms}}^2 = \frac{1}{12\pi} \left[ (6\pi + 3\pi) + (6\pi + 3\pi) \right] \\ = \frac{18\pi}{12\pi} = \frac{3}{2}$$

Therefore,

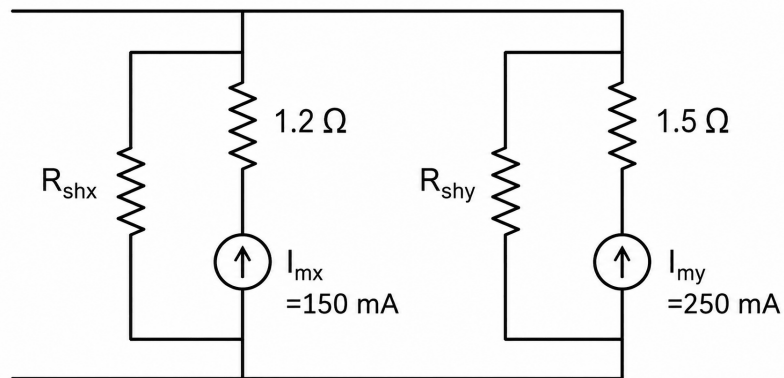
$$V_{\text{rms}} = \sqrt{\frac{3}{2}}$$

$$\boxed{\text{A. } \sqrt{\frac{3}{2}}}$$

### Key Points

- RMS value of any periodic signal is:

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

**Q2.14.5**
**NAT**
**2014**
**2M**
**Ans: 10.14**

**Fig. Solution Q2.14.5**
**Ammeter X:**

$$R_X = 1.2 \, \Omega, \quad I_{mx} = 0.15 \, \text{A}$$

At full-scale 15 A,

$$V_X = I_{mx} R_X = 0.15 \times 1.2 = 0.18 \, \text{V}$$

So the equivalent resistance of extended ammeter X is

$$R_{eqX} = \frac{V_X}{15} = \frac{0.18}{15} = 0.012 \, \Omega$$

**Ammeter Y:**

$$R_Y = 1.5 \, \Omega, \quad I_{my} = 0.25 \, \text{A}$$

At full-scale 15 A,

$$V_Y = I_{my} R_Y = 0.25 \times 1.5 = 0.375 \, \text{V}$$

So the equivalent resistance of extended ammeter Y is

$$R_{eqY} = \frac{V_Y}{15} = \frac{0.375}{15} = 0.025 \, \Omega$$

Since the two extended ammeters are in **parallel**, current division gives

$$I_X = 15 \frac{R_{eqY}}{R_{eqX} + R_{eqY}} = 15 \frac{0.025}{0.012 + 0.025} = 10.135 \, \text{A}$$

Hence, ammeter X reads

10.14

**Mistakes to be avoided**

- Read the question carefully to know how the ammeters are connected.

**Q2.14.6** NAT 2014 1M Ans: 57.7

A moving-iron (MI) voltmeter measures the **RMS value** since

$$T_d \propto I^2$$

From the graph, the sawtooth waveform rises linearly from

$$0 \rightarrow 100 \text{ V}$$

during

$$T = 20 \text{ ms}$$

Hence,

$$v(t) = \frac{100}{T}t, \quad 0 \leq t \leq T$$

RMS value:

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

Substituting,

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_0^T \left( \frac{100}{T}t \right)^2 dt} \\ &= \sqrt{\frac{10000}{T^3} \int_0^T t^2 dt} \end{aligned}$$

Since,

$$\int_0^T t^2 dt = \frac{T^3}{3}$$

Therefore,

$$V_{\text{rms}} = \sqrt{\frac{10000}{3}} = \frac{100}{\sqrt{3}} = 57.7 \text{ V}$$

Hence, MI voltmeter reading is

$$57.7 \text{ V}$$

**Q2.14.7** NAT 2014 2M Ans: 3.2

Given for the **PMMC ammeter**:

$$n = 100, \quad B = 0.2 \text{ Wb/m}^2, \quad A = 80 \text{ mm}^2 = 80 \times 10^{-6} \text{ m}^2$$

For the **electrodynamometer ammeter**:

$$\frac{dM}{d\theta} = 0.5 \text{ mH/rad} = 0.5 \times 10^{-3} \text{ H/rad}$$

Since both meters have the same spring constant, their deflections are equal when

$$\theta_1 = \theta_2$$

**PMMC meter:**

$$T_d = nBAI, \quad K\theta_1 = nBAI$$

**Electrodynamometer meter:**

$$T_d = I^2 \frac{dM}{d\theta}, \quad K\theta_2 = I^2 \frac{dM}{d\theta}$$

For equal deflection,

$$K\theta_1 = K\theta_2 \Rightarrow nBAI = I^2 \frac{dM}{d\theta} \Rightarrow I = \frac{nBA}{dM/d\theta}$$

Substituting,

$$I = \frac{100 \times 0.2 \times 80 \times 10^{-6}}{0.5 \times 10^{-3}}$$

$$I = \frac{1.6 \times 10^{-3}}{0.5 \times 10^{-3}} = 3.2 \text{ A}$$

3.2 A

### Key Points

- PMMC deflection grows linearly with current:  $\theta \propto I$

Electrodynamometer deflection grows quadratically with current:  $\theta \propto I^2$

At one particular current, these two curves intersect. For this question, that current is: 3.2 A

### Q2.13.1

MCQ

2013

1M

Ans: A

● ○ ○

For an **ideal PMMC voltmeter**, the reading is proportional to the **average (DC) value**, not the RMS value.

Given,

$$v(t) = 14.14 \sin(314t)$$

so the peak value is

$$V_m = 14.14 \text{ V}$$

The diode is ideal, so the waveform becomes **half-wave rectified**. Also,

$$100 \text{ k}\Omega \gg 1 \text{ k}\Omega$$

hence the voltage division effect is negligible.

For a half-wave rectified sine wave,

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

Therefore,

$$V_{\text{avg}} = \frac{14.14}{\pi} = 4.50 \text{ V}$$

Hence, the correct option is

A. 4.46

### Key Points

- For the diode :

Positive half-cycle  $\rightarrow$  ON

Negative half-cycle  $\rightarrow$  OFF

- Since the input waveform is sinusoidal, we can directly use the standard half-wave rectified average value:

$$V_{\text{avg}} = \frac{V_m}{\pi}$$

**Q2.13.2**
**MCQ**
**2013**
**1M**
**Ans: D**
☒ ☐ ☐

Since all are **moving-iron voltmeters**, they measure RMS magnitude.

Given,

$$X_C = -j1 \Omega, \quad X_L = j2 \Omega$$

Since both are in series, the same current  $I$  flows through them.

Voltage across capacitor:

$$V_1 = I(-j1) \Rightarrow |V_1| = I$$

Hence,

$$V_1 = I$$

Voltage across inductor:

$$V_2 = I(j2) \Rightarrow |V_2| = 2I$$

Hence,

$$V_2 = 2I$$

Total impedance:

$$Z = -j1 + j2 = j1$$

Therefore,

$$V = IZ = I(j1) \Rightarrow |V| = I$$

Hence,

$$V = I$$

And,

$$V_2 - V_1 = 2I - I = I (= V)$$

Therefore, the correct option is

$$\text{D. } V = V_2 - V_1$$

**Q2.12.1**
**MCQ**
**2012**
**1M**
**Ans: C**
☒ ☒ ☐

Given,

$$v(t) = E_1 \sin \omega t + E_3 \sin 3\omega t, \quad i(t) = I_1 \sin(\omega t - \phi_1) + I_3 \sin(3\omega t - \phi_3) + I_5 \sin 5\omega t$$

Average power is

$$P = \overline{v(t)i(t)}$$

On expanding  $v(t)i(t)$ , all cross-frequency terms such as

$$\sin \omega t \sin 3\omega t, \quad \sin \omega t \sin 5\omega t, \quad \sin 3\omega t \sin 5\omega t$$

have zero average over one cycle.

Instantaneous power:

$$p(t) = v(t)i(t)$$

Substituting,

$$p(t) = (E_1 \sin \omega t + E_3 \sin 3\omega t) [I_1 \sin(\omega t - \phi_1) + I_3 \sin(3\omega t - \phi_3) + I_5 \sin 5\omega t]$$

Expanding,

$$p(t) = E_1 I_1 \sin \omega t \sin(\omega t - \phi_1) + E_1 I_3 \sin \omega t \sin(3\omega t - \phi_3) + E_1 I_5 \sin \omega t \sin 5\omega t \\ + E_3 I_1 \sin 3\omega t \sin(\omega t - \phi_1) + E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3) + E_3 I_5 \sin 3\omega t \sin 5\omega t$$

Now use

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

First term:

$$E_1 I_1 \sin \omega t \sin(\omega t - \phi_1) = \frac{1}{2} E_1 I_1 [\cos \phi_1 - \cos(2\omega t - \phi_1)]$$

$$\Rightarrow \overline{E_1 I_1 \sin \omega t \sin(\omega t - \phi_1)} = \frac{1}{2} E_1 I_1 \cos \phi_1$$

Second term:

$$E_1 I_3 \sin \omega t \sin(3\omega t - \phi_3) = \frac{1}{2} E_1 I_3 [\cos(2\omega t - \phi_3) - \cos(4\omega t - \phi_3)]$$

$$\Rightarrow \text{average} = 0$$

Third term:

$$E_1 I_5 \sin \omega t \sin 5\omega t = \frac{1}{2} E_1 I_5 [\cos 4\omega t - \cos 6\omega t]$$

$$\Rightarrow \text{average} = 0$$

Fourth term:

$$E_3 I_1 \sin 3\omega t \sin(\omega t - \phi_1) = \frac{1}{2} E_3 I_1 [\cos(2\omega t + \phi_1) - \cos(4\omega t - \phi_1)]$$

$$\Rightarrow \text{average} = 0$$

Fifth term:

$$E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3) = \frac{1}{2} E_3 I_3 [\cos \phi_3 - \cos(6\omega t - \phi_3)]$$

$$\Rightarrow \overline{E_3 I_3 \sin 3\omega t \sin(3\omega t - \phi_3)} = \frac{1}{2} E_3 I_3 \cos \phi_3$$

Sixth term:

$$E_3 I_5 \sin 3\omega t \sin 5\omega t = \frac{1}{2} E_3 I_5 [\cos 2\omega t - \cos 8\omega t]$$

$$\Rightarrow \text{average} = 0$$

Thus the average power is

$$P = \frac{1}{2} E_1 I_1 \cos \phi_1 + \frac{1}{2} E_3 I_3 \cos \phi_3$$

Therefore, the correct option is

**C.**  $\frac{1}{2} (E_1 I_1 \cos \phi_1 + E_3 I_3 \cos \phi_3)$

### Key Points

- For the same-frequency term,

$$E_1 I_1 \sin \omega t \sin(\omega t - \phi_1) = \frac{1}{2} E_1 I_1 [\cos \phi_1 - \cos(2\omega t - \phi_1)]$$

Hence, after taking average  $\overline{\cos(2\omega t - \phi_1)} = 0$  over a complete cycle.

$$\overline{E_1 I_1 \sin \omega t \sin(\omega t - \phi_1)} = \frac{1}{2} E_1 I_1 \cos \phi_1$$

- For the cross-frequency term,

$$\frac{1}{2} E_1 I_3 [\cos(2\omega t - \phi_3) - \cos(4\omega t - \phi_3)]$$

both cosine terms have zero average over one cycle, so the average is

$$0$$

- Therefore, only same-frequency terms contribute to average power.

### Mistakes to be avoided

- The 5<sup>th</sup> harmonic current does not contribute because there is no corresponding 5<sup>th</sup> harmonic voltage.
- Do not use RMS power formula directly here; the signals are expressed in peak form.

**Q2.12.2**

**MCQ**

**2012**

**2M**

**Ans: D**



For an analog voltmeter, the reading is proportional to the meter current:

$$I_m = \frac{V}{R_v + R_s}$$

Hence, for the same applied voltage,

$$\text{Reading} \propto \frac{1}{R_v + R_s}$$

Given:

$$R_{s1} = 20 \text{ k}\Omega, \quad V_1 = 440 \text{ V}$$

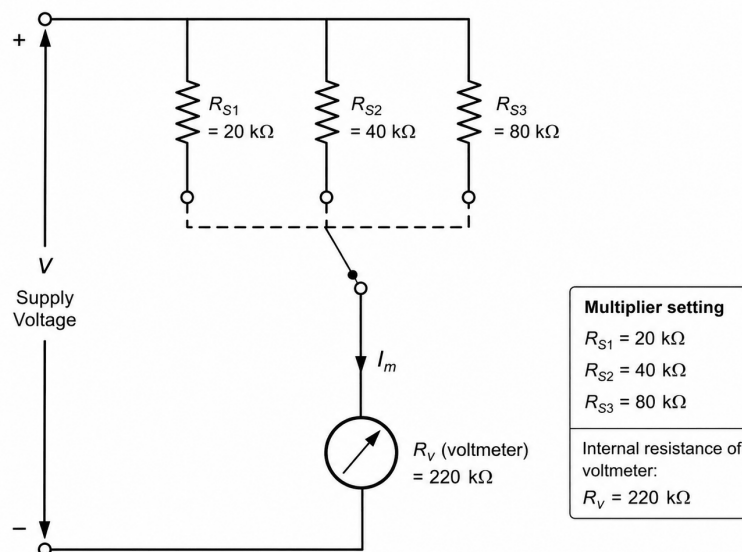
$$R_{s3} = 80 \text{ k}\Omega, \quad V_2 = 352 \text{ V}$$

So,

$$\frac{440}{352} = \frac{R_v + 80}{R_v + 20}$$

$$1.25 = \frac{R_v + 80}{R_v + 20} \Rightarrow 1.25(R_v + 20) = R_v + 80$$

$$1.25R_v + 25 = R_v + 80 \Rightarrow 0.25R_v = 55 \Rightarrow R_v = 220 \text{ k}\Omega$$



**Fig. Solution Q2.12.2**

Now,

$$440 = \frac{K}{220 + 20} \Rightarrow K = 440 \times 240 = 105600$$

For  $R_{s2} = 40 \text{ k}\Omega$ ,

$$V_3 = \frac{K}{220 + 40} = \frac{105600}{260} = 406.15 \text{ V}$$

Therefore, the correct option is

**D. 406 V**

## Key Points

### • Intuition:

Increasing multiplier resistance:

$$R_m \uparrow$$

causes meter current

$$I = \frac{V}{R_v + R_m}$$

to decrease.

Thus, the meter reads lower.

Therefore:

$$20 \text{ k}\Omega \rightarrow 440 \text{ V}$$

$$80 \text{ k}\Omega \rightarrow 352 \text{ V}$$

$$40 \text{ k}\Omega \rightarrow 406 \text{ V}$$

- The voltmeter becomes less sensitive as multiplier resistance increases.

### Q2.12.3

MCQ

2012

1M

Ans: A

☒ ☐ ☐

For a **PMMC meter**, the reading is proportional to the **average (DC) value**.

Hence,

$$V_{\text{avg}} = \frac{\text{area under one cycle}}{T}$$

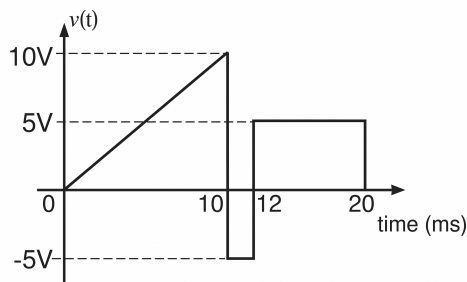


Fig. Solution Q2.12.3

From the waveform:

**Segment 1:** 0 to 10 ms

A triangle from 0 to 10 V, so

$$A_1 = \frac{1}{2} \times 10 \times 10 = 50 \text{ V} \cdot \text{ms}$$

**Segment 2:** 10 to 12 ms

A negative rectangle:

$$A_2 = -5 \times 2 = -10 \text{ V} \cdot \text{ms}$$

**Segment 3:** 12 to 20 ms

A positive rectangle:

$$A_3 = 5 \times 8 = 40 \text{ V} \cdot \text{ms}$$

Total area:

$$A = 50 - 10 + 40 = 80 \text{ V} \cdot \text{ms}$$

Total period:

$$T = 20 \text{ ms}$$

Therefore,

$$V_{\text{avg}} = \frac{80}{20} = 4 \text{ V}$$

Hence, the correct option is

**A. 4 V**

### Key Points

- The DC (average) value of a waveform is equal to the area under one complete cycle divided by the time period:

$$f_{\text{avg}} = \frac{1}{T} \int_0^T f(t) dt$$

### Mistakes to be avoided

- Do not forget the minus sign for the negative area portion of the waveform.

**Q2.11.1**

MCQ

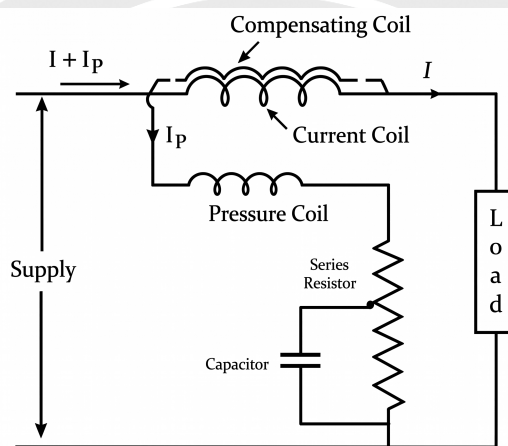
2011

1M

Ans: B



In a **Low Power Factor (LPF) wattmeter**, errors become significant because the pressure (voltage) coil circuit has inductance.



**Fig. Solution Q2.11.1**

Hence pressure coil current  $I_p$  does not remain exactly in phase with the applied voltage.

The pressure coil impedance is:

$$Z_p = R_p + jX_p$$

Thus,  $I_p$  lags the applied voltage, causing error in wattmeter reading, especially at low power factor.

To reduce this error, a **compensating coil** is used.

The compensating coil:

- is connected in series with the pressure coil circuit,
- carries pressure coil current  $I_p$ ,
- is wound coaxially over the main current coil,
- produces magnetic field opposite to the unwanted field due to pressure coil current.

Hence,

$$\text{Current coil field} \propto (I + I_p)$$

$$\text{Compensating coil field} \propto I_p$$

and acts in opposition.

Therefore, net field becomes:

$$(I + I_p) - I_p = I$$

Thus, the resultant field depends only on the load current  $I$ .  
Hence, the error due to pressure coil current is neutralized.  
Therefore, the compensating coil compensates the effect of:

Pressure (voltage) coil circuit impedance

and not the impedance of the current coil.

**Statement (i):**

“The compensating coil compensates the effect of impedance of the current coil.”

This is **false**, because the compensating coil does not compensate current coil impedance.

(i) False

**Statement (ii):**

“The compensating coil compensates the effect of impedance of the voltage coil circuit.”

This is **true**, because it neutralizes the phase error caused by:

$$R_p + jX_p$$

(ii) True

Therefore, the correct option is

**B. (i) is False but (ii) is True**

### Key Points

- The series resistance and capacitor in the pressure coil circuit of an LPF wattmeter are used to make the pressure coil current  $I_p$  nearly in phase with the applied voltage.

Pressure coil impedance is:  $Z_p = R_p + jX_L$

Hence, pressure coil current lags the applied voltage.

The series resistance makes the pressure coil circuit more resistive:

$$\tan \theta = \frac{X_L}{R}$$

Increasing  $R$  reduces  $\theta$ , so  $I_p$  becomes more in phase with voltage.

The capacitor provides a leading effect:

$$X_C = \frac{1}{\omega C}$$

Thus,

$$X_L - X_C \approx 0$$

making

$$I_p \approx \text{in phase with } V$$

### Q2.10.1

MCQ

2010

1M

Ans: D

● ○ ○

Given:

$$\text{Original range} = 0 - 5 \text{ A}, \quad R_m = 0.2 \Omega$$

Required new range:

$$0 - 25 \text{ A}$$

To increase ammeter range, a **shunt resistance** is connected in parallel.

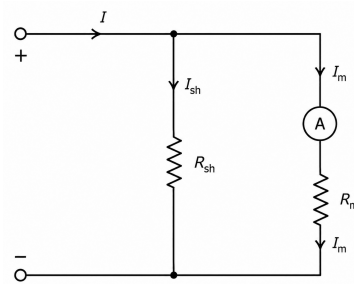


Fig. Solution Q2.10.1

At full-scale:

$$I_m = 5 \text{ A}, \quad I = 25 \text{ A}$$

Hence shunt current:

$$I_{sh} = 25 - 5 = 20 \text{ A}$$

Since meter and shunt are in parallel, voltage across both is same.

Voltage across meter:

$$V_m = I_m R_m = 5 \times 0.2 = 1 \text{ V}$$

Therefore,

$$V_{sh} = 1 \text{ V}$$

Shunt resistance:

$$R_{sh} = \frac{V_{sh}}{I_{sh}} = \frac{1}{20} = 0.05 \Omega$$

Therefore, the correct option is

**D.  $0.05 \Omega$  in parallel with the meter**

### Key Points

- **Increase ammeter range:**

Add shunt resistance in parallel

- **Decrease ammeter range:**

Add resistance in series

- For voltmeters, the logic is reversed:

Increase voltmeter range → Series resistance

Decrease voltmeter range → Parallel resistance (shunt)

### Q2.10.2

MCQ

2010

1M

Ans: D



A wattmeter measures

$$P = V_{PC} I_{CC} \cos \phi$$

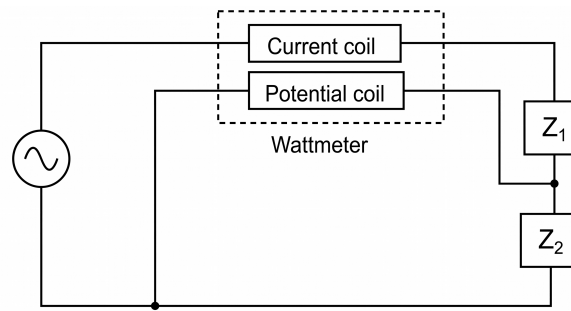


Fig. Solution Q2.10.2

where:

$V_{PC}$  = Voltage across the pressure (potential) coil

$I_{CC}$  = Current through the current coil

Here, the **current coil** is in series with  $Z_1 + Z_2$  (no current enters pressure coil), hence

$$I_{CC} = I$$

The **pressure coil** is connected only across  $Z_2$ , therefore

$$V_{PC} = V_{Z_2}$$

Thus wattmeter reading becomes

$$W = V_{Z_2} I \cos \phi$$

which is exactly the power absorbed by  $Z_2$  only.

Hence,

Wattmeter reads power absorbed by  $Z_2$  only

Therefore, the correct option is

**D. Power consumed by  $Z_2$**

#### Mistakes to be avoided

- Properly observe the connection of the **pressure coil** and **current coil** before deciding what power the wattmeter measures.
- Do not assume the wattmeter always measures total power of the entire circuit.

**Q2.09.1**

MCQ

2009

2M

Ans: C



For a wattmeter,

$$W = V_{PC} I_{CC} \cos \phi$$

where

$V_{PC}$  = voltage across pressure coil,  $I_{CC}$  = current through current coil

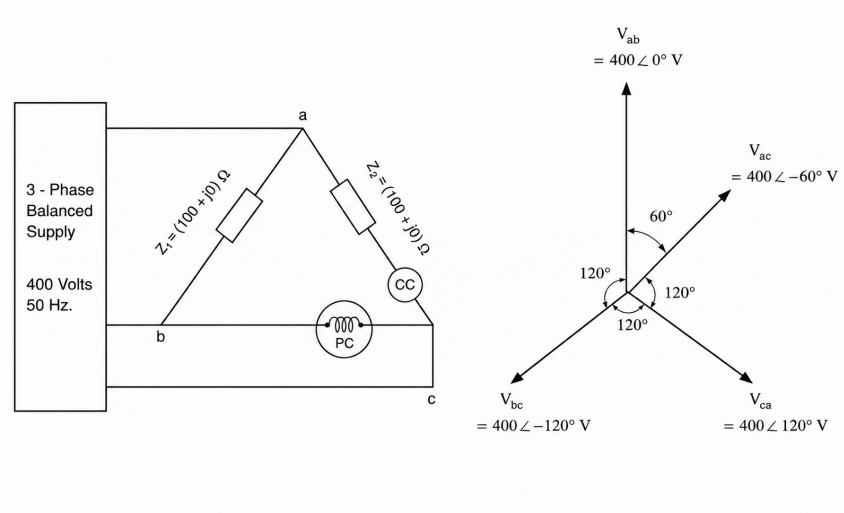


Fig. Solution Q2.09.1

From the circuit, the load is delta connected and

$$Z_1 = Z_2 = (100 + j0) \Omega$$

The source voltage is

$$V_{Line} = 400 \text{ V}$$

For a delta load,

$$V_{phase} = V_{Line}$$

Thus, branch voltage across  $Z_2$  is

$$V_{ac} = 400 \text{ V}$$

Hence current through branch  $Z_2$  (and through CC) is

$$I_{CC} = \frac{400}{100} = 4 \text{ A}$$

Since  $Z_2$  is resistive,  $I_{CC}$  is in phase with  $V_{ac}$ .  
The pressure coil is connected across  $b - c$ , so

$$V_{PC} = V_{bc}$$

Taking balanced sequence ( $V_{ab}$  as reference)

$$V_{ab} = 400 \angle 0^\circ, \quad V_{bc} = 400 \angle -120^\circ, \quad V_{ac} = 400 \angle -60^\circ$$

Thus,

$$I_{CC} = 4 \angle -60^\circ$$

Angle between  $V_{PC}$  and  $I_{CC}$  is

$$\phi = (-60^\circ) - (-120^\circ) = 60^\circ$$

Therefore,

$$W = 400 \times 4 \times \cos 60^\circ = 800 \text{ W}$$

Therefore, the correct option is

$$\boxed{\text{C. } 800 \text{ W}}$$

### Mistakes to be avoided

- Do phase calculation properly.

### Q2.09.2

MCQ

2009

1M

Ans: B



In a dynamometer type wattmeter, the pressure coil is connected across the supply voltage.

To measure true power,

$$P = VI \cos \phi$$

the pressure coil current must remain nearly in phase with the applied voltage.

If the pressure coil were inductive,

$$Z_p = R_p + jX_p$$

then pressure coil current  $I_p$  would lag the voltage, causing phase error and incorrect wattmeter reading.

To reduce this error, a large non-inductive resistance is connected in series with the pressure coil so that

$$R_p \gg X_p$$

Hence,

$$Z_p \approx R_p$$

Therefore, pressure coil current becomes nearly in phase with voltage.

Thus, the pressure coil behaves as:

Highly resistive

Hence, the correct option is

**B. Highly resistive**

### Q2.06.1

MCQ

2006

2M

Ans: B



For the **immersion heater** (purely resistive), the energy meter reads 2.3 kWh in 1 hour. Hence initial power consumed is

$$P_1 = \frac{2.3 \text{ kWh}}{1 \text{ h}} = 2.3 \text{ kW} = 2300 \text{ W}$$

For a resistive load,

$$P = \frac{V_{\text{rms}}^2}{R}$$

With initial RMS voltage  $V_1 = 230 \text{ V}$ ,

$$2300 = \frac{230^2}{R} \Rightarrow R = 23 \Omega$$

Now the input is a symmetric square wave of  $400 \text{ V}_{pp}$ , so

$$V_p = \frac{400}{2} = 200 \text{ V}$$

For a symmetric square wave,

$$V_{\text{rms}} = V_p = 200 \text{ V}$$

Therefore new power is

$$P_2 = \frac{V_{\text{rms}}^2}{R} = \frac{200^2}{23} = 1739 \text{ W} = 1.739 \text{ kW}$$

Therefore, the correct option is

**B. 1.739**

### Key Points

- For a purely resistive load:

$$P = \frac{V_{\text{rms}}^2}{R}$$

- Frequency does not affect power consumed by a pure resistor.

### Mistakes to be avoided

- Do not confuse peak voltage with peak-to-peak voltage:

$$V_p = \frac{V_{PP}}{2}$$

**Q2.06.2**

MCQ

2006

2M

Ans: C



Given,

$$i(t) = -8 + 6\sqrt{2} \sin(\omega t + 30^\circ)$$

This has:

$$I_{DC} = -8 \text{ A}$$

and sinusoidal AC component

$$i_{AC} = 6\sqrt{2} \sin(\omega t + 30^\circ)$$

#### 1. Centre-zero PMMC meter

A PMMC meter responds to the average value. The average of the sinusoidal term over one cycle is zero,

$$I_{avg} = -8 + 0 = -8 \text{ A}$$

Hence, PMMC reading is

$$-8 \text{ A}$$

#### 2. True RMS meter

For a DC plus sinusoidal current,

$$I_{rms} = \sqrt{I_{DC}^2 + I_{AC,rms}^2}$$

Here,

$$I_{AC,rms} = \frac{6\sqrt{2}}{\sqrt{2}} = 6 \text{ A}$$

Therefore,

$$I_{rms} = \sqrt{(-8)^2 + 6^2} = \sqrt{64 + 36} = 10 \text{ A}$$

So true RMS meter reads

$$10 \text{ A}$$

#### 3. Moving iron instrument

A moving iron instrument responds to RMS value, hence

$$10 \text{ A}$$

Therefore, the correct option is

$$\text{C. } -8, 10, 10$$

### Key Points

- Both **True RMS meters** and **Moving Iron (MI)** instruments measure RMS values, but their working principles are different.
- True RMS meter:**

A true RMS meter computes:

$$X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

It performs:

Square → Average → Square root

Hence, it gives accurate RMS value for almost any waveform

• **Moving Iron (MI) instrument:**

Deflecting torque:

$$T_d \propto I^2$$

Hence,

$$\theta \propto I_{\text{rms}}^2$$

So MI instruments effectively respond to RMS value.

• MI instruments are not ideal RMS meters because of:

Nonlinear scale, hysteresis, eddy current and frequency errors

• A **centre-zero PMMC** can indicate both positive and negative DC values.

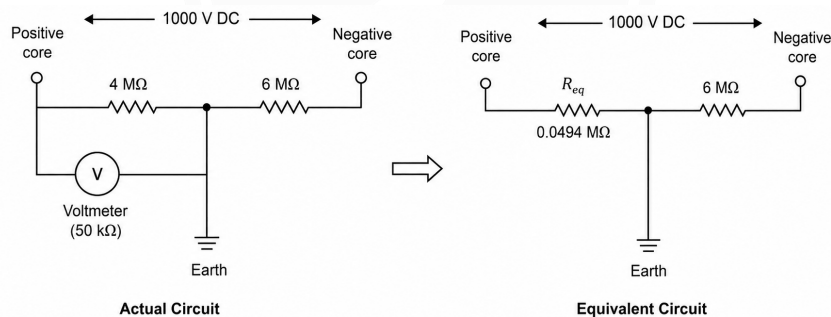
**Q2.05.1**

MCQ

2005

2M

Ans: A



**Fig. Solution Q2.05.1**

**Given:**

$$V_s = 1000 \text{ V}, \quad R_1 = 4 \text{ M}\Omega, \quad R_2 = 6 \text{ M}\Omega, \quad R_v = 50 \text{ k}\Omega = 0.05 \text{ M}\Omega$$

The voltmeter is connected between the positive core and earth, so it is in parallel with  $R_1$ .

**Step 1: Equivalent resistance of  $R_1$  and voltmeter**

$$R_{eq} = R_1 \parallel R_v = \frac{4 \times 0.05}{4 + 0.05} = 0.0494 \text{ M}\Omega$$

Thus the circuit becomes  $R_{eq}$  in series with  $R_2$ .

**Step 2: Total resistance**

$$R_T = R_{eq} + R_2 = 0.0494 + 6 = 6.0494 \text{ M}\Omega$$

**Step 3: Circuit current**

$$I = \frac{V_s}{R_T} = \frac{1000}{6.0494 \times 10^6} = 165.3 \mu\text{A}$$

**Step 4: Voltmeter reading (across  $R_{eq}$ )**

$$V_v = IR_{eq} = 165.3 \times 10^{-6} \times 0.0494 \times 10^6 = 8.16 \text{ V}$$

Therefore, the correct option is

**A. 8 V**

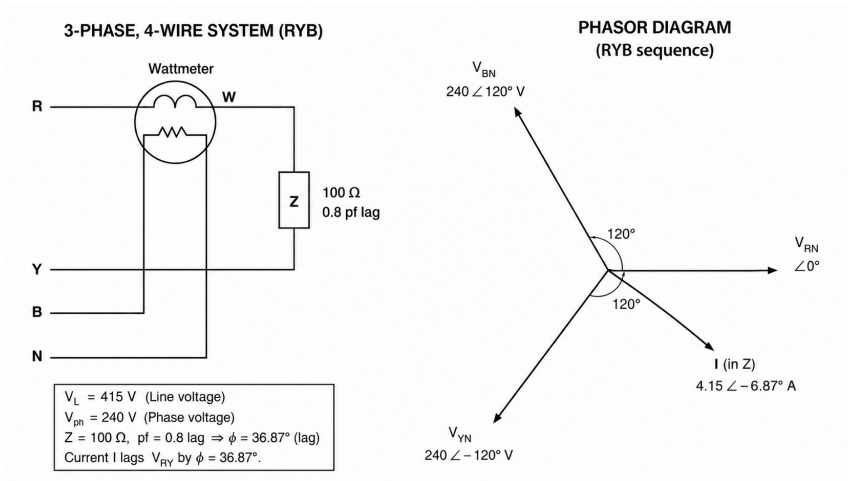
**Q2.04.1**

**MCQ**

**2004**

**2M**

**Ans: B**



**Fig. Solution Q2.04.1**

**Given:**

$$V_L = 415 \text{ V}, \quad Z = 100 \Omega, \quad \cos \phi = 0.8$$

Hence,

$$\phi = 36.87^\circ \text{ (lagging)}$$

**Step 1: Load current**

Since the load is connected between R and Y,

$$V_{RY} = 415 \angle 30^\circ \text{ V}$$

$$I = \frac{415}{100} = 4.15 \text{ A}$$

So the current coil current is

$$I_{CC} = 4.15 \angle (30^\circ - 36.87^\circ) = 4.15 \angle (-6.87^\circ)$$

**Step 2: Pressure coil voltage**

The pressure coil is connected between B and N, so

$$V_{PC} = V_{BN} = 240 \angle 120^\circ \text{ V}$$

**Step 3: Angle between  $V_{PC}$  and  $I_{CC}$**

$$\theta = 120^\circ - (-6.87^\circ) = 126.87^\circ$$

$$\cos \theta = \cos 126.87^\circ = -0.6$$

**Step 4: Wattmeter reading**

$$W = V_{PC} I_{CC} \cos \theta$$

$$W = 240 \times 4.15 \times (-0.6) = -597.6 \text{ W}$$

Therefore, the correct option is

**B. - 597 W**

### Mistakes to be avoided

- Be careful with signs.

**Q2.04.2**

MCQ

2004

2M

Ans: B



Given:

$$T_d = 240 \mu\text{Nm} = 240 \times 10^{-6} \text{ Nm}, \quad I = 10 \text{ A}$$

For a moving iron instrument,

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

Hence,

$$\frac{dL}{d\theta} = \frac{2T_d}{I^2}$$

Substituting,

$$\frac{dL}{d\theta} = \frac{2(240 \times 10^{-6})}{10^2} = \frac{480 \times 10^{-6}}{100} = 4.8 \times 10^{-6} \text{ H/rad}$$

Hence, the correct option is

**B. 4.8  $\mu\text{H/radian}$**

### Key Points

- Moving iron instruments work on the principle of change in self inductance with angular displacement.

### Mistakes to be avoided

- The given deflection angle  $120^\circ$  is not needed in this problem.

**Q2.04.3**

MCQ

2004

2M

Ans: C



Given:

$$V = 250 \text{ V}, \quad I = 15 \text{ A}$$

Meter constant:

$$14.4 \text{ A s/rev}$$

This means charge per revolution is

$$Q = 14.4 \text{ A} \cdot \text{s}$$

Energy consumed per revolution:

$$E_{\text{rev}} = VQ = 250 \times 14.4 = 3600 \text{ W} \cdot \text{s}$$

Since

$$1 \text{ Wh} = 3600 \text{ W} \cdot \text{s}$$

therefore,

$$E_{\text{rev}} = 1 \text{ Wh}$$

Hence,

$$1 \text{ rev} = 1 \text{ Wh}$$

Therefore,

$$1000 \text{ rev} = 1000 \text{ Wh} = 1 \text{ kWh}$$

So the meter constant is

**1000 rev/kWh**

Hence, the correct option is

**C. 1000 rev/kWh**

### Key Points

- Important conversion:

$$1 \text{ Wh} = 3600 \text{ J} = 3600 \text{ W} \cdot \text{s}$$

- Meter constant:

$$\text{rev/kWh} = \frac{\text{Total energy in 1 kWh}}{\text{Energy per revolution}}$$

### Mistakes to be avoided

- Do not forget that the question asks for:

$$\text{rev/kWh}$$

#### Q2.04.4

MCQ

2004

2M

Ans: A



Given:

$$N = 100, \quad B = 200 \text{ mT} = 0.2 \text{ T}$$

$$l = 10 \text{ mm} = 10 \times 10^{-3} \text{ m}$$

$$d = 20 \text{ mm} = 20 \times 10^{-3} \text{ m}$$

$$I = 50 \text{ mA} = 0.05 \text{ A}$$

For a PMMC instrument,

$$T = NBIA$$

where coil area is

$$A = l \times d$$

Thus,

$$A = (10 \times 10^{-3})(20 \times 10^{-3}) = 2 \times 10^{-4} \text{ m}^2$$

Substituting,

$$T = 100 \times 0.2 \times 0.05 \times 2 \times 10^{-4}$$

Hence,

$$T = 2 \times 10^{-4} \text{ Nm}$$

Converting into  $\mu\text{Nm}$ ,

$$T = 2 \times 10^{-4} \times 10^6 = 200 \mu\text{Nm}$$

Hence, the correct option is

$$\boxed{\text{A. } 200 \mu\text{Nm}}$$

#### Q2.04.5

MCQ

2004

1M

Ans: C



Given:

$$I_g = 10 \text{ mA} = 0.01 \text{ A}, \quad R_g = 1000 \Omega, \quad R_{sh} = 100 \Omega$$

Since galvanometer and shunt are in parallel,

$$I_g R_g = I_{sh} R_{sh}$$

Substituting,

$$0.01 \times 1000 = I_{sh} \times 100$$

$$10 = 100 I_{sh} \Rightarrow I_{sh} = 0.1 \text{ A}$$

Hence total measurable current is

$$I = I_g + I_{sh} = 0.01 + 0.1 = 0.11 \text{ A}$$

Multiplying power:

$$m = \frac{I}{I_g} = \frac{0.11}{0.01} = 11$$

Hence, the correct option is

**C. 11**

### Key Points

- Multiplying power of an ammeter is:

$$m = \frac{I}{I_g}$$

where:

$I$  = maximum measurable current

**Q2.04.6**

**MCQ**

**2004**

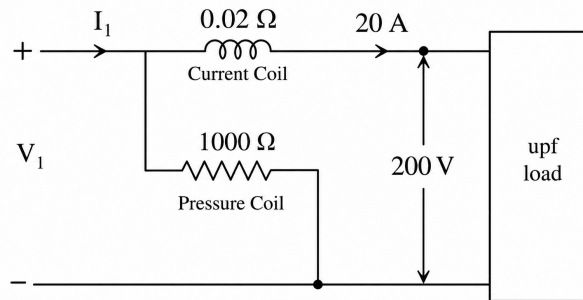
**1M**

**Ans: C**



**Given:**

$$R_{PC} = 1000 \Omega, \quad R_{CC} = 0.02 \Omega, \quad V_{\text{load}} = 200 \text{ V}, \quad I_{\text{load}} = 20 \text{ A}$$



**Fig. Solution Q2.04.6**

From the figure, the pressure coil is connected before the current coil, so the pressure coil voltage is

$$V_1 = V_{\text{load}} + I_{\text{load}} R_{CC}$$

$$V_1 = 200 + (20)(0.02) = 200.4 \text{ V}$$

Current through the pressure coil is

$$I_{PC} = \frac{V_1}{R_{PC}} = \frac{200.4}{1000} = 0.2004 \text{ A}$$

Hence current through the current coil is

$$I_1 = I_{\text{load}} + I_{PC} = 20 + 0.2004 = 20.2004 \text{ A}$$

However, the wattmeter reading is

$$P_m = V_1 I_1$$

Since pressure coil current contribution is negligible, error comes mainly from the current coil loss:

$$P_m = V_{\text{load}} I_{\text{load}} + I_{\text{load}}^2 R_{CC}$$

$$P_m = (200)(20) + 20^2(0.02) = 4000 + 8 = 4008 \text{ W}$$

True load power:

$$P_{\text{load}} = 200 \times 20 = 4000 \text{ W}$$

Percentage error:

$$\% \text{ error} = \frac{4008 - 4000}{4000} \times 100 = 0.2\%$$

Thus, the measured power is **0.2% more** than the load power.

Hence, the correct option is

C. 0.2% more

### Key Points

- Current coil resistance is very small, but at high current its power loss may still cause noticeable error.

**Q2.03.1**

MCQ

2003

2M

Ans: C



**Given:**

$$V = 220 \text{ V}, \quad I = 5 \text{ A}$$

Flux angle adjusted to:  $85^\circ$

Ideal flux angle:  $90^\circ$

Hence phase error:

$$\alpha = 90^\circ - 85^\circ = 5^\circ$$

For an induction energy meter,

$$T_d \propto VI \sin \delta$$

where:

$\delta$  = angle between voltage flux and load current

Under ideal condition,

$$\delta = 90^\circ - \phi$$

Therefore,

$$T_{\text{ideal}} \propto VI \sin(90^\circ - \phi)$$

$$T_{\text{ideal}} \propto VI \cos \phi$$

Because of phase error, actual flux angle becomes;  $85^\circ$

Hence,

$$\delta = 85^\circ - \phi$$

Thus actual torque becomes:

$$T_{\text{actual}} \propto VI \sin(85^\circ - \phi)$$

Therefore, power error is:

$$\Delta P = VI \left[ \sin(85^\circ - \phi) - \cos \phi \right]$$

**Case 1: Power factor = 1**

Hence,

$$\phi = 0^\circ$$

Substituting,

$$\Delta P = 220 \times 5 [\sin 85^\circ - 1]$$

$$= 1100(-0.0038) = -4.18$$

Therefore,

$$\Delta P \approx -4.2$$

### Case 2: Power factor = 0.5 lagging

$$\cos \phi = 0.5 \Rightarrow \phi = 60^\circ$$

Substituting,

$$\Delta P = 220 \times 5 [\sin 25^\circ - 0.5]$$

$$= 1100(-0.0774) = -85.1$$

Therefore,

$$\Delta P \approx -85.1$$

Therefore, the correct option is

$$\text{C. } -4 \text{ mW, } -85.1 \text{ mW}$$

### Key Points

- For ideal operation:

$$\delta = 90^\circ - \phi$$

hence,

$$T_d \propto VI \cos \phi$$

### Mistakes to be avoided

- Be careful with sign of error:

$$\text{Error} = P_{\text{actual}} - P_{\text{ideal}}$$

- At low power factor, even small phase-angle error can produce large measurement error.

### Q2.03.2

MCQ

2003

2M

Ans: C



Given,

$$L = \left( 10 + 3\theta - \frac{\theta^2}{4} \right) \mu\text{H}$$

Control spring torque:

$$T_c = 25 \times 10^{-6} \theta$$

Current:

$$I = 5 \text{ A}$$

For a moving iron instrument,

$$T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

At equilibrium,

$$T_d = T_c$$

Hence,

$$\frac{1}{2} I^2 \frac{dL}{d\theta} = 25 \times 10^{-6} \theta$$

Differentiating inductance,

$$\frac{dL}{d\theta} = \left( 3 - \frac{\theta}{2} \right) \mu\text{H/rad} = \left( 3 - \frac{\theta}{2} \right) \times 10^{-6} \text{ H/rad}$$

Substituting,

$$\frac{1}{2} (5)^2 \left( 3 - \frac{\theta}{2} \right) 10^{-6} = 25 \times 10^{-6} \theta$$

$$12.5 \left( 3 - \frac{\theta}{2} \right) = 25\theta$$

$$37.5 - 6.25\theta = 25\theta \Rightarrow 37.5 = 31.25\theta \Rightarrow \theta = \frac{37.5}{31.25} = 1.2$$

$$\theta = 1.2 \text{ rad}$$

Hence, the correct option is

**C. 1.2**

### Key Points

- The controlling torque in a Moving Iron (MI) instrument is:

$$T_c = K\theta$$

where:

$K$  = spring constant,  $\theta$  = pointer deflection

### Mistakes to be avoided

- The controlling torque is given in:

N-m/rad

or equivalently as:

$$T_c = K\theta$$

Read the units carefully before substituting values.

### Q2.03.3

MCQ

2003

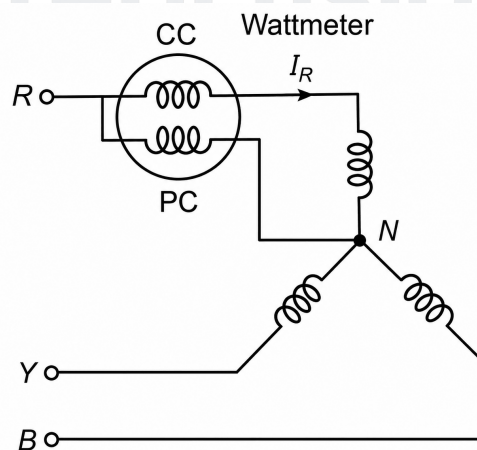
2M

Ans: B



Given a **balanced star load**,

$$\cos \phi = 0.8 \Rightarrow \phi = 36.87^\circ$$



**Fig. Solution Q2.03.3**

Initially, the current coil is in  $R$  phase and the pressure coil is between  $R$  and neutral, so the wattmeter reads

$$W_1 = 400 \text{ W}$$

Hence,

$$W_1 = V_{ph} I \cos \phi = V_{ph} I (0.8) \Rightarrow V_{ph} I = \frac{400}{0.8} = 500$$

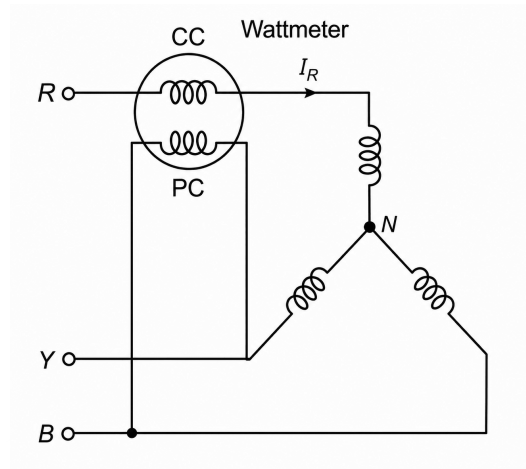


Fig. Solution Q2.03.3

Now the pressure coil is connected between  $B$  and  $Y$ , while the current coil remains in  $R$ . For  $RYB$  sequence,

$$V_{BY} = \sqrt{3} V_{ph} \angle 90^\circ, \quad I_R = I \angle (-36.87^\circ)$$

So the angle between  $V_{BY}$  and  $I_R$  is

$$90^\circ + 36.87^\circ = 126.87^\circ$$

Therefore,

$$W_2 = \sqrt{3} V_{ph} I \cos 126.87^\circ$$

Using  $V_{ph} I = 500$ ,

$$W_2 = \sqrt{3} \times 500 \times (-0.6) = -519.6 \text{ W}$$

Thus the wattmeter gives reverse deflection, and the reading is

$$519.6 \text{ W}$$

Hence, the correct option is

$$\text{B. } 519.6$$

#### Q2.03.4

MCQ

2003

1M

Ans: B



The effect of stray magnetic fields depends on how they combine with the instrument's operating magnetic field.

If the stray field is **parallel** to the operating field, then

$$B_{\text{net}} = B_{\text{operating}} \pm B_{\text{stray}}$$

Thus the resultant magnetic field changes maximum, causing maximum change in torque and maximum error.

If the two fields are perpendicular,

$$B_{\text{net}} = \sqrt{B_{\text{operating}}^2 + B_{\text{stray}}^2}$$

so the change in resultant field is comparatively smaller.

Hence, maximum effect occurs when:

$$\text{Operating field} \parallel \text{Stray field}$$

Therefore, the correct option is

$$\text{B. parallel}$$

### Key Points

- Stray magnetic field error depends on both:

Magnitude and direction of external magnetic field

- PMMC instruments are less affected by stray magnetic fields because they use strong permanent magnets.
- MI instruments are more affected because their operating magnetic field is comparatively weak.
- Stray magnetic field error can be reduced using:

Steel shielding or sheet-steel lining

**Q2.03.5**

MCQ

2003

1M

Ans: A



The moving coil of a PMMC instrument is generally made of copper, whose resistance increases with temperature:

$R_{\text{coil}} \uparrow$  as temperature increases

Hence meter current changes:

$$I_m = \frac{V}{R_{\text{meter}}}$$

causing inaccurate readings.

To reduce this effect, a **swamping resistance** made of manganin is connected in series with the moving coil. Since manganin has very low temperature coefficient,

$$R_{SW} \approx \text{constant}$$

and usually,

$$R_{SW} \gg R_{\text{coil}}$$

Therefore total meter resistance becomes:

$$R_{\text{meter}} = R_{SW} + R_{\text{coil}}$$

As temperature changes, variation in  $R_{\text{coil}}$  becomes negligible compared to the large swamping resistance, hence:

$$R_{\text{meter}} \approx \text{constant}$$

Thus meter current remains nearly constant and calibration remains stable.

Hence, the purpose of swamping resistance is:

to minimise the effect of temperature variation

Therefore, the correct option is

**A.** minimise the effect of temperature variation.

### Key Points

- A manganin shunt is also connected across the meter branch.

Since both  $R_{sh}$  and  $R_{SW}$  are made of manganin (very low temperature coefficient), the ratio:

$$\frac{I_{sh}}{I_m}$$

remains nearly constant with temperature.

**Q2.03.6**
**MCQ**
**2003**
**2M**
**Ans: C**


Given,

$$V_{\text{rms}} = 100 \text{ V}, \quad I_{fs} = 1 \text{ mA}, \quad R_m = 100 \Omega$$

A PMMC instrument responds to the **average** value, so the AC input is first full-wave rectified.

For a sinusoidal input,

$$V_m = \sqrt{2}V_{\text{rms}} = \sqrt{2} \times 100 = 141.4 \text{ V}$$

Average value of a full-wave rectified sine wave:

$$V_{\text{avg}} = \frac{2V_m}{\pi} = \frac{2 \times 141.4}{\pi} = 90.03 \text{ V}$$

At full-scale deflection,

$$I_{fs} = \frac{V_{\text{avg}}}{R_s + R_m}$$

Hence,

$$R_s + R_m = \frac{90.03}{0.001} = 90030 \Omega$$

Therefore,

$$R_s = 90030 - 100 = 89930 \Omega = 89.93 \text{ k}\Omega$$

Hence, the correct option is

**C. 89.93 kΩ**

**Q2.02.1**
**MCQ**
**2002**
**1M**
**Ans: C**


Given,

$$V_L = 100 \text{ V}, \quad Z = 5 \angle 60^\circ \Omega$$

Since the load is balanced and star connected,

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{100}{\sqrt{3}} = 57.7 \text{ V}$$

Phase current:

$$I_{ph} = \frac{V_{ph}}{|Z|} = \frac{57.7}{5} = 11.54 \text{ A}$$

Hence,

$$I_L = I_{ph} = 11.54 \text{ A}$$

For the two-wattmeter method,

$$W_1 = V_L I_L \cos(30^\circ + \phi), \quad W_2 = V_L I_L \cos(30^\circ - \phi)$$

Here,

$$\phi = 60^\circ$$

Thus,

$$W_1 = 100 \times 11.54 \cos 90^\circ = 0$$

and

$$W_2 = 100 \times 11.54 \cos(-30^\circ) = 1154 \times 0.866 \approx 1000 \text{ W}$$

Hence, the correct option is

**C.  $W_1 = 0, \quad W_2 = 1000 \text{ W}$**

### Key Points

- In star connection:

$$I_{Line} = I_{phase}$$

- In delta connection:

$$V_{Line} = V_{phase}$$

#### Q2.01.1

MCQ

2001

1M

Ans: B



For a 3-phase, 3-wire system, the minimum number of wattmeters required to measure total power for both balanced and unbalanced loads is:

2

In a 3-phase, 3-wire system:

$$I_R + I_Y + I_B = 0$$

Thus only two line currents are independent, so only two wattmeters are sufficient to measure total power.

Using the two-wattmeter method:

$$P_{total} = W_1 + W_2$$

This method works for:

Balanced and unbalanced loads

and for:

Leading as well as lagging power factor

Therefore, two wattmeters are sufficient for any 3-phase, 3-wire system.

Hence, the correct option is

B. 2

### Key Points

- According to **Blondel's theorem**, the number of wattmeters required to measure total power in an  $N$ -phase system is:

$$N \quad \text{or} \quad (N - 1)$$

- If neutral wire is available:

$$\text{Number of wattmeters required} = N$$

- If neutral wire is not available:

$$\text{Number of wattmeters required} = N - 1$$

### Mistakes to be avoided

- Carefully read whether the question asks for **balanced** or **unbalanced** load.
- Although one wattmeter can measure total power in a **balanced** load using special connections, the question here asks for the general minimum number required, which must also work for unbalanced loads.

### Q2.01.2

MCQ

2001

1M

Ans: D



Given,

$$P = 450 \text{ W}, \quad t = 100 \text{ s}, \quad N = 10$$

Energy consumed:

$$E = Pt = 450 \times 100 = 45000 \text{ J}$$

Since,

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

Energy in kWh:

$$E = \frac{45000}{3.6 \times 10^6} = 0.0125 \text{ kWh}$$

Thus,

$$10 \text{ revolutions} \equiv 0.0125 \text{ kWh}$$

Hence meter constant is

$$K = \frac{10}{0.0125} = 800 \text{ rev/kWh}$$

Therefore,

$$800 \text{ rev/kWh}$$

Hence, the correct option is

$$\text{D. } 800$$

#### Key Points

- Meter constant:

$$K = \frac{N \times 3600 \times 1000}{Pt}$$

where  $K = \text{rev/kWh}$ ,  $N = \text{number of revolutions}$ ,  $P = \text{power in watts}$ ,  $t = \text{time in seconds}$ .

### Q2.00.1

MCQ

2000

2M

Ans: D



**Explanation:**

The two-wattmeter method measures total active power in a 3-phase, 3-wire system as:

$$P = W_1 + W_2$$

This relation is valid irrespective of whether the voltages are balanced or unbalanced.

For a 3-wire system:

$$I_R + I_Y + I_B = 0$$

The total instantaneous power is:

$$p = v_R i_R + v_Y i_Y + v_B i_B$$

Using:

$$i_Y = -(i_R + i_B)$$

Substituting,

$$\begin{aligned} p &= v_R i_R + v_Y [-(i_R + i_B)] + v_B i_B \\ &= (v_R - v_Y) i_R + (v_B - v_Y) i_B \end{aligned}$$

Thus,

$$p = V_{RY} i_R + V_{BY} i_B = W_1 + W_2$$

Hence the total power measured by the two-wattmeter method equals the true total power, independent of voltage unbalance.

Therefore, neither negative-sequence voltages, nor zero-sequence voltages affect the total power measured by

the two-wattmeter method.

The correct answer is:

**D. not affected by negative or zero sequence voltages**

### Key Points

- Any unbalanced 3-phase voltage/current set can be decomposed into:

Positive sequence + Negative sequence + Zero sequence

This is called **symmetrical components**.

| Sequence | Magnitude | Phase Order | Example          |
|----------|-----------|-------------|------------------|
| Positive | Equal     | RYB         | Normal system    |
| Negative | Equal     | RBY         | Reverse sequence |
| Zero     | Equal     | Same angle  | All overlap      |

**Q2.00.2**

MCQ

2000

1M

Ans: C

☒ ☐ ☐

A true RMS measuring circuit computes:

$$V_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

The process is:

- Square the input:

$$v(t) \rightarrow v^2(t)$$

- Average over time:

$$\frac{1}{T} \int_0^T v^2(t) dt$$

- Take square root.

The slow response occurs because of the averaging stage.

To obtain a stable average value, the circuit uses an RC integrator / low-pass filter having time constant:

$$\tau = RC$$

This averaging smooths fluctuations, so sudden input changes take time to appear at the output.

Hence:

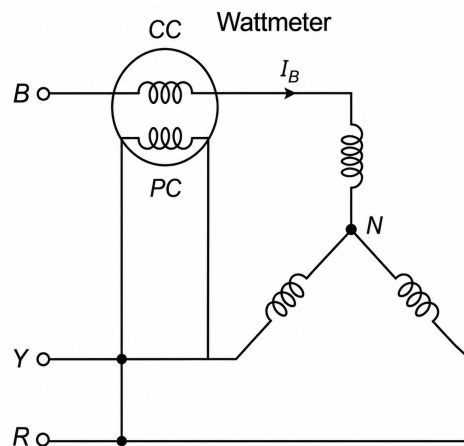
Averaging  $\Rightarrow$  Slow response

The correct answer is:

**C. the averaging function built into the circuit**

**Q2.00.3**
**NAT**
**2000**
**5M**
**Ans: \***
☒ ☒ ☐

For a balanced 3-phase load with phase sequence: RYB



**Fig. Solution Q2.00.3**

Pressure coil (PC) is connected across: R-Y

Hence,

$$V_{PC} = V_{RY}$$

Current coil (CC) is connected in phase B

Hence,

$$I_{CC} = I_B$$

Need to show:

Wattmeter reading  $\propto$  Reactive power

### Step 1: Phase and line voltages

For a balanced system:

$$V_R = V_{ph} \angle 0^\circ, \quad V_Y = V_{ph} \angle -120^\circ, \quad V_B = V_{ph} \angle 120^\circ$$

Line voltage:

$$V_{RY} = \sqrt{3} V_{ph} \angle 30^\circ$$

Thus,

$$V_{RY} = V_L \angle 30^\circ$$

where

$$V_L = \sqrt{3} V_{ph}$$

### Step 2: Current in B phase

If load power factor angle is  $\phi$ , then:

$$I_B = I \angle (120^\circ - \phi)$$

### Step 3: Angle between $V_{RY}$ and $I_B$

$$V_{RY} = V_L \angle 30^\circ, \quad I_B = I \angle (120^\circ - \phi)$$

Hence,

$$\theta = (120^\circ - \phi) - 30^\circ = 90^\circ - \phi$$

### Step 4: Wattmeter reading

Wattmeter measures:

$$W = VI \cos \theta$$

Substituting,

$$W = V_L I \cos(90^\circ - \phi)$$

Using,

$$\cos(90^\circ - \phi) = \sin \phi$$

we get:

$$W = V_L I \sin \phi$$

### Step 5: Compare with reactive power

Reactive power of a balanced 3-phase load is:

$$Q = \sqrt{3} V_L I \sin \phi$$

Therefore,

$$Q = \sqrt{3} W \Rightarrow W = \frac{Q}{\sqrt{3}}$$

Hence,

$$W \propto Q$$

Thus, the wattmeter indication is proportional to the reactive power of the balanced 3-phase load.

**Q2.99.1**

MCQ

1999

1M

Ans: C

☒ ☐ ☐

### 1: Ideal wattmeter operation

In a dynamometer wattmeter:

- i) Current coil (CC) carries load current  $I$
- ii) Pressure coil (PC) carries current proportional to voltage  $I_p \propto V$

Ideally,

$$I_p \parallel V$$

Then wattmeter reads:

$$W = VI \cos \phi$$

which is the true power.

### 2: Practical situation

Pressure coil has:

Resistance + inductance

Hence pressure coil current lags applied voltage by angle  $\alpha$

Thus,

$$I_p = V \angle (-\alpha)$$

So wattmeter actually measures:

$$W' = VI \cos(\phi + \alpha)$$

instead of:

$$W = VI \cos \phi$$

Error:

$$= VI[\cos(\phi + \alpha) - \cos \phi]$$

### 3: Why error becomes large at low power factor

At low power factor:

$$\phi \rightarrow 90^\circ$$

Hence  $\cos \phi$  is already very small.

So even a small phase error  $\alpha$  causes large percentage error.

Example:

$$\cos 80^\circ = 0.174$$

and,

$$\cos 85^\circ = 0.087$$

which gives nearly:

$$50\% \text{ error}$$

Thus large errors occur because:

Pressure coil current is not in phase with voltage

The correct answer is:

C. pressure coil currents are not in phase

**Q2.99.2**

**NAT**

**1999**

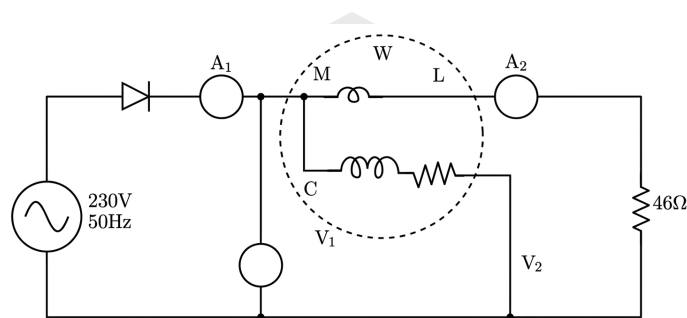
**5M**

**Ans: \***



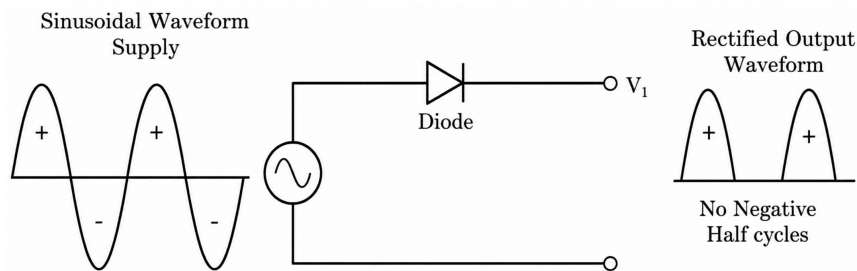
Given,

$$V_{\text{rms}} = 230 \text{ V}, \quad f = 50 \text{ Hz}, \quad R = 46 \Omega$$



**Fig. Solution Q2.99.2**

The diode gives **half-wave rectification**, so each instrument must be interpreted according to its principle.



**Fig. Solution Q2.99.2**

Peak voltage:

$$V_m = \sqrt{2} V_{\text{rms}} = 1.414 \times 230 = 325 \text{ V}$$

Peak current:

$$I_m = \frac{V_m}{R} = \frac{325}{46} = 7.07 \text{ A}$$

### 1. Reading of $A_1$ (moving-iron ammeter)

Moving-iron instruments respond to **RMS value**. For a half-wave rectified sine wave,

$$I_{\text{rms}} = \frac{I_m}{2} = \frac{7.07}{2} = 3.535 \text{ A}$$

Hence,

$$A_1 = 3.54 \text{ A}$$

## 2. Reading of $A_2$ (full-wave rectifier type ammeter)

Rectifier-type instruments respond to **average value** and are calibrated to indicate RMS for a sine wave. Average current for half-wave rectified waveform:

$$I_{\text{avg}} = \frac{I_m}{\pi} = \frac{7.07}{\pi} = 2.25 \text{ A}$$

Indicated reading:

$$A_2 = 1.11 I_{\text{avg}} = 1.11 \times 2.25 = 2.50 \text{ A}$$

## 3. Reading of $V_1$ (peak-response voltmeter)

Peak-response voltmeter measures the **peak value**:

$$V_1 = V_m = 325 \text{ V}$$

## 4. Reading of $V_2$ (PMMC voltmeter)

PMMC measures:

Average DC value

Average current:

$$I_{\text{avg}} = 2.25 \text{ A}$$

Therefore,

$$V_2 = I_{\text{avg}} R = 2.25 \times 46 = 103.5 \text{ V}$$

Hence,

$$V_2 = 103.5 \text{ V}$$

**Important:**

Do not use:

$$V_2 = A_2 \times 46$$

because  $A_2$  is a calibrated indication, not the actual average current.

## 5. Reading of wattmeter $W$

The electrodynameometer wattmeter measures **average power**:

$$W = I_{\text{rms}}^2 R = (3.535)^2 \times 46 = 575 \text{ W}$$

Therefore,

$$A_1 = 3.54 \text{ A}, \quad A_2 = 2.50 \text{ A}, \quad V_1 = 325 \text{ V}, \quad V_2 = 103.5 \text{ V}, \quad W = 575 \text{ W}$$

**Q2.98.1**

MCQ

1998

1M

Ans: C

● ○ ○

Given,

$$S = 1000 \Omega/\text{V}, \quad V_{fs} = 100 \text{ V}$$

Total voltmeter resistance:

$$R_v = S \times V_{fs} = 1000 \times 100 = 100000 \Omega = 100 \text{ k}\Omega$$

The meter reads half of full-scale value, hence applied voltage is:

$$V = \frac{100}{2} = 50 \text{ V}$$

Using Ohm's law,

$$I = \frac{V}{R_v} = \frac{50}{100000} = 0.0005 \text{ A} = 0.5 \text{ mA}$$

Hence, the correct option is

**C. 0.5 mA**

### Key Points

- Shortcut:

$$S = 1000 \Omega/V \Rightarrow I_{fs} = \frac{1}{1000} = 1 \text{ mA}$$

At half-scale reading:

$$I = 0.5 \text{ mA}$$

**Q2.98.2**

**NAT**

**1998**

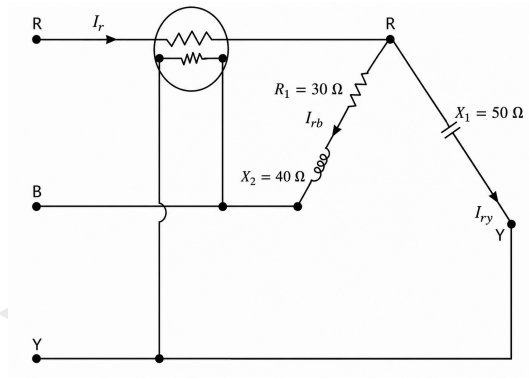
**5M**

**Ans: -404.8**



Given,

$$V_L = 400 \text{ V}, \quad \text{phase sequence } RYB$$



**Fig. Solution Q2.98.2**

Load impedances:

$$Z_{RB} = 30 + j40 = 50 \angle 53.13^\circ, \quad Z_{RY} = -j50 = 50 \angle (-90^\circ)$$

Line voltages:

$$V_{RB} = 400 \angle (-30^\circ), \quad V_{RY} = 400 \angle 30^\circ$$

Current through branch RB:

$$I_{RB} = \frac{V_{RB}}{Z_{RB}} = \frac{400 \angle (-30^\circ)}{50 \angle 53.13^\circ} = 8 \angle (-83.13^\circ) = 0.96 - j7.94$$

Current through branch RY:

$$I_{RY} = \frac{V_{RY}}{-j50} = \frac{400 \angle 30^\circ}{50 \angle (-90^\circ)} = 8 \angle 120^\circ = -4 + j6.93$$

Current through wattmeter current coil:

$$I_R = I_{RB} + I_{RY} = (0.96 - j7.94) + (-4 + j6.93) = -3.04 - j1.01 = 3.20 \angle (-161.6^\circ)$$

Pressure coil voltage:

$$V_{PC} = V_{RY} = 400 \angle 30^\circ$$

Angle between  $V_{PC}$  and  $I_R$ :

$$\phi = 30^\circ - (-161.6^\circ) = 191.6^\circ$$

Wattmeter reading:

$$W = V_{PC} I_R \cos \phi = 400 \times 3.20 \times \cos 191.6^\circ = -404.8 \text{ W}$$

**-404.8**

### Key Points

- Negative wattmeter reading indicates:  
Reverse deflection
- Negative sign occurs when current coil and pressure coil polarities produce:

$$\cos \phi < 0$$

### Mistakes to be avoided

- During complex number calculations, sign and angle mistakes are very common.

Q2.97.1

MCQ

1997

1M

Ans: A



In a dynamometer type wattmeter:

- Current coil carries load current:

$$i = I_m \sin \omega t$$

- Pressure coil carries current proportional to voltage:

$$v = V_m \sin(\omega t + \phi)$$

Deflecting torque:

$$T_d \propto vi$$

Thus,

$$T_d \propto V_m I_m \sin \omega t \sin(\omega t + \phi)$$

Using:

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

we get:

$$T_d \propto \frac{1}{2} V_m I_m [\cos \phi - \cos(2\omega t + \phi)]$$

This contains:

Constant term  $\propto VI \cos \phi$ , which represents active power.

Oscillating term  $\propto \cos(2\omega t + \phi)$ , whose average over one cycle is zero.

Hence average torque:

$$T_{\text{avg}} \propto VI \cos \phi$$

Therefore, a dynamometer wattmeter measures:

Average active power

The correct answer is:

A. average value of active power

**Q2.96.1**
**MCQ**
**1996**
**1M**
**Ans: B**


In a PMMC (Permanent Magnet Moving Coil) instrument, deflecting torque is:

$$T_d = NBIA$$

Strong permanent magnets produce high magnetic flux density(B): Hence large torque is produced even for small current.

Also, the moving system (coil + pointer) is very light.

Therefore:

$$\frac{\text{Torque}}{\text{Weight of moving parts}}$$

is high.

This gives:

High sensitivity, quick response and better accuracy

Hence, an important advantage of PMMC instruments is:

High torque/weight ratio

**Why other options are wrong:**

**A. Free from friction error** ×

PMMC instruments use pivots and bearings, so friction error is reduced but not completely eliminated.

**C. Low torque/weight ratio** ×

This is opposite to the actual PMMC characteristic.

**D. Can be used on both AC and DC** ×

PMMC instruments work only on DC because for AC torque reverses every half cycle.

Hence average torque becomes zero.

**B. has high (torque/weight of moving parts) ratio**

**Q2.96.2**
**MCQ**
**1996**
**1M**
**Ans: C**


A uniform scale means:

Equal change in input  $\Rightarrow$  Equal change in deflection

i.e.,

$$\theta \propto X$$

For a PMMC instrument, deflecting torque is:

$$T_d = NBIA$$

Hence,

$$T_d \propto I$$

Controlling torque:

$$T_c = K\theta$$

At equilibrium:

$$T_d = T_c \Rightarrow I \propto \theta \Rightarrow \theta \propto I$$

Thus PMMC instruments have:

Uniform (linear) scale

**Why other options are wrong:**

**A. Moving iron** ×

For moving iron instruments:

$$T_d \propto I^2 \Rightarrow \theta \propto I^2$$

Hence scale is nonlinear and crowded near zero.

**B. Induction** ×

Induction instruments are mainly used for energy meters.

**D. Dynamometer** ×

Scale is generally non-uniform because torque depends on product of currents.

The correct answer is:

**C. moving coil permanent magnet**

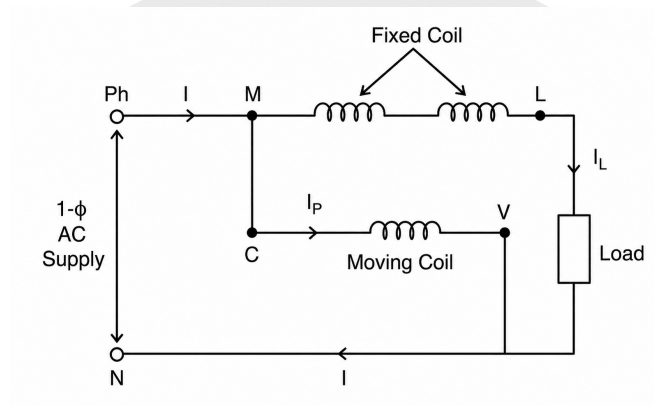
**Q2.96.3**

**MCQ**

**1996**

**1M**

**Ans: B**



**Fig. Solution Q2.96.3**

A dynamometer wattmeter has:

- Fixed coil (Current coil, CC)

Connected in series with the load, hence:

$$I_{CC} = I_{load}$$

- Moving coil (Pressure/Potential coil, PC)

Connected across the supply through a high resistance, hence:

$$I_{PC} \propto V$$

The interaction of:

$$\text{Magnetic field due to CC} \propto I$$

and

$$\text{Current in PC} \propto V$$

produces deflecting torque:

$$T_d \propto VI \cos \phi$$

Thus the wattmeter measures power.

**Why other options are wrong:**

**A. In series with fixed coil** ×

Then the moving coil would carry load current instead of voltage-proportional current.

**C. In series with load** ×

This is the connection of the current coil.

**D. Across the load**

Practically load and supply voltages are nearly same, but standard wattmeter terminology states:

Pressure coil is connected across the supply

The correct answer is:

**B. across the supply**

### Key Points

- The two fixed coils are connected in series with the load and carry full load current:

$$I_{CC} = I_L$$

**Q2.96.4**

MCQ

1996

1M

Ans: A

● ○ ○

In an induction type ammeter, deflecting torque is produced due to interaction between:

- induced eddy currents
- alternating magnetic flux

Deflecting torque:

$$T_d \propto \phi_1 \phi_2 \sin \delta$$

For a fixed frequency:

$$\delta = \text{constant}$$

Also, magnetic flux is proportional to current:

$$\phi \propto I$$

Therefore,

$$T_d \propto I^2$$

The correct answer is:

**A. current<sup>2</sup>**

**Q2.95.1**

NAT

1995

1M

Ans: 150

● ○ ○

Given two PMMC voltmeters:

**Voltmeter 1:**

$$FOM_1 = 10 \text{ k}\Omega/\text{V}, \quad V_1 = 100 \text{ V}$$

Hence,

$$R_1 = 100 \times 10 \text{ k}\Omega = 1000 \text{ k}\Omega = 1 \text{ M}\Omega$$

**Voltmeter 2:**

$$FOM_2 = 20 \text{ k}\Omega/\text{V}, \quad V_2 = 100 \text{ V}$$

Hence,

$$R_2 = 100 \times 20 \text{ k}\Omega = 2000 \text{ k}\Omega = 2 \text{ M}\Omega$$

When connected in series,

$$R_T = R_1 + R_2 = 1 + 2 = 3 \text{ M}\Omega$$

For a PMMC voltmeter,

$$I_{fs} = \frac{1}{\text{FOM}}$$

So,

$$I_{fs1} = \frac{1}{10 \text{ k}\Omega/\text{V}} = 100 \mu\text{A}$$

$$I_{fs2} = \frac{1}{20 \text{ k}\Omega/\text{V}} = 50 \mu\text{A}$$

Since the series current is the same, the lower full-scale current limits the combination:

$$I_{\max} = 50 \mu\text{A}$$

Therefore, maximum measurable voltage is

$$V_{\max} = I_{\max} R_T = 50 \times 10^{-6} \times 3 \times 10^6 = 150 \text{ V}$$

150

**Q2.94.1**

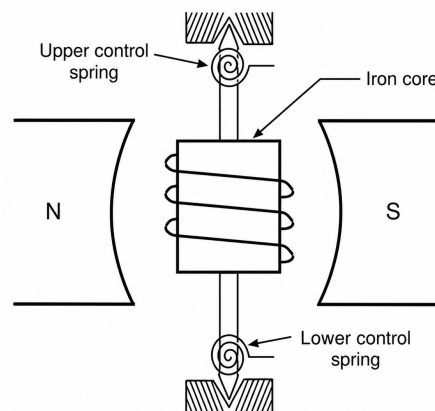
MCQ

1994

1M

Ans: D

● ○ ○



**Fig. Solution Q2.94.1**

A PMMC instrument has two control springs which:

- i) provide controlling torque
- ii) carry current to the moving coil

Current path:

Terminal → Top spring → Moving coil → Bottom spring → Terminal

If the bottom spring snaps, the moving coil circuit becomes open.

Hence,

$$I_{\text{coil}} = 0$$

Therefore deflecting torque:

$$T_d = NBIA$$

becomes:

$$T_d = 0$$

Thus no torque acts on the pointer, and it returns to zero position.

Hence meter reading becomes:

0 mA

Therefore,

D. zero

**Q2.93.1**

MCQ

1993

1M

Ans: C

● ○ ○

Given,

$$GF = 2, \quad R = 120 \Omega, \quad \epsilon = 10^{-5}$$

Gauge factor relation:

$$GF = \frac{\Delta R/R}{\epsilon}$$

Hence,

$$\Delta R = GF \times R \times \epsilon$$

Substituting,

$$\begin{aligned} \Delta R &= 2 \times 120 \times 10^{-5} \\ &= 240 \times 10^{-5} = 2.4 \times 10^{-3} \Omega \end{aligned}$$

Therefore,

$$\Delta R = 2.4 \times 10^{-3} \Omega$$

Hence, the correct option is

$$\text{C. } \Delta R = 2.4 \times 10^{-3} \Omega$$

### Key Points

- Gauge factor (GF) of a strain gauge is defined as:

$$GF = \frac{\Delta R/R}{\epsilon}$$

- It represents:

Fractional change in resistance per unit strain

- Higher gauge factor  $\Rightarrow$  Higher sensitivity

**Q2.92.1**

MCQ

1992

1M

Ans: D

● ○ ○

A moving iron (MI) instrument works on:

$$T_d \propto \phi^2$$

where:

$\phi$  = magnetic flux produced by the coil current

Since torque depends on  $\phi^2$

MI instruments can operate on both AC and DC.

Now suppose a stray DC magnetic field  $B_s$  appears along the meter axis.

This stray field either adds to or opposes the instrument magnetic field  $B_i$ .

Hence total magnetic field becomes:

$$B_{\text{total}} = B_i \pm B_s$$

Therefore torque becomes:

$$T_d \propto (B_i \pm B_s)^2$$

Expanding,

$$(B_i \pm B_s)^2 = B_i^2 + B_s^2 \pm 2B_iB_s$$

If the stray field aids the instrument field:

$$(B_i + B_s)^2$$

torque increases, hence meter reading increases.

If the stray field opposes the instrument field:

$$(B_i - B_s)^2$$

torque decreases, hence meter reading decreases.

Hence,

**D.** either decreased or increased depending on the di- rection of the DC field

**Q2.91.1**

**NAT**

**1991**

**1M**

**Ans: 85**



Given,

$$R = 10 \Omega$$

The current waveform is:

$$12 \text{ A for } T, \quad 5 \text{ A for next } T$$

and repeats every

$$2T$$

A DC voltmeter measures the **average** voltage across the resistor.

Average current:

$$I_{\text{avg}} = \frac{12T + 5T}{2T} = 8.5 \text{ A}$$

Hence average voltage:

$$V_{\text{avg}} = I_{\text{avg}}R = 8.5 \times 10 = 85 \text{ V}$$

85 V

# DEBUG INFO

## Chapter II

Analog Measuring Instruments

**Total Questions : 65**

MCQ : 49

MSQ : 0

NAT : 16



PrepFusion

## SOLUTIONS

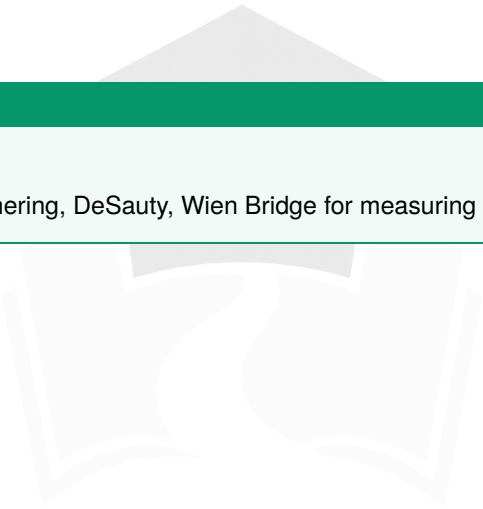


# DC And AC Bridges

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### Topics

1. Wheatstone Bridge
2. AC Bridges: Maxwell, Hay, Schering, DeSauty, Wien Bridge for measuring Inductance and Capacitance



PrepFusion

**Q3.26.1**
**NAT**
**2026**
**2M**
**Ans: 750**


Given,

$$P = 2000 \Omega, \quad Q = 500 \Omega, \quad R = 3000 \Omega$$

Battery voltage:

$$E = 50 \text{ V}$$

Battery internal resistance:

$$r = 1 \Omega$$

Galvanometer resistance:

$$G = 50 \Omega$$

For a balanced Wheatstone bridge,

$$\frac{Q}{P} = \frac{S}{R}$$

Hence,

$$\frac{500}{2000} = \frac{S}{3000} \Rightarrow 2000S = 500 \times 3000 \Rightarrow S = 750 \Omega$$

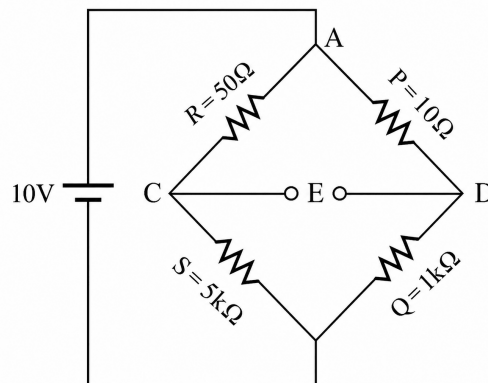
At balance, no current flows through the galvanometer, so  $G$  has no effect. The battery internal resistance  $r$  also does not affect the balance condition.

750

**Q3.25.1**
**NAT**
**2025**
**2M**
**Ans: 1.96**


Given,

$$R = 50 \Omega, \quad P = 10 \Omega, \quad Q = 1 \text{ k}\Omega, \quad S = 5 \text{ k}\Omega, \quad V_s = 10 \text{ V}$$



**Fig. Solution Q3.25.1**

The output voltage across the galvanometer terminals is

$$E = V_C - V_D$$

where

$$V_C = 10 \frac{S}{R + S}, \quad V_D = 10 \frac{Q}{P + Q}$$

Initially,

$$E = 10 \left( \frac{5000}{5050} - \frac{1000}{1010} \right) \approx 0$$

so the bridge is balanced.

Now increase  $R$  from  $50\ \Omega$  to  $51\ \Omega$ :

$$V'_C = 10 \frac{5000}{5051}, \quad V'_D = 10 \frac{1000}{1010}$$

Hence,

$$E' = 10 \left( \frac{5000}{5051} - \frac{1000}{1010} \right) \approx -0.00196\text{ V}$$

So,

$$|E'| = 1.96\text{ mV}$$

Since  $\Delta R = 1\ \Omega$ ,

$$\frac{\Delta E}{\Delta R} = \frac{1.96}{1} = 1.96\text{ mV}/\Omega$$

1.96

### Key Points

- There is a unit change in resistance: so the resistance can become:

$$50 \rightarrow 51\ \Omega \quad \text{or} \quad 50 \rightarrow 49\ \Omega$$

- The magnitude of sensitivity remains same in both cases, only the sign of output voltage changes.
- Hence both answers  $+1.96\text{ mV}/\Omega$  and  $-1.96\text{ mV}/\Omega$  are valid according to the final answer key.

Q3.21.1

MCQ

2021

1M

Ans: B

● ○ ○

Different AC bridges are used for measurement of different electrical quantities:

| Bridge          | Measures                        |
|-----------------|---------------------------------|
| Schering bridge | Capacitance and dielectric loss |
| Maxwell bridge  | Inductance                      |
| Kelvin bridge   | Very low resistance             |
| Wien bridge     | Frequency                       |

Hence, the bridge used for measurement of inductance is:

B. Maxwell bridge

Q3.15.1

MCQ

2015

2M

Ans: A

● ● ○

Given circuit,

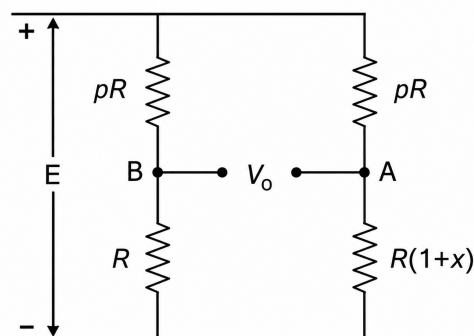


Fig. Solution Q3.15.1

Need to maximize:

$$|V_0| = |V_A - V_B|$$

Supply voltage = E

**Step 1: Find node voltage  $V_A$**

Using voltage division in the right branch:

Upper resistance = pR

Lower resistance = R(1+x)

Hence,

$$V_A = E \frac{R(1+x)}{pR + R(1+x)} = E \frac{1+x}{p+1+x}$$

**Step 2: Find node voltage  $V_B$**

Similarly for the left branch:

Upper resistance = pR

Lower resistance = R

Thus,

$$V_B = E \frac{R}{pR + R} = E \frac{1}{p+1}$$

**Step 3: Find bridge output voltage**

$$V_0 = V_A - V_B$$

Substituting,

$$V_0 = E \left( \frac{1+x}{p+1+x} - \frac{1}{p+1} \right)$$

Taking LCM,

$$V_0 = E \frac{(1+x)(p+1) - (p+1+x)}{(p+1)(p+1+x)}$$

Simplifying numerator,

$$(1+x)(p+1) - (p+1+x) = px$$

Hence,

$$V_0 = E \frac{px}{(p+1)(p+1+x)}$$

Since E and x are constants, maximize:

$$f(p) = \frac{p}{(p+1)(p+1+x)}$$

**Step 4: Differentiate**

$$f(p) = \frac{p}{p^2 + p(2+x) + (1+x)}$$

Using quotient rule,

$$f'(p) = \frac{[p^2 + p(2+x) + (1+x)] - p(2p + 2 + x)}{[p^2 + p(2+x) + (1+x)]^2}$$

For maximum value:

$$f'(p) = 0$$

Thus,

$$p^2 + p(2+x) + (1+x) = p(2p + 2 + x)$$

Simplifying,

$$1+x = p^2$$

Therefore,

$$\boxed{\text{A. } \sqrt{1+x}}$$

### Mistakes to be avoided

- Do not forget that only positive value is physically meaningful:

$$p = \sqrt{1+x}$$

**Q3.14.1**

**NAT**

**2014**

**2M**

**Ans: 0.3**



At balance,

$$\frac{Z_2}{Z_1} = \frac{Z_4}{Z_3}$$

Here,

$$Z_1 = R_1 = 35 \text{ k}\Omega, \quad Z_2 = R_2 = 105 \text{ k}\Omega$$

$$Z_3 = \frac{1}{j\omega C_1}, \quad Z_4 = \frac{1}{j\omega C_2}$$

Hence,

$$\frac{R_2}{R_1} = \frac{Z_4}{Z_3} = \frac{C_1}{C_2}$$

So,

$$\frac{105}{35} = \frac{C_1}{0.1} \Rightarrow 3 = \frac{C_1}{0.1} \Rightarrow C_1 = 0.3 \mu\text{F}$$

0.3

**Q3.14.2**

**NAT**

**2014**

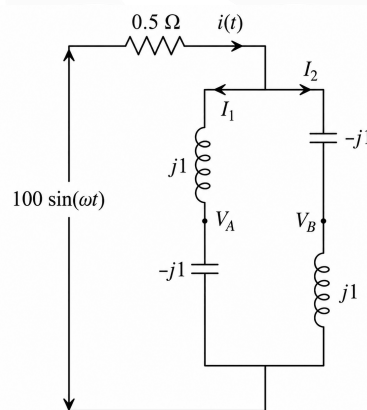
**2M**

**Ans: 141.4**



Given source:

$$v(t) = 100 \sin \omega t$$



**Fig. Solution Q3.14.2**

Hence RMS source voltage:

$$V_s = \frac{100}{\sqrt{2}} = 70.71 \text{ V}$$

Series resistance:

$$R = 0.5 \Omega$$

The bridge has two branches:

$$Z_1 = j1 + (-j1) = 0, \quad Z_2 = -j1 + j1 = 0$$

Thus the bridge network behaves as a short circuit, so the current through the circuit is very large, but the

voltmeter measures the potential difference between the midpoints of the two branches.

Taking current in each branch:

$$I_1 = I_2 = \frac{100/2}{0.5\sqrt{2}} = 70.71 \text{ A}$$

Voltage drop across the inductor:

$$V_L = I_b(j1) = j70.71$$

Voltage drop across the capacitor:

$$V_C = I_b(-j1) = -j70.71$$

So midpoint potentials are

$$V_A = -j70.71, \quad V_B = +j70.71$$

Hence voltmeter voltage:

$$V_V = V_A - V_B = -j70.71 - j70.71 = -j141.42$$

Therefore,

$$|V_V| = 141.42 \text{ V}$$

141.4

### Key Points

- Even though:

$$\frac{-j}{j} = \frac{j}{-j}$$

the bridge does not give zero output.

- The midpoint voltages are:

$$V_A = -j70.7, \quad V_B = +j70.7$$

- These voltages have equal magnitude but are  $180^\circ$  out of phase
- Hence,  $V_A \neq V_B$
- The bridge arms contain purely imaginary impedances with opposite signs, causing opposite phasor voltages.
- The balance condition of Wheatstone/AC bridge assumes finite impedances in all four arms and zero current through detector.

### Q3.13.1

MCQ

2013

2M

Ans: C



Given,

$$R_s = 300 \Omega, \quad R_1 = R_2 = R_3 = 300 \Omega$$

Maximum permissible current through strain gauge:

$$I_{\max} = 20 \text{ mA}$$

Resistance increases by 1%

Hence,

$$R'_s = 300(1 + 0.01) = 303 \Omega$$

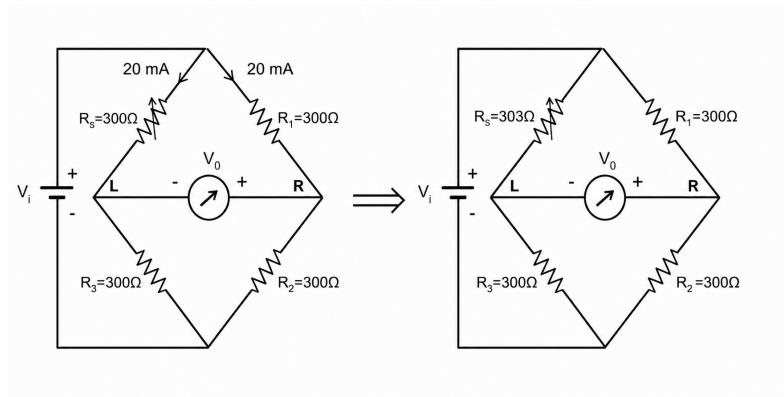


Fig. Solution Q3.13.1

### Step 1: Find excitation voltage

Initially the bridge is balanced, so all resistances are:

$$300\ \Omega$$

Current through strain gauge arm:

$$I = \frac{V_i}{300 + 300} = \frac{V_i}{600}$$

Given:

$$I = 20\ \text{mA} = 0.02\ \text{A}$$

Thus,

$$0.02 = \frac{V_i}{600} \Rightarrow V_i = 12\ \text{V}$$

### Step 2: Find midpoint voltages after strain

Left midpoint voltage:

$$V_L = 12 \times \frac{300}{303 + 300} = 12 \times \frac{300}{603} = 5.970\ \text{V}$$

Right midpoint voltage:

$$V_R = 12 \times \frac{300}{300 + 300} = 6\ \text{V}$$

### Step 3: Bridge output voltage

$$V_0 = V_R - V_L = 6 - 5.970 = 0.02985\ \text{V}$$

Converting to mV:

$$V_0 = 29.85\ \text{mV}$$

$$\boxed{\text{C. } 29.85\ \text{mV}}$$

### Mistakes to be avoided

- Carefully see the voltmeter connection and identify where the positive node and negative node are.

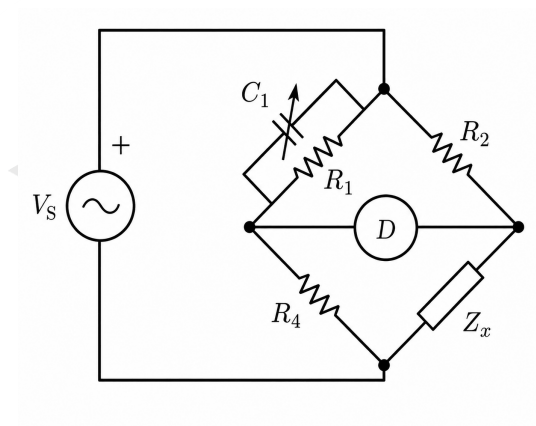
**Q3.12.1**
**MCQ**
**2012**
**1M**
**Ans: A**


Different AC bridges are used for measurement of different electrical quantities:

| Bridge                    | Used for / Measures                 |
|---------------------------|-------------------------------------|
| Heaviside-Campbell bridge | Mutual inductance                   |
| Schering bridge           | Capacitance and dielectric loss     |
| De Sauty bridge           | Comparison of lossless capacitances |
| Wien bridge               | Frequency measurement               |

Hence, the bridge used for measurement of mutual inductance is:

**A. Heaviside-Campbell bridge**

**Q3.11.1**
**MCQ**
**2011**
**1M**
**Ans: C**


**Fig. Solution Q3.11.1**

The given circuit is a **Maxwell bridge** (Maxwell inductance-capacitance bridge), identified by:

Variable capacitor,  $C_1$

in parallel with,  $R_1$

Resistive ratio arms,  $R_2, R_4$

Unknown impedance,  $Z_x$

A Maxwell bridge is used for measurement of:

Inductance of low-Q coils

It is most suitable for:

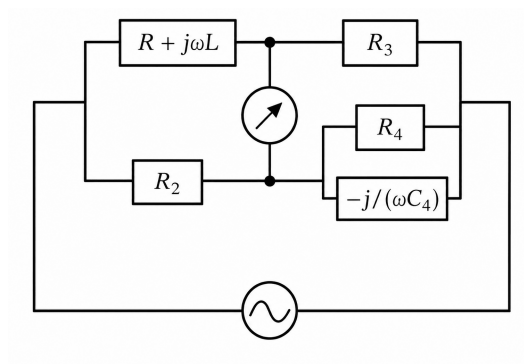
$$1 < Q < 10$$

Hence:

**C. low-Q inductor**

**Q3.10.1**
**MCQ**
**2010**
**2M**
**Ans: A**


This is a **Maxwell inductance-capacitance bridge**.



**Fig. Solution Q3.10.1**

Unknown inductive arm:

$$Z_1 = R + j\omega L$$

Other arms:

$$Z_2 = R_2, \quad Z_3 = R_3, \quad Z_4 = R_4 \parallel \frac{1}{j\omega C_4}$$

For balance,

$$\frac{Z_2}{Z_1} = \frac{Z_4}{Z_3}$$

Now,

$$Z_4 = \frac{R_4}{1 + j\omega C_4 R_4}$$

Hence,

$$\frac{R_2}{R + j\omega L} = \frac{R_4}{R_3} \cdot \frac{1}{1 + j\omega C_4 R_4}$$

So,

$$R + j\omega L = \frac{R_2 R_3}{R_4} (1 + j\omega C_4 R_4)$$

Comparing real and imaginary parts,

$$R = \frac{R_2 R_3}{R_4}$$

and

$$L = C_4 R_2 R_3$$

Hence, the correct option is

$$\text{A. } R = \frac{R_2 R_3}{R_4}, \quad L = C_4 R_2 R_3$$

### Key Points

- Maxwell bridge uses a:
 

Standard capacitor instead of a standard inductor
- This is advantageous because, Precision capacitors are cheaper and more stable and High-quality standard inductors are difficult to manufacture.

Q3.08.1

MCQ

2008

1M

Ans: A

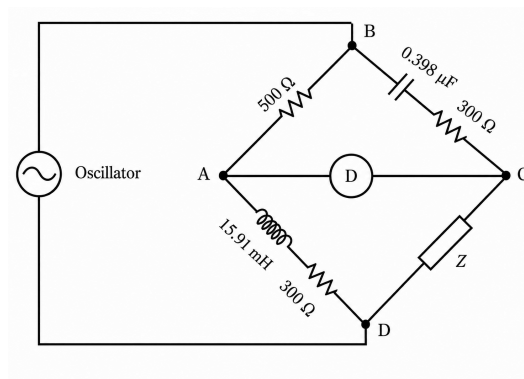


Fig. Solution Q3.08.1

Given bridge balance condition,

$$\frac{Z_{AB}}{Z_{AD}} = \frac{Z_{BC}}{Z_{CD}} \Rightarrow Z_{CD} = \frac{Z_{AB}Z_{BC}}{Z_{AD}}$$

Now,

$$Z_{AB} = 500 \Omega$$

$$Z_{AD} = 300 + j\omega L = 300 + jX_L$$

with

$$L = 15.91 \text{ mH}, \quad f = 2 \text{ kHz}$$

so

$$\omega = 2\pi f = 4000\pi, \quad X_L = \omega L \approx 4000\pi(15.91 \times 10^{-3}) \approx 200 \Omega$$

hence

$$Z_{AD} = 300 + j200$$

Also,

$$Z_{BC} = 300 - \frac{j}{\omega C} = 300 - jX_C$$

so

$$X_C = \frac{1}{\omega C} \approx \frac{1}{4000\pi(0.398 \times 10^{-6})} \approx 200 \Omega$$

hence

$$Z_{BC} = 300 - j200$$

Therefore,

$$Z_{CD} = \frac{500(300 - j200)}{300 + j200}$$

Multiplying by the conjugate,

$$Z_{CD} = 500 \frac{(300 - j200)^2}{300^2 + 200^2}$$

Since the product is purely real,

$$Z_{CD} = 260 + j0$$

$$\boxed{\text{A. } (260 + j0) \Omega}$$

Q3.07.1

MCQ

2007

1M

Ans: A

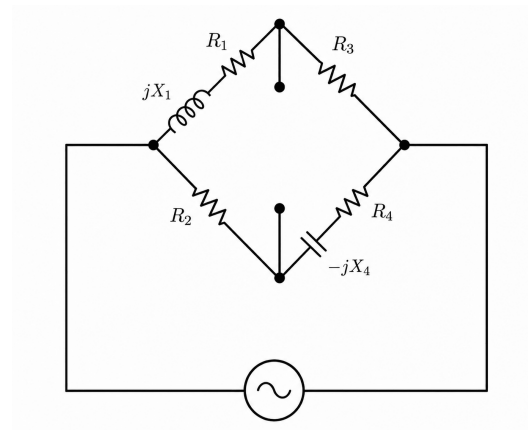


Fig. Solution Q3.07.1

The bridge arms are:

$$Z_1 = R_1 + jX_1, \quad Z_2 = R_2, \quad Z_3 = R_3, \quad Z_4 = R_4 - jX_4$$

At balance,

$$Z_1 Z_4 = Z_2 Z_3$$

So,

$$(R_1 + jX_1)(R_4 - jX_4) = R_2 R_3$$

Expanding,

$$R_1 R_4 + X_1 X_4 + j(X_1 R_4 - R_1 X_4) = R_2 R_3$$

Equating real and imaginary parts,

$$R_1 R_4 + X_1 X_4 = R_2 R_3, \quad X_1 R_4 = R_1 X_4$$

The imaginary equation must be satisfied first, so the phase balance is set by adjusting  $R_4$ . After that,  $R_1$  is adjusted to obtain final balance.

The solution where any one or both of  $R_1$  or  $R_4$  is/are 0 is also possible as the branch will not get shorted due to inductor and capacitor.

Hence, the correct sequence is:

First adjust  $R_4$ , then adjust  $R_1$ 

**Why others are wrong**

**B. First adjust  $R_2$ , then adjust  $R_3$  ×**

Both appear only in the real-balance term  $R_2 R_3$ . They cannot satisfy

$$X_1 R_4 = R_1 X_4$$

so phase balance is ignored.

**C. First adjust  $R_2$ , then adjust  $R_4$  ×**

The branch resistance  $R_2$  should not be zero while satisfying the real part equation/

**D. First adjust  $R_4$ , then adjust  $R_2$  ×**

The first step is correct, but the second step the branch resistance  $R_2$  should not be zero while satisfying the real part equation.

Therefore, the correct option is

**A. First adjust  $R_4$ , then adjust  $R_1$**

**Q3.03.1**
**MCQ**
**2003**
**2M**
**Ans: A**
☒ ☐ ☐

Different bridges are used for measurement of different electrical quantities:

| Group | Measurement                            | Correct Bridge       |
|-------|----------------------------------------|----------------------|
| $P$   | Resistance in milliohm range           | Kelvin double bridge |
| $Q$   | Low values of capacitance              | Schering bridge      |
| $R$   | Comparison of nearly equal resistances | Carey-Foster bridge  |
| $S$   | Inductance of high-Q coil              | Hay's bridge         |

Therefore, the correct option is

**A.  $P - 2, Q - 3, R - 6, S - 5$**



# DEBUG INFO

## Chapter III

DC And AC Bridges

**Total Questions : 13**

MCQ : 9

MSQ : 0

NAT : 4



PrepFusion

## SOLUTIONS

# IV

## Measurement of Resistance

### Topics

1. Kelvin Double Bridge
2. V-A, A-V method
3. Loss of Charge Method
4. Ohmmeter, Potentiometer

PrepFusion

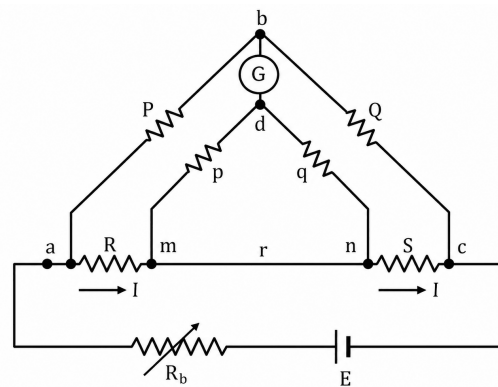
Q4.02.1

MCQ

2002

1M

Ans: A



Kelvin Double Bridge

Fig. Solution Q4.02.1

Kelvin double bridge is a modified Wheatstone bridge used for measuring:

Very low resistances

typically in:

milliohm ( $m\Omega$ ) and micro-ohm ( $\mu\Omega$ ) range

It eliminates errors caused by Lead resistance and Contact resistance which become significant while measuring very low resistances.

Therefore,

**A. Resistance of very low value**

Q4.97.1

MCQ

1997

1M

Ans: B

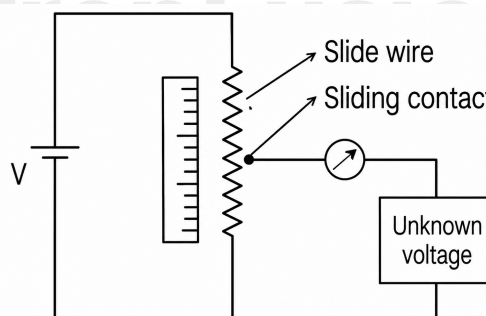


Fig. Solution Q4.97.1

A potentiometer works on the **null balance principle**.

It measures an unknown voltage by comparing it with a known voltage and adjusting until:

$$\text{Unknown voltage} = \text{Known voltage}$$

At balance:

$$I_G = 0$$

where  $I_G$  is galvanometer current.

Since measurement is obtained under zero-current condition, a potentiometer is called a:

Deflection instruments depend on pointer movement proportional to the measured quantity. Potentiometer works at zero deflection.

A potentiometer may be analog or digital; its defining feature is the null method.

Therefore,

**B. Null type instrument**

Q4.97.2

MCQ

1997

1M

Ans: B

● ○ ○

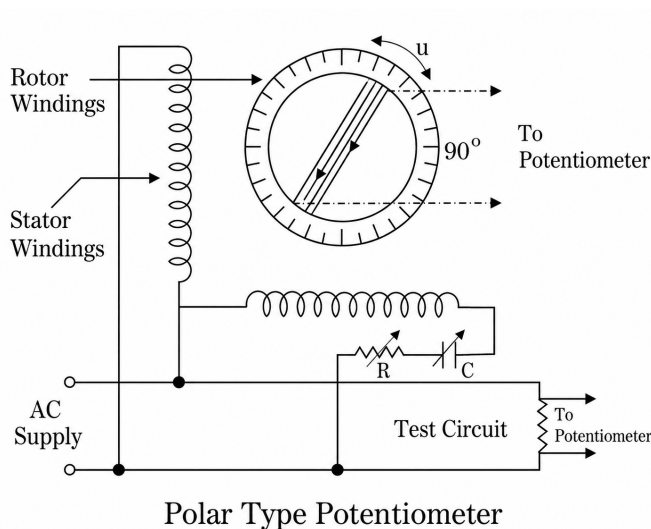


Fig. Solution Q4.97.2

A polar type AC potentiometer is standardized using a **transfer instrument**.

A transfer instrument should:

(i) work accurately on both AC and DC.

(ii) give the same indication for equal RMS values of AC and DC.

Thermal instruments satisfy these conditions because they operate on

$$\text{Heating effect} \propto I^2 R,$$

which is independent of waveform.

| Property                                       | Electrostatic       | Thermal         | Dynamometer       | PMMC                 |
|------------------------------------------------|---------------------|-----------------|-------------------|----------------------|
| Works on AC                                    | ✓                   | ✓               | ✓                 | ×                    |
| Works on DC                                    | ✓                   | ✓               | ✓                 | ✓                    |
| Measures true RMS                              | ×                   | ✓               | Approx.           | ×                    |
| Depends on waveform                            | Yes                 | No              | Yes               | –                    |
| Frequency effect                               | Present             | Negligible      | Present           | –                    |
| Phase error                                    | No                  | No              | Yes               | –                    |
| Suitable for AC–DC transfer                    | ×                   | ✓ Best          | Possible          | ×                    |
| Principle                                      | Electrostatic force | $I^2 R$ heating | EM torque         | Magnetic interaction |
| Main use                                       | High voltage        | RMS             | Power measurement | DC measurement       |
| Preferred for AC potentiometer standardization | ×                   | ✓               | ×                 | ×                    |

Therefore,

**B. a thermal instrument**

## Key Points

| Feature             | Polar Type AC Potentiometer          | Normal DC Potentiometer     |
|---------------------|--------------------------------------|-----------------------------|
| Measured quantity   | AC magnitude + phase angle           | DC magnitude only           |
| Standardization     | Requires AC & DC standardization     | Only DC standardization     |
| Balancing condition | Magnitude + phase balance            | Only magnitude balance      |
| Detector used       | AC detector / vibration galvanometer | DC galvanometer             |
| Application         | AC impedance, phase, pf measurement  | DC emf, resistance, current |

Q4.94.1

MCQ

1994

2M

Ans: B

● ○ ○

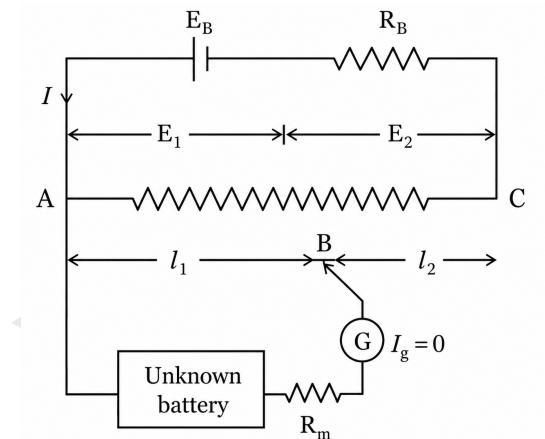


Fig. Solution Q4.94.1

For a potentiometer operating under null balance condition:

$$\frac{E_2}{E_1} = \frac{l_2}{l_1}$$

where:

$E_1$  = known (standard) cell emf,  $E_2$  = unknown/test cell emf

$l_1$  = balancing length corresponding to  $E_1$

$l_2$  = balancing length corresponding to  $E_2$

Given:

$$E_1 = 1.18 \text{ V}, \quad l_1 = 600 \text{ mm}, \quad l_2 = 680 \text{ mm}$$

Substituting,

$$\frac{E_2}{1.18} = \frac{680}{600}$$

Hence,

$$E_2 = 1.18 \times \frac{680}{600}$$

$$= 1.18 \times 1.1333 = 1.337 \text{ V}$$

Therefore,

$$E_2 \approx 1.34 \text{ V}$$

Hence, the correct option is:

**B. 1.34 V**

### Mistakes to be avoided

- Do not use the nominal potentiometer range directly for calculation.
- Think of it this way:

2 V over 800 mm  $\Rightarrow$  instrument specification      1.18 V at 600 mm  $\Rightarrow$  actual calibrated condition

- During measurement, the calibrated standard cell condition overrides the nominal range.
- Hence use:

$$\frac{1.18}{600} \quad \text{and not} \quad \frac{2}{800}$$

**Q4.92.1**

MCQ

1992

1M

Ans: C

☒ ☐ ☐

In a DC potentiometer, unwanted thermal emf (thermoelectric emf) may be generated at junctions of dissimilar metals due to temperature differences.

Let:

$$E_x = \text{unknown emf}, \quad E_t = \text{stray thermal emf}$$

Initially,

$$E_1 = E_x + E_t$$

After reversing the polarities of both the battery and galvanometer connections,

$$E_2 = E_x - E_t$$

Taking the average,

$$\frac{E_1 + E_2}{2} = \frac{(E_x + E_t) + (E_x - E_t)}{2} = \frac{2E_x}{2} = E_x$$

Thus,

Thermal emf cancels out

Therefore, the correct option is:

C. Stray thermal emf's

## DEBUG INFO

### Chapter IV

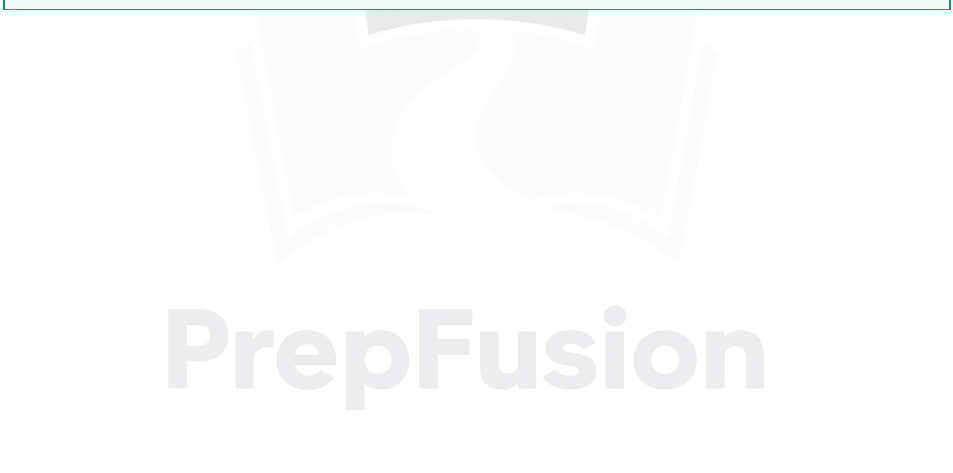
Measurement of Resistance

**Total Questions : 5**

MCQ : 5

MSQ : 0

NAT : 0



PrepFusion

SOLUTIONS



# Electronic Measurement and Instrument Transformer

## Topics

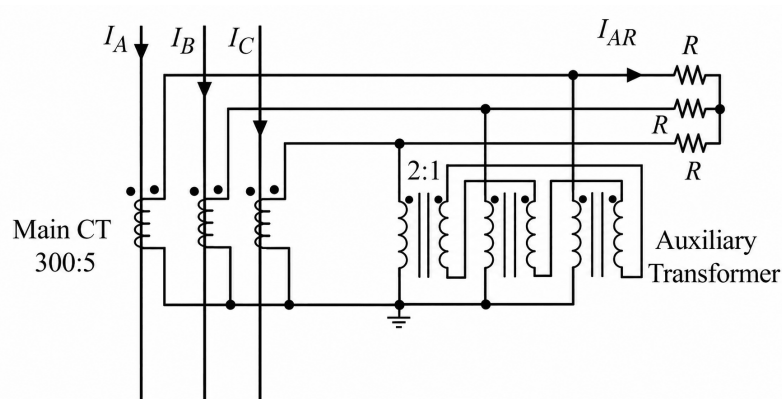
1. CRO
2. Current Transformer CT, Potential Transformer PT
3. Digital Voltmeter - DVM
4. Q-Meter

PrepFusion

**Q5.26.1**
**MCQ**
**2026**
**2M**
**Ans: D**


Given,

$$I_A = 572.812 + j50.115, \quad I_B = -254.525 - j459.175, \quad I_C = -207.083 + j444.091$$



**Fig. Solution Q5.26.1**

Main CT ratio:

$$300 : 5 \Rightarrow \text{secondary current} = \frac{1}{60} \times \text{primary current}$$

Hence, the **secondary currents of Main CT** are:

$$I_a = \frac{I_A}{60} = 9.583 \angle 5^\circ, \quad I_b = \frac{I_B}{60} = 8.75 \angle (-119^\circ), \quad I_c = \frac{I_C}{60} = 8.167 \angle 115^\circ$$

Zero-sequence current:

$$I_{a0} = \frac{I_a + I_b + I_c}{3}$$

Now,

$$I_a + I_b + I_c = 1.943 \angle 16.01^\circ$$

So,

$$I_{a0} = 0.648 \angle 16.01^\circ \approx 0.65 \angle 16^\circ$$

For the auxiliary transformer,

$$I_{AR} = I_a - I_{a0}$$

Thus,

$$I_{AR} = 9.583 \angle 5^\circ - 0.65 \angle 16^\circ \approx 8.954 \angle 4.105^\circ \text{ A}$$

**D.  $8.954 \angle 4.105^\circ \text{ A}$**

### Key Points

- Zero-sequence current:

$$I_0 = \frac{I_a + I_b + I_c}{3}$$

- For a  $Yn\Delta$  auxiliary transformer, the delta side blocks zero-sequence current.
- In symmetrical components:

$$I_a = I_{a1} + I_{a2} + I_{a0}$$

so the final delta-side current is:

$$I_{AR} = I_{a1} + I_{a2}$$

### Mistakes to be avoided

- Do not assume

$$I_A + I_B + I_C = 0$$

unless the system is perfectly balanced.

Q5.24.1

MCQ

2024

1M

Ans: C

● ○ ○

Comparison between CT and PT:

| Feature                                  | CT | PT | Reason                                 |
|------------------------------------------|----|----|----------------------------------------|
| Primary connected in parallel to grid    | ×  | ✓  | PT measures voltage                    |
| Open-circuited secondary not desirable   | ✓  | ×  | CT secondary should never be open      |
| Primary current is line current          | ✓  | ×  | CT primary carries actual line current |
| Secondary burden affects primary current | ×  | ✓  | PT behaves like a voltage transformer  |

Hence, the correct option is:

C. X matches with Q and R; Y matches with P and S.

### Key Points

- CT secondary should never be open-circuited.

$$I_s = 0 \Rightarrow \phi \uparrow \Rightarrow E_s \uparrow$$

where:

$$I_s = \text{secondary current, } \phi = \text{core flux, } E_s = \text{secondary induced emf}$$

- Hence dangerously high voltage appears across CT secondary.
- PT behaves like a normal transformer.
- Increasing secondary burden increases:

$$I_s \uparrow \Rightarrow I_p \uparrow$$

where:

$$I_p = \text{primary current}$$

- Therefore PT primary current depends on secondary loading.

Q5.17.1

MCQ

2017

1M

Ans: D

● ○ ○

For a stationary closed Lissajous pattern,

$$\frac{f_h}{f_v} = \frac{\text{Number of vertical tangencies}}{\text{Number of horizontal tangencies}}$$

Given,

$$N_H = 3, \quad N_V = 2, \quad f_h = 3 \text{ kHz}$$

Thus,

$$\frac{3}{f_v} = \frac{2}{3}$$

Hence,

$$f_v = 3 \times \frac{3}{2} = 4.5 \text{ kHz}$$

Therefore,

**D. 4.5 kHz**

**Q5.17.2**

**MCQ**

**2017**

**1M**

**Ans: A**



In a CRO:

Delay due to slope and level detector:

100 ns

Delay due to start-stop sweep generator:

50 ns

Hence total delay before horizontal sweep starts:

$$100 + 50 = 150 \text{ ns}$$

Therefore the input signal in the Y-channel must be delayed by the same amount so that the initial portion of the waveform is not lost.

Thus the required delay line is:

150 ns

inserted in:

Y-channel

Therefore, the correct option is:

**A. 150 ns inserted in Y-channel**

### Key Points

- Delay line in CRO delays the Y-signal so that the X-sweep gets enough time to start.
- This ensures the leading edge (initial portion) of the waveform becomes visible on the screen.

**Q5.14.1**

**MCQ**

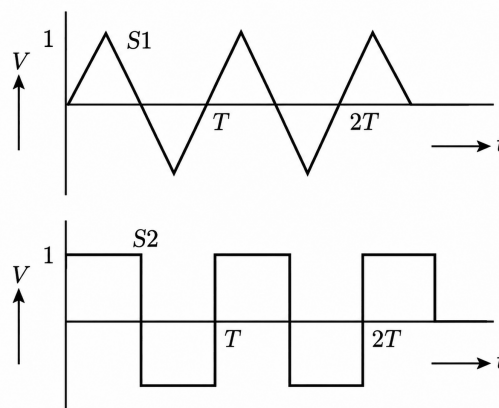
**2014**

**1M**

**Ans: A**



Given,  $S_1$  (triangular wave) is applied to the Y-plates and  $S_2$  (square wave) is applied to the X-plates.



**Fig. Solution Q5.14.1**

Thus,

$$X = S_2, \quad Y = S_1$$

The beam position over one cycle is:

| Time                | X from $S_2$ | Y from $S_1$ | Coordinate |
|---------------------|--------------|--------------|------------|
| $t = 0$             | +1           | 0            | (+1, 0)    |
| $t = \frac{T}{4}$   | +1           | +1           | (+1, +1)   |
| $t = \frac{T}{2}^-$ | +1           | 0            | (+1, 0)    |
| $t = \frac{T}{2}^+$ | -1           | 0            | (-1, 0)    |
| $t = \frac{3T}{4}$  | -1           | -1           | (-1, -1)   |
| $t = T^-$           | -1           | 0            | (-1, 0)    |
| $t = T^+$           | +1           | 0            | (+1, 0)    |

So the beam moves vertically upward at  $X = +1$ , returns to  $(+1, 0)$ , jumps horizontally to  $(-1, 0)$ , moves vertically downward, and then jumps back.

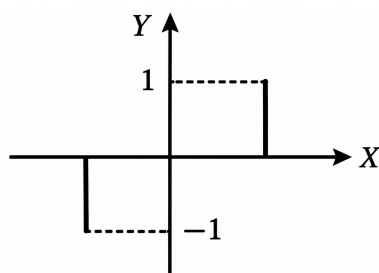


Fig. Solution Q5.14.1

This pattern matches **Option A**.

**A**

#### Mistakes to be avoided

- Follow waveform progression with time; wrong time ordering gives incorrect CRO pattern and lead answer to Option D.

#### Q5.14.2

MCQ

2014

1M

Ans: B

● ○ ○ ○

In a CRO/oscilloscope, the linear sweep (time-base signal) is applied to the horizontal deflection plates.

The purpose of the sweep voltage is to move the electron beam uniformly from left to right across the screen.

Thus,

$$V_x \propto t$$

Hence the horizontal direction represents time axis

**B. horizontal axis**

#### Key Points

- Time-base sweep  $\rightarrow$  X-plates (horizontal)
- Input signal  $\rightarrow$  Y-plates (vertical)
- Therefore CRO displays:

Voltage vs Time

### Q5.11.1

MCQ

2011

2M

Ans: C



Given:

Accuracy specification:

$$\pm(0.2\% \text{ of reading} + 10 \text{ counts})$$

Reading = 100 V

Full-scale range = 200 V

DMM =  $4\frac{1}{2}$ -digit

#### Method 1: Using Resolution of DMM

Error due to percentage of reading:

$$0.2\% \text{ of } 100 \text{ V} = \frac{0.2}{100} \times 100 = 0.2 \text{ V}$$

A  $4\frac{1}{2}$ -digit DMM on 200 V range generally displays:

$$199.99 \text{ V}$$

Hence resolution (least count):

$$0.01 \text{ V}$$

Therefore,

$$10 \text{ counts} = 10 \times 0.01 = 0.1 \text{ V}$$

Thus total error:

$$\pm(0.2 + 0.1) = \pm 0.3 \text{ V}$$

Percentage error:

$$\frac{0.3}{100} \times 100 = 0.3\%$$

#### Method 2: Using Maximum Count

For a  $4\frac{1}{2}$ -digit DMM:

$$\text{Maximum count} = 19999$$

Full-scale range:

$$200 \text{ V}$$

Hence,

$$1 \text{ count} = \frac{200}{19999} \approx 0.01 \text{ V}$$

Therefore,

$$10 \text{ counts} = 10 \times \frac{200}{19999} \approx 0.100005 \text{ V} \approx 0.1 \text{ V}$$

Percentage error component:

$$0.2\% \text{ of } 100 \text{ V} = 0.2 \text{ V}$$

Thus total error:

$$\pm(0.1 + 0.2) = \pm 0.3 \text{ V}$$

Percentage error:

$$\frac{0.3}{100} \times 100 = 0.3\%$$

Therefore,

$$\boxed{\text{C. } \pm 0.3\%}$$

## Key Points

- A  $4\frac{1}{2}$ -digit DMM means:

First digit: 0 or 1,      remaining digits: 0 to 9

- Common DMM counts:

| DMM Type              | Maximum Count |
|-----------------------|---------------|
| $3\frac{1}{2}$ -digit | 1999          |
| $4\frac{1}{2}$ -digit | 19999         |
| $5\frac{1}{2}$ -digit | 199999        |

### Q5.11.2

MCQ

2011

1M

Ans: C



In a dual-trace CRO, both input channels share the same vertical amplifier through an electronic switch (multiplexer). In **ALTERNATE** mode:

- Channel 1 is displayed during one sweep.
- Channel 2 is displayed during the next sweep.

Hence the multiplexer must switch channels once every sweep cycle.

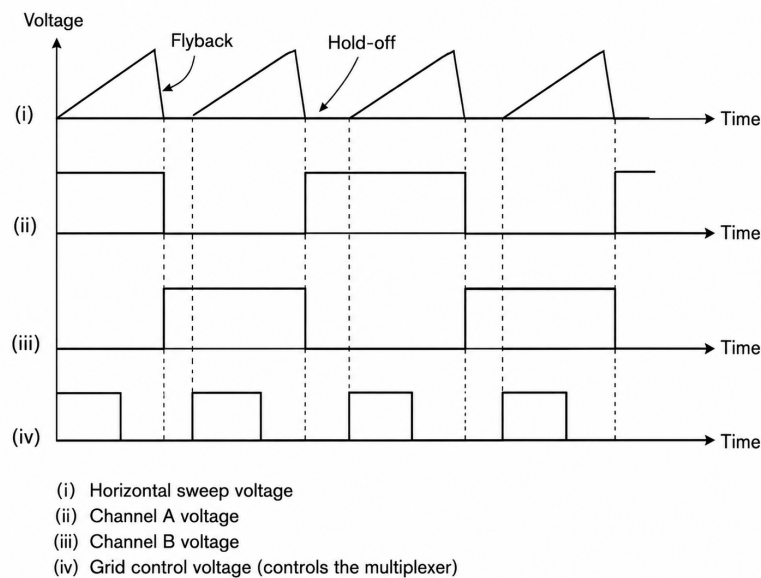


Fig. Solution Q5.11.2

The figure represents the waveform timing in a dual-trace oscilloscope operating in **ALTERNATE** mode. It shows how the oscilloscope alternately displays Channel A and Channel B during successive sweeps. Therefore, the switching frequency must be equal to the sweep frequency. Therefore, the correct option is:

**C. the frequency of the time-base (sweep) oscillator**

### Key Points

- In **Alternate mode**, channel switching occurs once per sweep.
- In **Chop mode**, channel switching occurs at very high frequency independent of sweep
- Alternate mode is preferred for high-frequency signals.
- Chop mode is preferred for low-frequency signals.

**Q5.09.1**

MCQ

2009

2M

Ans: A



Given:

The DMM is an average-reading type and indicates 10 V

For a symmetric triangular wave:

$$V_{avg} = \frac{V_m}{2}$$

Since the average-reading meter reads 10 V,

$$\frac{V_m}{2} = 10$$

Hence,

$$V_m = 20 \text{ V}$$

For a triangular waveform:

$$V_{rms} = \frac{V_m}{\sqrt{3}}$$

Substituting  $V_m = 20 \text{ V}$ ,

$$V_{rms} = \frac{20}{\sqrt{3}} \text{ V}$$

Therefore, the correct option is:

$$\text{A. } \frac{20}{\sqrt{3}} \text{ V}$$

**Q5.09.2**

MCQ

2009

1M

Ans: D



In the X-Y mode of a CRO:

- one signal is applied to the X-plates
- the other signal is applied to the Y-plates

If both signals are sinusoidal and have the same frequency, a Lissajous figure is obtained.

For:

$$x = A \sin \omega t, \quad y = B \sin(\omega t + \phi)$$

the shape depends on the phase difference  $\phi$ .

| Phase Difference              | Pattern                      |
|-------------------------------|------------------------------|
| $\phi = 0^\circ$              | Straight line                |
| $0^\circ < \phi < 90^\circ$   | Tilted ellipse               |
| $\phi = 90^\circ$             | Circle (if amplitudes equal) |
| $90^\circ < \phi < 180^\circ$ | Ellipse with opposite tilt   |

The figure changes:

ellipse  $\rightarrow$  circle  $\rightarrow$  ellipse

with changing orientation, indicating a varying phase difference between two sinusoidal signals of the same frequency.

Hence the correct option is:

**D.** There is a constant but small phase difference between the signals

### Key Points

- A stationary Lissajous figure implies:

$$f_1 = f_2$$

If  $f_1 \neq f_2$ , the pattern continuously drifts and becomes non-stationary.

**Q5.08.1**

MCQ

2008

2M

Ans: D



Given,

$$p(\omega_1 t) = A \sin \omega_1 t$$

applied to the X-input and the unknown signal  $q(\omega_2 t)$  applied to the Y-input. The CRO is in X-Y mode and shows a Lissajous figure.

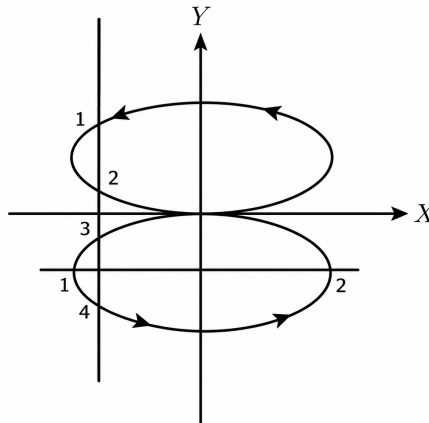


Fig. Solution Q5.08.1

For Lissajous figures,

$$\frac{f_x}{f_y} = \frac{\text{Number of tangencies to vertical sides}}{\text{Number of tangencies to horizontal sides}}$$

From the figure, the horizontal lines cut the curve at 2 points and vertical line at 4 points. Hence,

$$\frac{f_x}{f_y} = \frac{4}{2} \Rightarrow f_y = \frac{f_x}{2}$$

Therefore,

$$\omega_2 = \frac{\omega_1}{2}$$

The figure is symmetric about both axes and corresponds to a signal in quadrature phase. Since a cosine wave is a sine wave shifted by  $90^\circ$ ,

$$q(\omega_2 t) = A \cos \omega_2 t$$

Therefore, the correct option is

**D.**  $q(\omega_2 t) = A \cos \omega_2 t$ ,  $\omega_2 = \frac{\omega_1}{2}$

### Mistakes to be avoided

- Draw vertical and horizontal line(to find number of tangencies) at the place where you will get maximum points of intersection.
- Frequency ratio in Lissajous figures is inverse of what many students intuitively assume.
- Symmetry alone is not sufficient; also check loop orientation and number of lobes.

Q5.07.1

MCQ

2007

1M

Ans: B

● ● ○

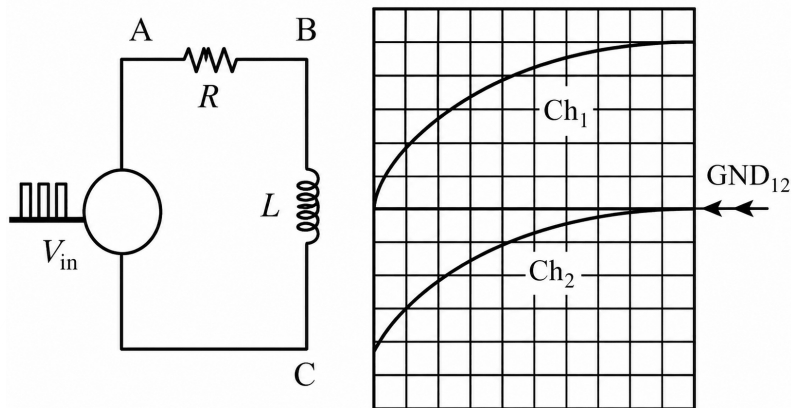


Fig. Solution Q5.07.1

The circuit is a series  $RL$  circuit driven by a square wave input.  
A non-isolated CRO has both channel grounds internally connected, so:

$$G_1 \equiv G_2$$

Hence both ground clips must be connected to the same circuit point. From the given waveforms, the common reference point is  $B$ . Therefore,

$$G_1 = G_2 = B$$

Now, channel 1 waveform starts from zero and rises exponentially, which is the voltage across the resistor in an  $RL$  transient. Hence the signal probe for Ch1 must be at  $A$ , so that

$$V_{AB} = V_R$$

Channel 2 waveform starts from high (in negative) and decays exponentially to zero, which is the voltage across the inductor. The ground is already at  $B$ , is the reason for getting negative value. The signal probe for Ch2 must be at  $C$ , so that

$$V_{BC} = V_L$$

Therefore the correct connection is:

**B. A, B, C, B**

### Q5.06.1

MCQ

2006

2M

Ans: A



Given:

$$K_n = \frac{200}{1} = 200, \quad I_P = 160 \text{ A}$$

Initial ratio error:

$$e_{r1} = -0.5\%$$

Initial phase angle error:

$$\delta_1 = 30'$$

Initial secondary turns:

$$N_{s1} = 200$$

After reducing one turn:

$$N_{s2} = 199$$

Let the actual transformation ratio be  $K_a$ .

For a current transformer,

$$K_a = n \left( 1 + \frac{I_e}{I_P} \right)$$

where  $n$  is the turns ratio and  $I_e$  is the excitation current component.

**Step 1: Find initial actual ratio**

Using the ratio error relation,

$$e_{r1} = \frac{K_n - K_{a1}}{K_{a1}} \times 100$$

So,

$$-0.5 = \frac{200 - K_{a1}}{K_{a1}} \times 100 \Rightarrow K_{a1} = 201$$

**Step 2: Find new actual ratio after reducing one turn**

Since actual ratio is proportional to secondary turns,

$$K_{a2} = K_{a1} \cdot \frac{N_{s2}}{N_{s1}} = 201 \cdot \frac{199}{200} \approx 200$$

**Step 3: Find new ratio error**

$$e_{r2} = \frac{K_n - K_{a2}}{K_{a2}} \times 100 = \frac{200 - 200}{200} \times 100 = 0\%$$

**Step 4: Phase angle error**

A small change of one secondary turn mainly corrects the ratio error, while the phase angle error remains practically unchanged.

Hence,

$$\delta_2 \approx 30'$$

Therefore,

$$\boxed{\text{A. } 0.0, 30'}$$

### Key Points

- Reducing secondary turns decreases turns ratio and is used for ratio-error compensation.
- Small reduction in turns usually has negligible effect on phase angle error.

### Mistakes to be avoided

- Do not ignore sign convention of ratio error.

**Q5.06.2**
**MCQ**
**2006**
**1M**
**Ans: A**
☒ ☐ ☐

A sampling wattmeter computes:

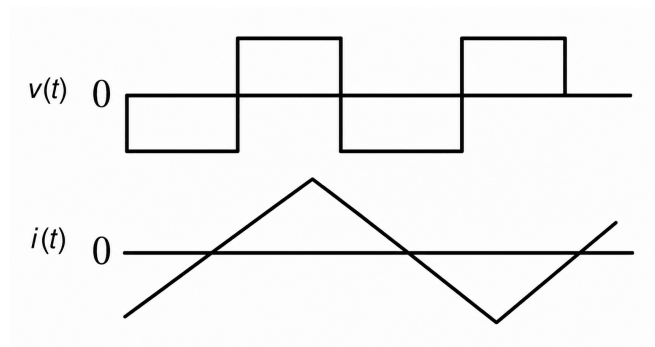
$$P_{\text{avg}} = \frac{1}{T} \int_0^T v(t)i(t) dt$$

Given square-wave voltage,

$$V_{pp} = 10 \text{ V} \Rightarrow V_m = 5 \text{ V}$$

So,

$$v(t) = \begin{cases} -5, & 0 < t < \frac{T}{2} \\ +5, & \frac{T}{2} < t < T \end{cases}$$



**Fig. Solution Q5.06.2**

Hence,

$$P_{\text{avg}} = \frac{1}{T} \left[ \int_0^{T/2} -5 i(t) dt + \int_{T/2}^T (5) i(t) dt \right]$$

$$P_{\text{avg}} = \frac{5}{T} \left[ \int_0^{T/2} -i(t) dt + \int_{T/2}^T i(t) dt \right]$$

From the given triangular waveform, the positive area during the first half cycle equals the negative area during the second half cycle, so

$$\int_0^{T/2} i(t) dt = \int_{T/2}^T i(t) dt$$

Therefore,

$$P_{\text{avg}} = \frac{5}{T} (A - A) = 0$$

**A. 0 W**

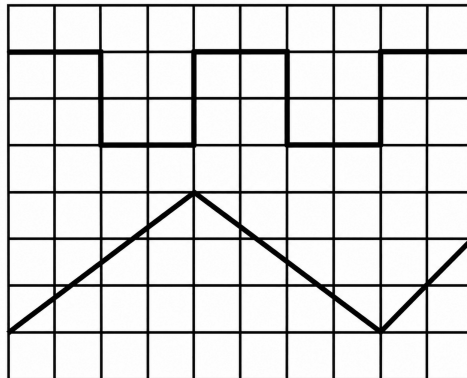
#### Mistakes to be avoided

- Be careful with signs and simplification of expression.

**Q5.06.3**
**MCQ**
**2006**
**1M**
**Ans: C**
☒ ☐ ☐

Given calibration signal:

$$f = 1 \text{ kHz} \Rightarrow T = \frac{1}{f} = 1 \text{ ms}$$



**Fig. Solution Q5.06.3**

From the upper square-wave trace, one complete cycle occupies 4 horizontal divisions. Hence,

$$\text{Time/div} = \frac{1 \text{ ms}}{4} = 0.25 \text{ ms/div}$$

The calibration pulse amplitude is:

$$V_{pp} = 5 \text{ V}$$

From the figure, this spans 2 vertical divisions. Hence,

$$\text{Voltage/div} = \frac{5}{2} = 2.5 \text{ V/div}$$

For the lower triangular waveform, peak-to-peak height is 3 vertical divisions. So,

$$V_{pp} = 3 \times 2.5 = 7.5 \text{ V}$$

One cycle of the triangular wave occupies 8 horizontal divisions. Hence,

$$T = 8 \times 0.25 = 2 \text{ ms}$$

Therefore,

$$\boxed{\text{C. } 7.5 \text{ V, } 2 \text{ ms}}$$

**Q5.05.1**
**MCQ**
**2005**
**2M**
**Ans: B**
☒ ☒ ☐

Given,

$$x(t) = P \sin(4t + 30^\circ)$$

So the angular frequency of X-input is

$$\omega_x = 4 \text{ rad/s}$$

For a stationary Lissajous figure,

$$\frac{f_x}{f_y} = \frac{\text{number of tangencies on vertical sides}}{\text{number of tangencies on horizontal sides}}$$

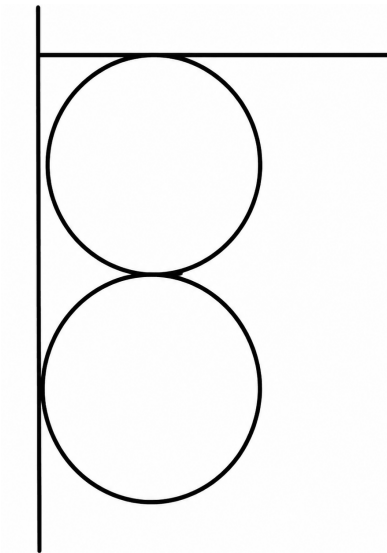


Fig. Solution Q5.05.1

From the given figure-8 pattern,

$$\frac{f_x}{f_y} = \frac{4}{2}$$

Hence,

$$f_y = \frac{f_x}{2} \Rightarrow \omega_y = \frac{\omega_x}{2} = \frac{4}{2} = 2 \text{ rad/s}$$

So the Y-input must have angular frequency 2 rad/s, and from the given options the matching signal is:

$$y(t) = Q \sin(2t + 15^\circ)$$

**B.  $Q \sin(2t + 15^\circ)$**

### Mistakes to be avoided

- Proper imagination of "a vertical figure-of-8 display" is important.

### Q5.05.2

MCQ

2005

1M

Ans: C

☒ ☐ ☐

A Q-meter is used to measure:

Quality factor ( $Q$ )

of coils and resonant circuits.

It operates on the principle of **series resonance**.

At resonance:

$$X_L = X_C$$

Hence the circuit impedance becomes minimum and current becomes maximum.

The quality factor is given by:

$$Q = \frac{V_C}{V}$$

where:

$V_C$  = voltage across capacitor,  $V$  = applied voltage

Thus the Q-meter measures voltage magnification in a series resonant circuit.

Therefore, the correct option is:

**C. Series resonance**

### Key Points

- Q-meter measurements become inaccurate for:

very low-Q coils, highly lossy inductors, and coils with large distributed capacitance

#### Q5.04.1

MCQ

2004

2M

Ans: A



Given,

$$f = 50 \text{ Hz}, \quad N_s = 500, \quad I_s = 5 \text{ A}, \quad R_b = 1 \Omega, \quad AT_m = 200$$

Since the burden is purely resistive, the secondary current is in phase with the secondary voltage and the magnetizing ampere-turns are perpendicular to the secondary ampere-turns.

Secondary ampere-turns:

$$AT_s = N_s I_s = 500 \times 5 = 2500$$

For a current transformer,

$$\tan \theta = \frac{AT_m}{AT_s}$$

Hence,

$$\tan \theta = \frac{200}{2500} = 0.08 \Rightarrow \theta = \tan^{-1}(0.08) \approx 4.6^\circ$$

Therefore, the correct option is

**A. 4.6°**

### Key Points

- For CT :

$$\tan \theta = \frac{\text{vertical component}}{\text{horizontal component}}$$

where,

$$I_c = \text{core-loss component}, \quad I_m = \text{magnetizing component}$$

- Therefore,

$$\tan \theta = \frac{I_m}{nI_s + I_c}$$

- For small angle and pure resistive load, we can prove

$$\tan \theta \approx \frac{I_m}{nI_s}$$

#### Q5.04.2

MCQ

2004

2M

Ans: B



Given,

$$f = 50 \text{ Hz}, \quad N_s = 500, \quad I_s = 5 \text{ A}, \quad R = 1 \Omega$$

Secondary voltage:

$$V_s = I_s R = 5 \times 1 = 5 \text{ V}$$

For sinusoidal flux, transformer emf equation is:

$$E_{\text{rms}} = 4.44 f N \phi_m$$

Neglecting leakage and the load is purely resistive, So;

$$E_s \approx V_s$$

Therefore,

$$5 = 4.44 \times 50 \times 500 \times \phi_m$$

Hence,

$$\phi_m = \frac{5}{4.44 \times 50 \times 500}$$

$$\phi_m = 4.5 \times 10^{-5} \text{ Wb}$$

**B.  $45 \mu\text{Wb}$**

### Key Points

- In a CT, secondary voltage is small and since the secondary turns are large,

$$E = 4.44 f N \phi_m$$

even a small voltage requires only a very small flux.

### Q5.04.3

MCQ

2004

2M

Ans: B



Given:

$$V_s = 10 \angle 0^\circ \text{ V}, \quad f = 100 \text{ kHz}$$

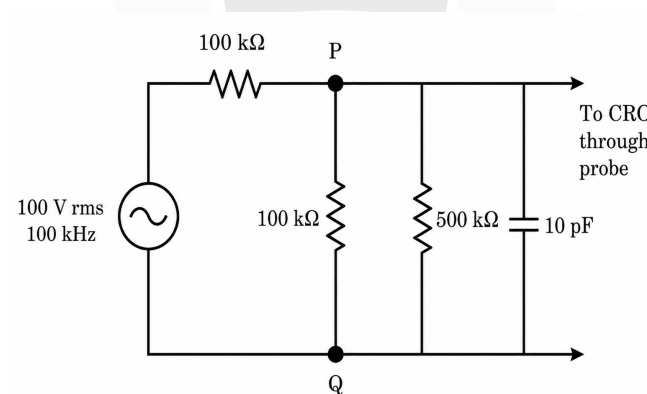


Fig. Solution Q5.04.3

Probe impedance:

$$500 \text{ k}\Omega \parallel 10 \text{ pF}$$

Capacitive reactance of probe:

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 100 \times 10^3 \times 10 \times 10^{-12}} = 159.15 \text{ k}\Omega$$

Applying KCL at node P,

$$\frac{V_p - 10}{100 \times 10^3} + \frac{V_p}{100 \times 10^3} + \frac{V_p}{500 \times 10^3} + \frac{V_p}{-j159.15 \times 10^3} = 0$$

Multiplying throughout by  $10^3$ ,

$$\frac{V_p - 10}{100} + \frac{V_p}{100} + \frac{V_p}{500} + \frac{V_p}{-j159.15} = 0$$

Rearranging,

$$\left( \frac{1}{100} + \frac{1}{100} + \frac{1}{500} + \frac{1}{-j159.15} \right) V_p = \frac{10}{100}$$

Now,

$$\frac{1}{100} = 0.01, \quad \frac{1}{500} = 0.002, \quad \frac{1}{-j159.15} = j0.006283$$

Hence,

$$(0.022 + j0.006283)V_p = 0.1$$

Therefore,

$$V_p = \frac{0.1}{0.022 + j0.006283}$$

$$V_p = 4.37 \angle -15.94^\circ \text{ V}$$

Therefore, the correct option is

**B. 4.37 V**

#### Mistakes to be avoided

- One can use KVL by simplifying the circuit instead of KCL, however answer will come but calculations might be tough in actual GATE paper, so try to understand circuit and what to apply where.

**Q5.03.1**

MCQ

2003

1M

Ans: D

● ○ ○

Given,

$$V_x = V_{xm} \sin \omega t, \quad V_y = V_{ym} \sin(\omega t + \phi)$$

applied to the X- and Y-plates of a CRO. The displayed Lissajous pattern depends on the phase difference  $\phi$ .

- Same-frequency sinusoidal signals produce Lissajous figures.
- Shape depends on phase difference.
- $\phi = 0^\circ$  gives a straight line with positive slope.
- $\phi = 180^\circ$  gives a straight line with negative slope.
- $\phi = 90^\circ$  gives a circle/ellipse with clockwise rotation.
- $\phi = 270^\circ$  gives a circle/ellipse with anti-clockwise rotation.
- Intermediate phase differences produce ellipses.

Hence, the correct option is

**D. P-1, Q-5, R-6, S-4**

**Q5.03.2**

MCQ

2003

2M

Ans: B

● ● ○

Given,

Rated CT:

$$\frac{500 \text{ A}}{5 \text{ A}}$$

Nominal ratio:

$$K_n = \frac{500}{5} = 100$$

Since it is a bar-primary CT,

$$N_p = 1$$

Hence,

$$N_s = 100$$

Secondary current:

$$I_s = 5 \text{ A}$$

Magnetizing ampere-turns:

$$AT_m = 250$$

Since the burden is purely resistive,

$$I_c = 0$$

Magnetizing current referred to the secondary side:

$$I_m = \frac{AT_m}{N_s} = \frac{250}{100} = 2.5 \text{ A}$$

$I_m$  is perpendicular to  $I_s$  (Since flux lags induced emf by  $90^\circ$ );

The actual primary current referred to the secondary side is:

$$I'_p = \sqrt{I_s^2 + I_m^2}$$

$$I'_p = \sqrt{5^2 + 2.5^2} = \sqrt{31.25} = 5.59 \text{ A}$$

Hence the actual transformation ratio is:

$$K_a = \frac{I'_p}{I_s} K_n$$

$$K_a = \frac{5.59}{5} \times 100 = 111.8$$

Therefore, percentage ratio error:

$$e_r = \frac{K_n - K_a}{K_a} \times 100$$

$$e_r = \frac{100 - 111.8}{111.8} \times 100 = -10.56\%$$

**B. - 10.56**

#### Mistakes to be avoided

- Be careful with the sign of ratio error. Options have both positive and negative values.

**Q5.03.3**

MCQ

2003

1M

Ans: A



Given,

$$Q = 120, \quad C_1 = 300 \text{ pF}$$

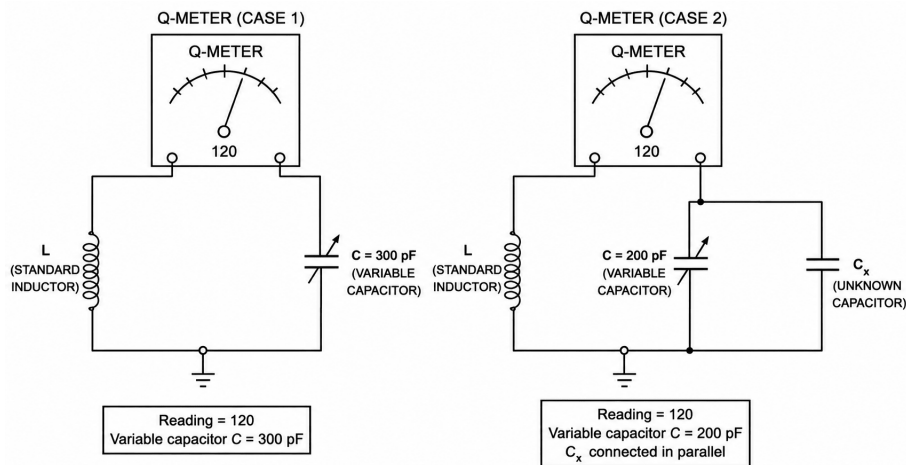


Fig. Solution Q5.03.3

When an unknown **lossless** capacitor  $C_x$  is connected in parallel with the variable capacitor, the same Q-meter reading is obtained after readjusting the variable capacitor to

$$C_2 = 200 \text{ pF}$$

Since the Q-meter operates at the same resonant condition, the total resonating capacitance remains unchanged. Initially,

$$C_{\text{total}} = 300 \text{ pF}$$

After connecting  $C_x$ ,

$$C_{\text{total}} = 200 + C_x$$

Equating the two:

$$200 + C_x = 300$$

Hence,

$$C_x = 100 \text{ pF}$$

A. 100

Q5.02.1

MCQ

2002

1M

Ans: D

● ○ ○

Given,

$$V_x = \sin(\omega t), \quad V_y = \sin(\omega t)$$

Since the two signals are identical and in phase,

$$V_y = V_x$$

In X-Y mode,

$$x = V_x, \quad y = V_y$$

Hence,

$$y = x$$

This is the equation of a straight line passing through the origin with slope

$$m = 1$$

Therefore,

$$\tan \theta = 1 \Rightarrow \theta = 45^\circ$$

So the trace observed on the CRO screen is a straight line inclined at  $45^\circ$  to the X-axis.

D. A straight line inclined at  $45^\circ$  with respect to the x-axis

**Q5.00.1**
**MCQ**
**2000**
**1M**
**Ans: C**
☒ ☐ ☐

Instrument transformers (CTs and PTs) are not ideal transformers.

They possess:

- winding resistance
- leakage reactance
- magnetizing reactance
- core-loss resistance

These are represented by the transformer's:

open-circuit parameters and short-circuit parameters

These non-ideal parameters cause:

- ratio (magnitude) error
- phase angle error

Hence, instrument transformer errors arise due to the open-circuit and short-circuit parameters of the transformer.

### C. Open and short circuit parameters of the instrument transformers

**Q5.99.1**
**MCQ**
**1999**
**2M**
**Ans: A**
☒ ☒ ☐

In a dual-slope ADC:

- The unknown input voltage  $V_{in}$  is integrated for a fixed time  $T_1$ .
- A known reference voltage  $V_{ref}$  of opposite polarity is then applied.
- The time taken for the integrator output to return to zero is the deintegration time  $T_2$ .

For a dual-slope ADC,

$$\frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1}$$

Hence,

$$V_{in} = V_{ref} \frac{T_2}{T_1}$$

Given,

$$V_{ref} = 100 \text{ mV}$$

$$T_1 = 300 \text{ ms}, \quad T_2 = 370.2 \text{ ms}$$

Therefore,

$$V_{in} = 100 \times \frac{370.2}{300}$$

$$V_{in} = 123.4 \text{ mV}$$

A  $3\frac{1}{2}$ -digit DVM can display values up to 199.9

**A. 123.4**

### Mistakes to be avoided

- Always compare the calculated value with the DVM/DMM display range. If the computed value exceeds the maximum displayable count, the instrument will indicate over-range (overflow) rather than the calculated value.

Q5.98.2

MCQ

1998

2M

Ans: C

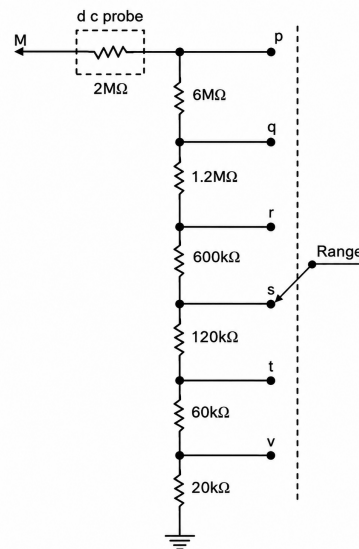


Fig. Solution Q5.98.2

Given,

$$V_M = 12 \text{ V}$$

with the switch at position  $q$ .

Total resistance from  $M$  to ground:

$$R_{\text{total}} = 2 + 6 + 12 + 0.6 + 0.12 + 0.06 + 0.02 = 10 \text{ M}\Omega$$

For switch position  $q$ , resistance from  $q$  to ground:

$$R_q = 1.2 + 0.6 + 0.12 + 0.06 + 0.02 = 2 \text{ M}\Omega$$

So the internal meter voltage at full-scale is:

$$V_{\text{meter}} = 12 \left( \frac{2}{10} \right) = 2.4 \text{ V}$$

Now for switch position  $s$ ,

$$R_s = 120 \text{ k}\Omega + 60 \text{ k}\Omega + 20 \text{ k}\Omega = 200 \text{ k}\Omega = 0.2 \text{ M}\Omega$$

At full-scale, the meter still requires 2.4 V, so

$$2.4 = V_M \left( \frac{0.2}{10} \right)$$

$$2.4 = 0.02 V_M \Rightarrow V_M = \frac{2.4}{0.02} = 120 \text{ V}$$

C. 120 V

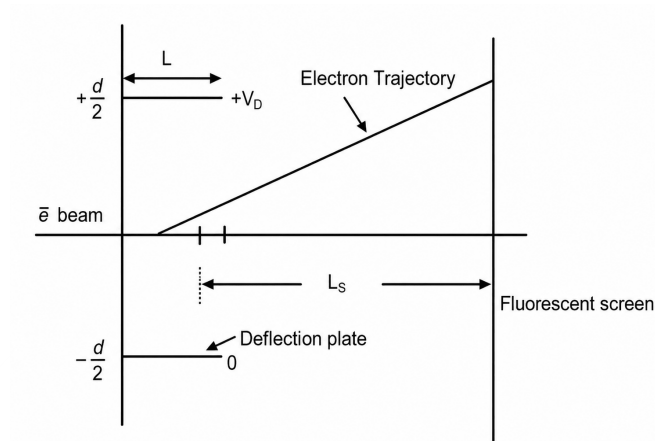
#### Mistakes to be avoided

- Be careful with units while adding resistances. Convert all values to a common unit ( $\text{M}\Omega$ ,  $\text{k}\Omega$ , etc.) before performing calculations.

**Q5.98.2**
**MCQ**
**1998**
**2M**
**Ans: A**


For an electrostatic deflection CRT:

- $V_A$  = accelerating voltage
- $V_D$  = deflecting voltage
- $d$  = separation between deflection plates
- $L_s$  = distance from centre of the plates to screen
- $L$  = length of deflection plates
- $m$  = mass of electron



**Fig. Solution Q5.98.2**

When an electron is accelerated through a voltage  $V_A$ ,

$$\frac{1}{2}mv_x^2 = eV_A \Rightarrow v_x^2 = \frac{2eV_A}{m}$$

The time spent inside the deflection plates is

$$t = \frac{L}{v_x}$$

The electric field between the plates is

$$E = \frac{V_D}{d}$$

Therefore, the vertical acceleration is

$$a_y = \frac{eE}{m} = \frac{eV_D}{md}$$

The vertical velocity acquired by the electron while traversing the plates is

$$v_y = a_y t = \left( \frac{eV_D}{md} \right) \left( \frac{L}{v_x} \right)$$

Hence, the deflection angle is

$$\tan \theta = \frac{v_y}{v_x} = \frac{eV_D L}{md v_x^2}$$

We get,

$$\tan \theta = \frac{V_D L}{2dV_A}$$

For a CRT, the electron trajectory appears to originate from the centre of the deflection plates. Therefore, the screen deflection is

$$D = L_s \tan \theta$$

Substituting for  $\tan \theta$ ,

$$D = L_s \left( \frac{V_D L}{2dV_A} \right) = \frac{LL_s V_D}{2dV_A}$$

Hence, the deflection sensitivity is

$$S = \frac{D}{V_D} = \frac{LL_s}{2dV_A}$$

Therefore,

$$S \propto \frac{LL_s}{dV_A}$$

Hence,

$$\mathbf{A. \frac{LL_s}{dV_A}}$$

**Q5.96.1**

**NAT**

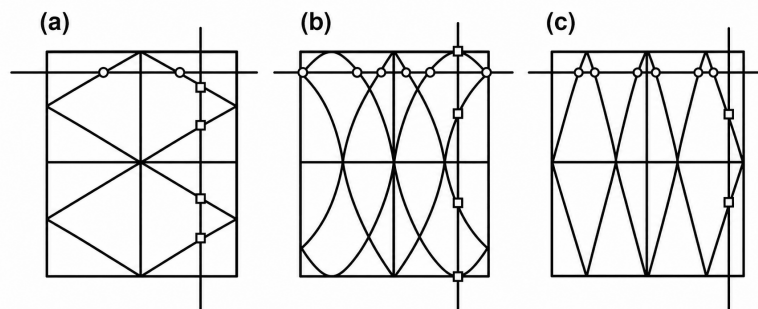
**1996**

**3M**

**Ans: \***



Given  $f_x = 100$  Hz.



**Fig. Solution Q5.96.1**

For a stationary Lissajous figure,

$$\frac{f_x}{f_y} = \frac{\text{Number of vertical tangencies}}{\text{Number of horizontal tangencies}}$$

**(a)** From the figure,

$$N_V = 4, \quad N_H = 2$$

Hence,

$$\frac{100}{f_y} = \frac{4}{2} = 2 \Rightarrow f_y = 50 \text{ Hz}$$

**(b)** From the figure,

$$N_V = 4, \quad N_H = 6$$

Hence,

$$\frac{100}{f_y} = \frac{4}{6} = \frac{2}{3} \Rightarrow f_y = 150 \text{ Hz}$$

(c) From the figure,

$$N_V = 2, \quad N_H = 6$$

Hence,

$$\frac{100}{f_y} = \frac{2}{6} = \frac{1}{3} \Rightarrow f_y = 300 \text{ Hz}$$

Therefore,

$$(a) \rightarrow 50 \text{ Hz (P)}, \quad (b) \rightarrow 150 \text{ Hz (S)}, \quad (c) \rightarrow 300 \text{ Hz (U)}$$

**Q5.95.1**

MCQ

1995

1M

Ans: B



The sweep generator of a CRO produces a sawtooth (ramp) voltage  $V_s(t)$ .

Normally, this voltage is applied to the horizontal plates.

If the sweep output is connected to the input channel, then:

$$V_x = V_s(t) \quad \text{and} \quad V_y = V_s(t)$$

Thus,

$$V_y = V_x$$

Hence,

$$y = x$$

This represents a straight line passing through the origin.

Since the X- and Y-channel sensitivities are equal, the line is inclined at  $45^\circ$ .

**B. Diagonal line**

**Q5.95.2**

MCQ

1995

1M

Ans: A



Given,

$$v(t) = 1 + \sin(314t) \text{ V}$$

Since,

$$314 = 2\pi \times 50$$

the signal can be written as

$$v(t) = 1 + \sin(2\pi 50t)$$

Thus,

$$\text{DC component} = 1 \text{ V}$$

and

$$\text{AC component} = \sin(2\pi 50t)$$

The integration time is:

$$T_{\text{int}} = 300 \text{ ms} \quad \text{and} \quad \text{signal frequency, } f = 50 \text{ Hz}$$

Hence the signal period is:

$$T = \frac{1}{50} = 20 \text{ ms}$$

Number of cycles integrated:

$$\frac{300}{20} = 15$$

Thus the ADC integrates over exactly 15 complete cycles.

A dual-slope ADC measures the average value of the input during the integration interval.

The average value of a sine wave over an integral number of cycles is:

$$\frac{1}{T} \int_0^T \sin(\omega t) dt = 0$$

Therefore,

$$V_{\text{avg}} = 1 + \text{Average}(\sin(314t)) = 1 + 0 = 1 \text{ V}$$

**A. 1.000**

### Key Points

- A dual-slope ADC measures the average value of the input over the integration interval; if the integration time contains an integer number of AC cycles, the average AC component is zero.

### Mistakes to be avoided

- Always check whether the integration time contains an integer number of cycles. If it does, the average value of the sinusoidal component is zero; otherwise it must be evaluated explicitly.

**Q5.94.1**

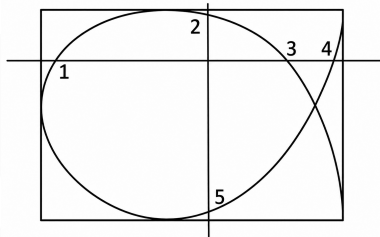
**MCQ**

**1994**

**1M**

**Ans: C**

☒ ☐ ☐



**Fig. Solution Q5.94.1**

For a **stationary Lissajous figure**,

$$\frac{f_x}{f_y} = \frac{\text{Number of tangencies to vertical sides}}{\text{Number of tangencies to horizontal sides}}$$

From the figure, tangencies to the vertical sides are (2 and 5):

$$N_v = 2$$

Similarly, tangencies to the horizontal sides are (1, 3 and 4):

$$N_h = 3$$

Therefore,

$$\frac{f_x}{f_y} = \frac{N_v}{N_h} = \frac{2}{3}$$

**C. 2 : 3**

**Q5.93.1**
**MCQ**
**1993**
**1M**
**Ans: B**
☒ ☐ ☐

Given,

$$v(t) = 5 \sin(314t + 45^\circ)$$

Comparing with

$$v(t) = V_m \sin(\omega t + \phi)$$

we get

$$\omega = 314 \text{ rad/s}$$

Hence,

$$f = \frac{\omega}{2\pi} = \frac{314}{2\pi} \approx 50 \text{ Hz}$$

Therefore,

$$T = \frac{1}{f} = \frac{1}{50} = 20 \text{ ms}$$

The CRO screen has 10 horizontal divisions and the time-base setting is 5 ms/div.

Hence total time displayed on the screen is

$$10 \times 5 = 50 \text{ ms}$$

Number of cycles displayed:

$$N = \frac{\text{Total screen time}}{\text{Time period}} = \frac{50}{20} = 2.5$$

Therefore,

**B. 2.5 cycles**

**Q5.92.1**
**NAT**
**1992**
**2M**
**Ans: 0.3**
☒ ☒ ☐

Given,

$$f = 50 \text{ Hz}$$

The first integration is carried out for 10 periods.

Hence,

$$T_1 = 10 \left( \frac{1}{f} \right) = 10 \left( \frac{1}{50} \right) = 0.2 \text{ s}$$

Reference voltage:  $V_{\text{ref}} = 2 \text{ V}$  and Input voltage:  $V_{\text{in}} = 1 \text{ V}$

For a dual-slope ADC,

$$\frac{V_{\text{in}}}{V_{\text{ref}}} = \frac{T_2}{T_1}$$

Therefore,

$$T_2 = T_1 \frac{V_{\text{in}}}{V_{\text{ref}}} = 0.2 \times \frac{1}{2} = 0.1 \text{ s}$$

Hence the total conversion time is

$$T_{\text{conv}} = T_1 + T_2 = 0.2 + 0.1 = 0.3 \text{ s}$$

**0.3**

### Mistakes to be avoided

- Do not report only the deintegration time  $T_2$ ; the conversion time is:

$$T_{\text{conv}} = T_1 + T_2$$

## DEBUG INFO

### Chapter V

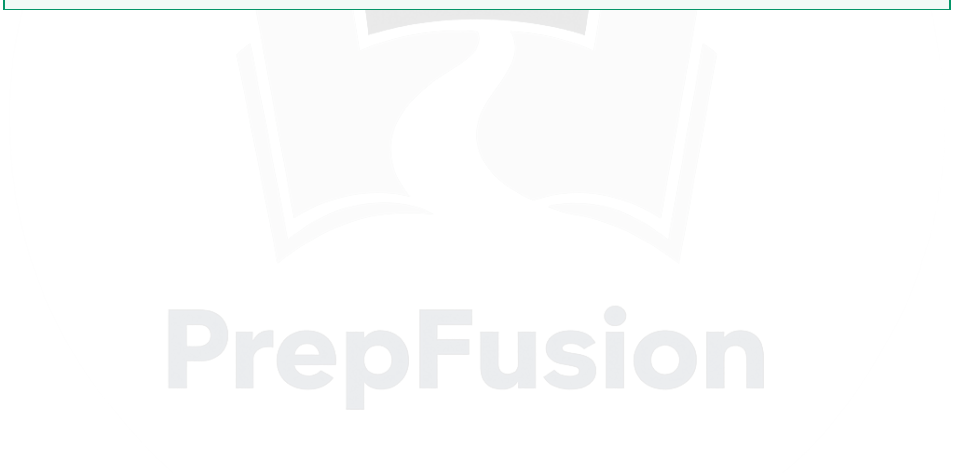
Electronic Measurement and Instrument Transformer

**Total Questions : 34**

MCQ : 32

MSQ : 0

NAT : 2



PrepFusion